

Chemistry

CYL1010



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Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

Contents

- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion

Band Theory of Solids

Introduction

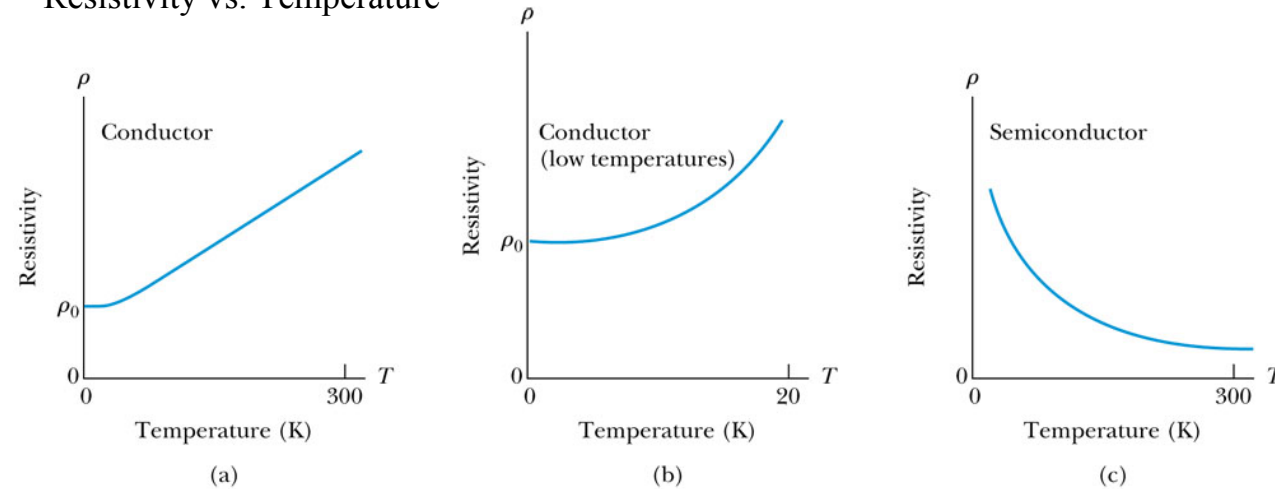
The band theory of solids is the basic principle of semiconductor physics and it is used to explain the differences in electrical properties between Metals, Insulators and Semiconductors.

What is Band Theory of Solids?

- ✓ In molecules, Two atomic orbitals combine together to form a molecular orbit with two distinct energy levels.
- ✓ In solids, 10^{23} stacked up lines confined in a tiny space would look like a band. Thereby forming energy continuum called energy bands.
- ✓ *This theory explains the quantum state that an electron takes inside metal solid. Every molecule comprises of various discrete energy levels. The way electrons behave inside a molecule is well explained through this theory.*
- ✓ *This theory helps to visualize the difference between conductor, semiconductor and an insulator by plotting available energies for an electron in a material.*

Band Theory of Solids

Resistivity vs. Temperature



- In order to account for *decreasing* resistivity with increasing temperature as well as other properties of semiconductors, a new theory known as the **band theory** is introduced.
- The essential feature of the band theory is that the allowed energy states for electrons are nearly continuous over certain ranges, called **energy bands**, with forbidden energy gaps between the bands.

Figure (a) Resistivity versus temperature for a typical conductor. Notice the linear rise in resistivity with increasing temperature at all but very low temperatures. (b) Resistivity versus temperature for a typical conductor at very low temperatures. Notice that the curve flattens and approaches a nonzero resistance as $T \rightarrow 0$. (c) Resistivity versus temperature for a typical semiconductor. The resistivity increases dramatically as $T \rightarrow 0$.

What is the Band Theory of Solids?

Band theory of solids describes the quantum state that an electron takes inside a metal solid. Every molecule comprises various discrete energy levels. The way electrons behave inside a molecule is well explained through band theory. **Band Theory was developed from the knowledge gained during the quantum revolution in science. In 1928, Felix Bloch applied quantum theory to solids.** In atoms, electrons are filled in respective energy orbits following Pauli's exclusion principle. Two atomic orbitals combine to form a molecular orbit with two distinct energy levels. In solids, 10^{23} stacked up lines confined in a tiny space would look like a band. Thereby forming an energy continuum called energy bands. Band theory helps to visualize the difference between a conductor, semiconductor, and an insulator by plotting available energies for an electron in a material.

Energy Bands in Solids

In the band theory of solids, there are many energy bands but the following are the three most important energy bands in solids:

- Valence Band
- Conduction Band
- Forbidden Band

Valence band

The energy band that consists of valence electrons energy levels, is known as the valence band. The valence band is present below the conduction band and the electrons of this band are loosely bound to the nucleus of the atom.

Conduction band

The energy band that consists of free electrons energy levels, is known as the conduction band. For electrons to be free, external energy must be applied such that the valence electrons get pushed to the conduction band and become free.

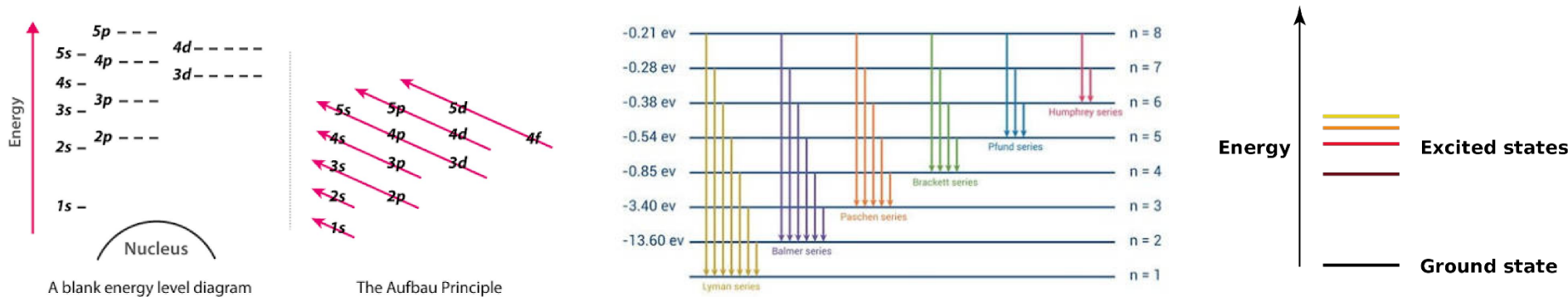
Forbidden band

The energy gap between the valence band and the conduction band is known as the forbidden band which is also known as the forbidden gap. The electrical conductivity of a solid is determined by the forbidden gap and also the classification of the materials as conductors, semiconductors, and insulators.

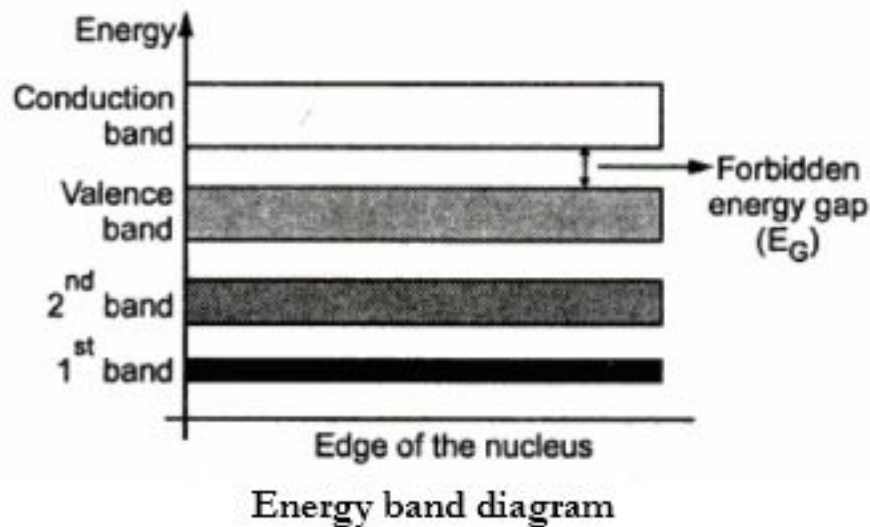
Band Theory of Solids

Atomic Energy Levels and Solids

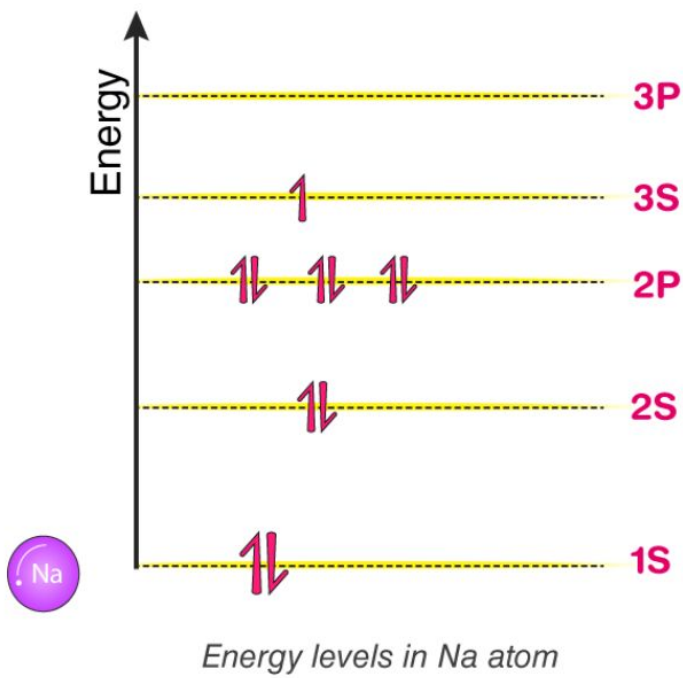
- In an **isolated atom**, electrons occupy **discrete energy levels** (like shells or orbitals).



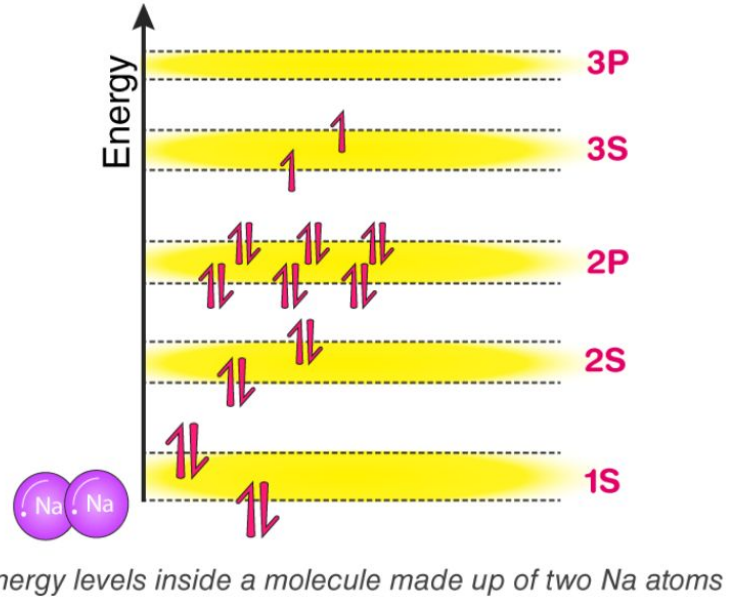
In a **solid**, atoms are **very closely packed** — sometimes on the order of 10^{23} atoms in a crystal. Because of this close packing, *the atomic energy levels interact (due to overlapping electron wavefunctions)*. These interactions **split the discrete levels** into a **large number of closely spaced levels** — forming what we call **energy bands**.



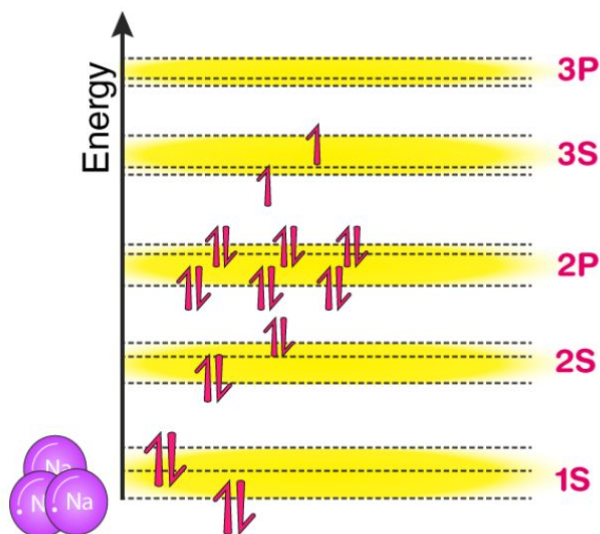
Energy Band inside an atom: Specific example with sodium metal



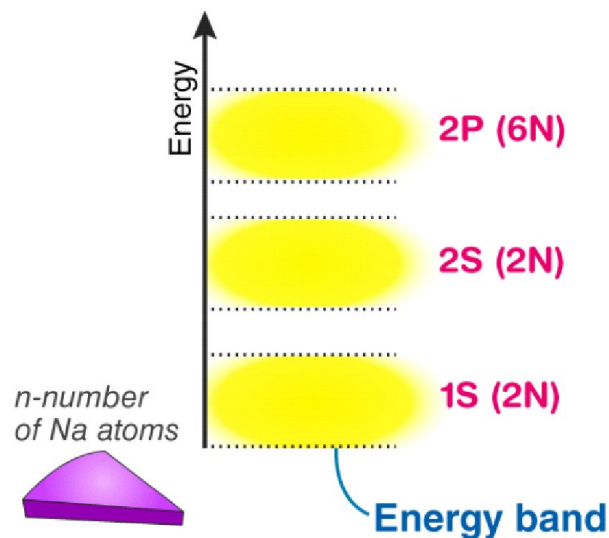
Consider a Sodium atom. It comprises 11 electrons. They fill up the energy level following Pauli's exclusion principle



Energy levels inside a molecule made up of two atoms
 What happens when two sodium atoms are very close to each other, almost forming a molecule? Now each atom cannot have the configuration as it follows as an individual atom. If they do, they will violate Pauli's exclusion principle and end up with many electrons of the same energy levels. When two atoms come very close to each other, What is going to happen to this system? into what we call Molecular orbit. That The answer is that their respective energy bands will overlap with each other and transform is, the 1s orbit of individual Sodium atoms combines to form 1s molecular orbital. As two atomic orbitals are overlapping, the molecular orbit ends up having two discrete energy levels. The Lower energy level is called a bonding orbital, and the Higher energy level is called an anti-bonding orbital.



Energy levels inside a molecule made up of three Na atoms



Energy levels inside a molecule made up of n-number of Na atoms

Energy levels inside a molecule made up of three atoms

Now try to picture what will happen if we add a third sodium atom to the mix? Well, according to the theory we learned just now. Here three atomic orbitals will be overlapping, forming a single molecular orbital with three discrete energy levels. Each molecular orbital here will inherit three energy levels. In general, the more we add atoms, the more energy levels the molecular orbit will have.

Energy levels inside a solid made up of n-number of atoms

In general, if there are n-number of atoms, there will be n discrete energy levels in each energy band. In such a system of n number of atoms, the molecular orbitals are called energy bands. Single 1s orbital and 2s orbital can fit 2 electrons each. Thus, the total number of electrons a 1s and 2s energy band can fit is 2n. A single 2p level can fit 6 electrons so that a 2p energy band can fit 6n electrons so on and so forth.

Why do we see energy gap between molecular orbitals?

Additional clarification: *“the atomic energy levels interact (due to overlapping electron wavefunctions)”*

•Atomic Orbitals Combine:

- When atoms combine to form a molecule, their atomic orbitals interact and combine to form molecular orbitals.

•Constructive and Destructive Interference:

- This interaction can be either constructive (leading to bonding orbitals with lower energy) or destructive (leading to antibonding orbitals with higher energy).

•Bonding Orbitals:

- In bonding orbitals, the electron density is concentrated between the nuclei, leading to increased stability and lower energy.

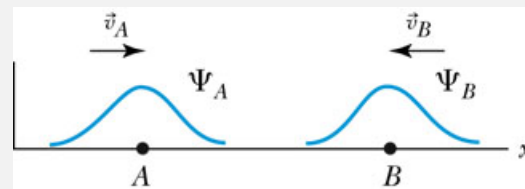
•Antibonding Orbitals:

- In antibonding orbitals, there is a node (a region of zero electron density) between the nuclei, leading to increased repulsion and higher energy.

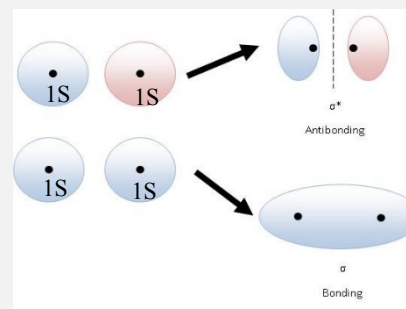
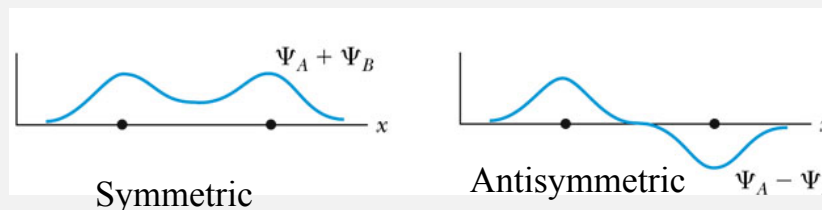
•Energy Gap:

- The difference in energy between the bonding and antibonding orbitals is the energy gap.

Consider initially the known wave functions of two hydrogen atoms far enough apart so that they do not interact.



Interaction of the wave functions occurs as the atoms get closer:



An electron in the symmetric state has a nonzero probability of being halfway between the two atoms, while an electron in the antisymmetric state has a zero probability of being at that location.

Band Theory of Solids

In the symmetric case the binding energy is slightly stronger resulting in a lower energy state. Thus there is a splitting of all possible energy levels (1s, 2s, and so on).

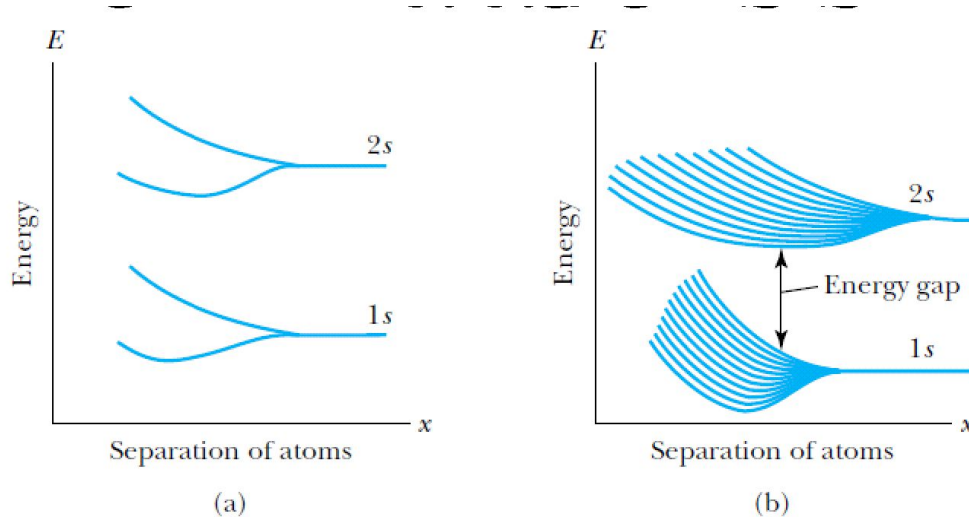
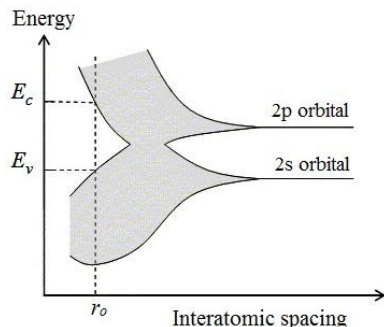


Figure 11.3 The 1s and 2s energy-level splittings of approaching hydrogen atoms for (a) 2 atoms and (b) 11 atoms. Notice the splitting of each energy level into a nearly continuous band.

- Electrons occupy energy states in atomic orbitals
- When several atoms are brought close to each other in a solid these energy states split in to a series of energy states (molecular orbitals).



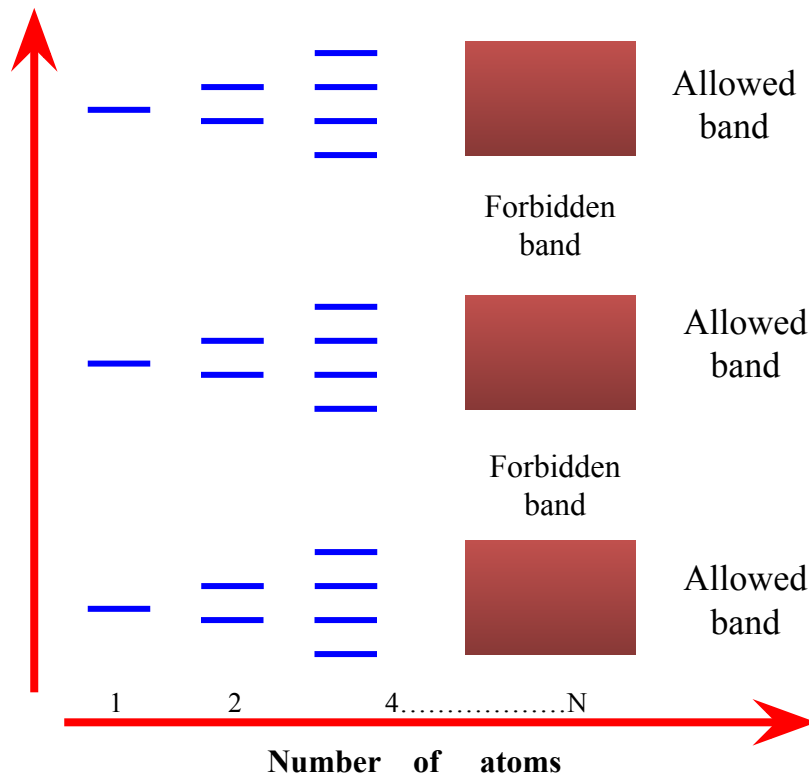
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If we now decrease the interatomic spacing of our imaginary, an atom will exert an electric force on its neighbors. Because of this coupling between atoms, the atomic-wave functions overlap

an energy band.

Band Formation in crystals

We know that the electrons in an atom are present in different energy level. When we try to assemble a lattice of a solid with N atoms, then each level of an atom must split up into N levels in the solid. This splitting up of sharp and tightly packed energy levels forms Energy Bands. The gap between adjacent bands representing a range of energies that possess no electron is called a Forbidden gap.



- The electrons surrounding a nucleus have certain well-defined energy-levels.
- Electrons don't like to have the same energy in the same potential system.
- The most we could get together in the same energy-level was two, provided that they had opposite spins. This is called *Pauli Exclusion Principle*.

In the solid, the atomic energy levels combine together to form energy bands, valence band and conduction band. Why do consider only these two bands, and don't consider so many other such continuum energy band?

- Yes, many energy bands form in a solid.
- But we usually talk about two main bands — valence and conduction — because:
- They're the most relevant for describing electrical, thermal, and optical properties.
- *Lower bands are filled and inert, and higher bands are often empty and not involved under normal conditions*

The **conduction and valence bands** are:

1. Closest to the Fermi level.
2. Most relevant to solid-state physics, semiconductors, and electrical engineering.

• **Only a few bands are near the Fermi level** (the highest energy electrons occupy at 0 K). These are the bands that participate in:

- Electrical conduction
- Thermal conduction
- Optical properties

• **The lower energy bands** (from inner core electrons) **also form bands**, but:

- They're **completely filled**.
- The energy gap to higher states is large, so they don't affect conduction.
- They are **chemically inert** in most cases.

• The **conduction and valence bands** are:

- Closest to the Fermi level.
- Most relevant to solid-state physics, semiconductors, and electrical engineering

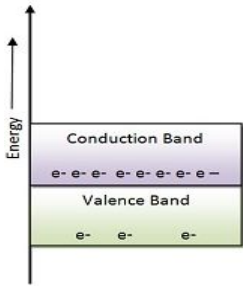
Classification of crystalline solids

Based on the width of forbidden band, solids are classified into insulators, Semiconductors and conductors.

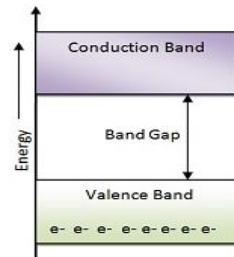
Conductors: The valence band and conduction bands are overlapped with each other and the energy gap E_g is zero. At room temperatures, free electrons already exist in huge number at conduction band. Hence these solids are good electrical conductors as well as good thermal conductors. The electrical resistivity increases at high temperature by collisions among the free electrons. Ex: Al, Cu, Ag, Au etc.,

Semiconductors: The valence band and conduction bands are separated with a small energy band gap $E_g \approx 1$ eV. At low temperatures (0 K), free electrons are not available in conduction band. Hence, they behave like insulators at low temperatures. The electrical conductivity increases at high temperatures by the transition of free electrons from valence band to conduction band. Thus, these solids behave like electrical conductors at high temperatures. Ex: Silicon $E_g = 1.1$ eV, Germanium $E_g = 0.7$ eV

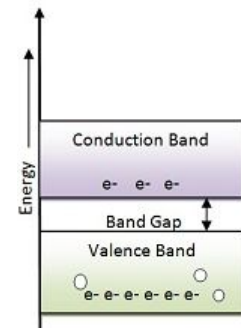
Insulators: The valence band and conduction bands are separated by a very large energy gap $E_g \geq 3$ eV. At room temperatures conduction band is empty and valence band is full of electrons. Hence, they these solids are electrical insulators. Even at high temperatures valence electrons are unable to jump into conduction band. Ex: Glass, Mica, Ebonite, Rubber etc.,



Conductor: have a very small energy gap
□ Result: current flows easily



Insulator: have a large energy gap
□ Result: no current flows

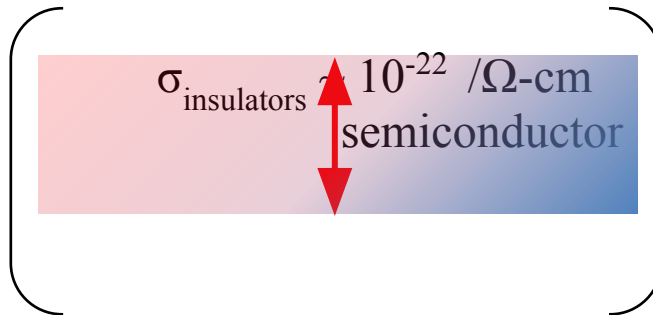


Semiconductor: have a medium energy gap
□ Result: only a small amount of current can flow

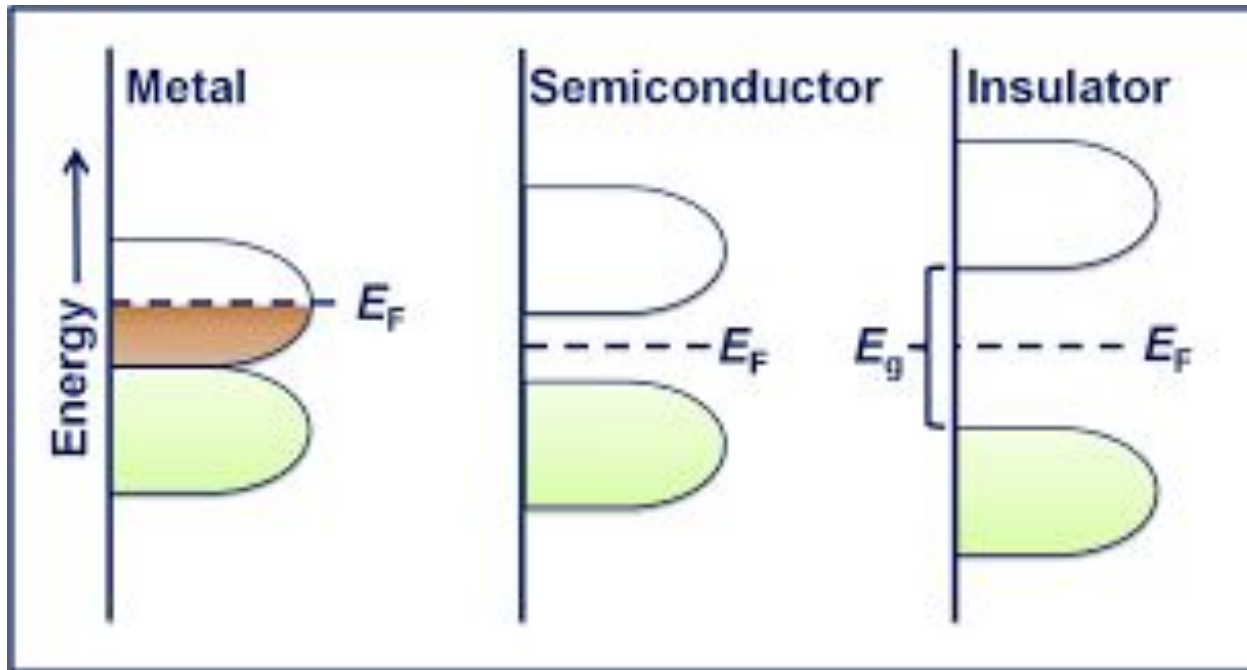
The name “**semiconductor**” implies that it conducts somewhere between the two cases (conductors or insulators)

Conductivity : σ

$$\sigma_{\text{metals}} \sim 10^{10} / \Omega\text{-cm}$$



The conductivity (σ) of a semiconductor lies between these two extreme cases.



Resistivity vs. Temperature

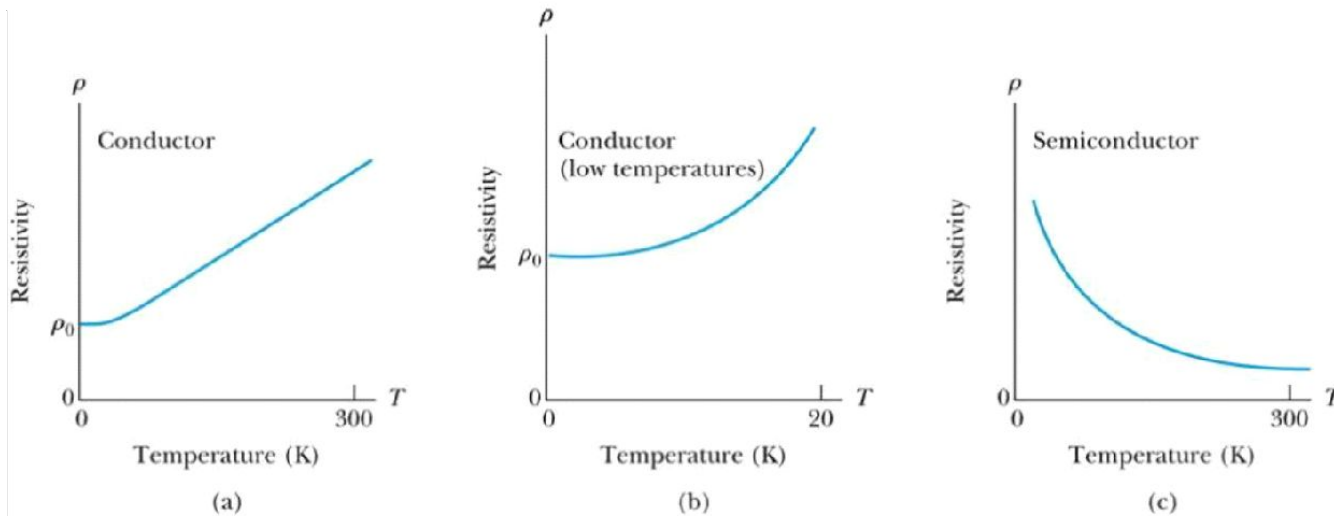
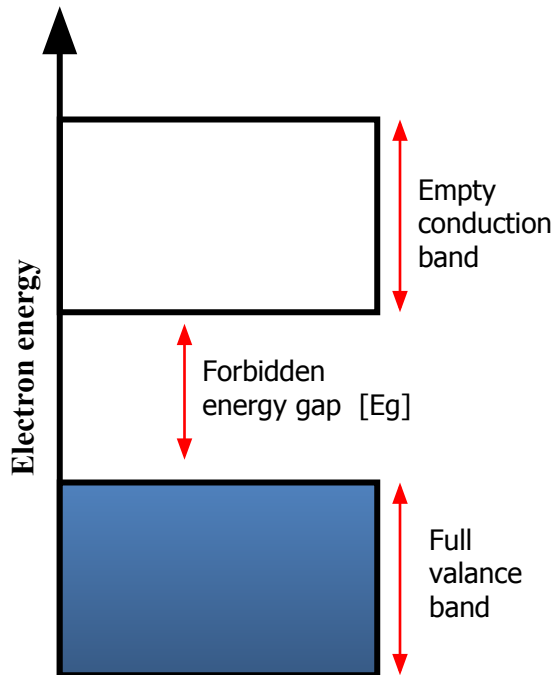


Figure : (a) Resistivity versus temperature for a typical conductor. Notice the linear rise in resistivity with increasing temperature at all but very low temperatures. (b) Resistivity versus temperature for a typical conductor at very low temperatures. Notice that the curve flattens and approaches a nonzero resistance as $T \rightarrow 0$. (c) Resistivity versus temperature for a typical semiconductor. The resistivity increases dramatically as $T \rightarrow 0$.

Note! The relationship between resistivity and conductivity is that conductivity is the reciprocal of resistivity ($\sigma = 1/\rho$).

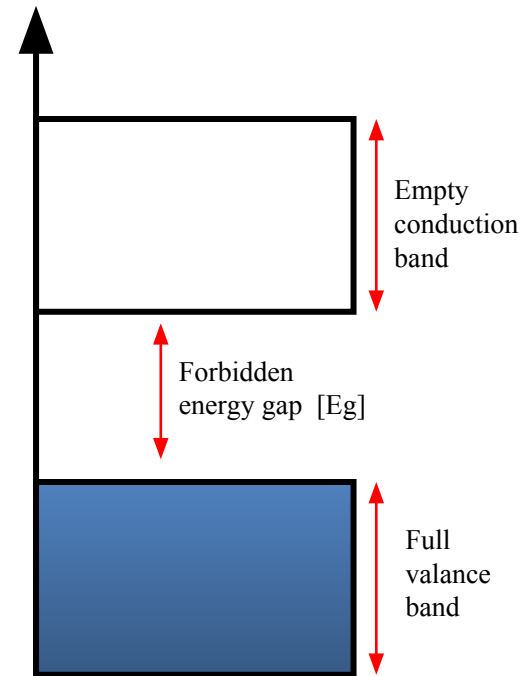
::: Semiconductor energy bands at low temperature :::



- At low temperatures the valance band is *full*, and the conduction band is empty.
- Recall that a full band can not conduct, and neither can an empty band.
- At low temperatures, semiconductor do not conduct, they behave like insulators.
- The *thermal energy* of the electrons sitting at the top of the full band is much lower than that of the *Eg* at *low temperatures*.

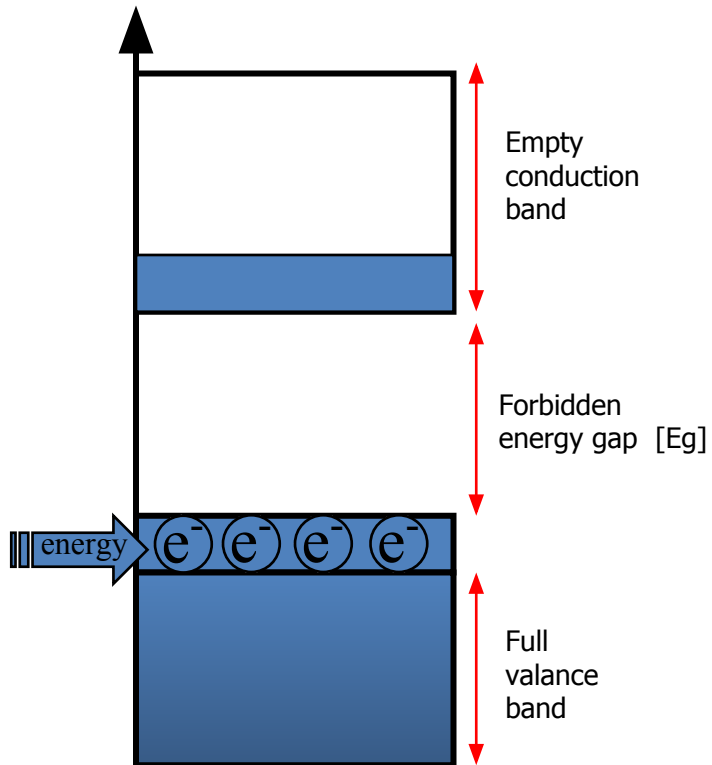
Conduction Electron :

- Assume some kind of energy is provided to the electron (**valence electron**) sitting at the top of the *valance band*.
- This electron gains energy from the applied field and it would like to move into higher energy states.
- This electron contributes to the *conductivity* and this electron is called as a *conduction electron*.
- At 0^0K , electron sits at the lowest energy levels. The valance band is the highest filled band at zero kelvin.



Semiconductor energy bands at room temperature

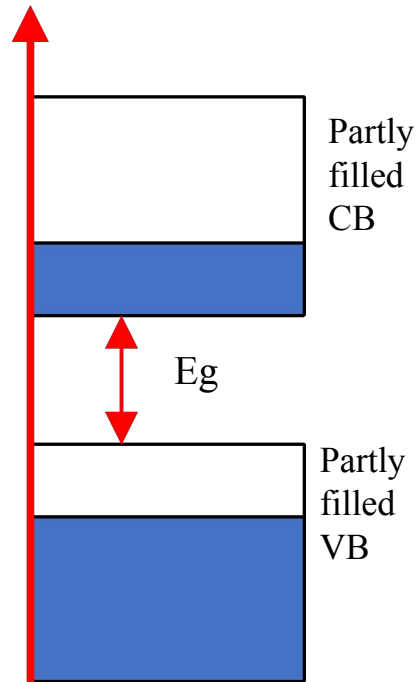
- When enough *energy* is supplied to the e^- sitting at the top of the valance band, e^- can make a transition to the bottom of the conduction band.
- When electron makes such a transition it leaves behind a *missing electron state*.
- This missing electron state is called as a *hole*.
- Hole behaves as a *positive charge carrier*.
- Magnitude of its charge is the same with that of the electron but with an opposite sign.



What kind of excitation mechanism can cause an e^- to make a transition from the top of the valence band (VB) to the minimum or bottom of the conduction band (CB) ?

Answer :

Thermal energy ?
Electrical field ?
Electromagnetic radiation ?



Energy band diagram of a semiconductor at a finite temperature.

Important Note!

Momentum Conservation:

When an electron transitions from the valence band to the conduction band, both energy and momentum must be conserved.

1- Electric field

For low fields, this mechanism doesn't promote electrons to the CB in common semiconductors such as Si and GaAs.

For low fields, this mechanism doesn't promote electrons to the CB in common semiconductors such as Si and GaAs.

✓ **Electric field provides energy slowly:**

An electric field applies a force on charged particles, accelerating electrons, but **does not provide discrete energy quanta** like photons do. So it's not easy to supply the full 1.1 eV energy needed to promote an electron across the band gap using just an electric field.

✓ An electric field of 10^{18} V/m can provide an energy of the order of 1 eV. This field is enormous.

let's **calculate the electric field strength** needed to excite an electron across the **1.1 eV band gap** of silicon using a simplified model.

We want to estimate the electric field E required to move an electron across the band gap energy $\Delta E = 1.1 \text{ eV}$ over a certain distance d . This distance can be taken as the **lattice constant** or the **distance over which the potential drops by 1.1 eV**.

let's **calculate the electric field strength** needed to excite an electron across the **1.1 eV band gap** of silicon using a simplified model.

Electric field required to produce a potential difference equal to a band gap E_g across a distance d is

$$E = \frac{V}{d} = \frac{E_g}{q d},$$

but for an electron 1 eV of energy = 1 V of potential difference, so you can simply use

$$E = \frac{E_g(\text{in eV})}{d}.$$

(That gives E in V/m when d is in meters.)

For silicon $E_g = 1.1$ eV (so = 1.1 V for a single electron). Here are numerical values for some typical distances:

- $d = 1 \text{ nm} = 1 \times 10^{-9} \text{ m}$:
$$E = \frac{1.1 \text{ V}}{1 \times 10^{-9} \text{ m}} = 1.10 \times 10^9 \text{ V/m}.$$
- $d = 10 \text{ nm} = 1 \times 10^{-8} \text{ m}$:
$$E = 1.10 \times 10^8 \text{ V/m}.$$

This field strength is **extremely high** and well above the **breakdown field** of silicon ($\sim 3 \times 10^5 \text{ V/cm}$). This will usually cause dielectric breakdown or induce tunneling/impact ionization long before a clean “band-gap excitation” occurs across the whole region. Applying such a field would likely destroy the material before significant excitation occurs.

2- Electromagnetic Radiation :

$$E = h\nu = h \frac{c}{\lambda} = (6.62 \times 10^{-34} \text{ J-s}) \times (3 \times 10^8 \text{ m/s}) / \lambda(\text{m}) \Rightarrow E(\text{eV}) = \frac{1.24}{\lambda(\text{in } \mu\text{m})}$$

$$\begin{aligned} h &= 6.62 \times 10^{-34} \text{ J-s} \\ c &= 3 \times 10^8 \text{ m/s} \\ 1 \text{ eV} &= 1.6 \times 10^{-19} \text{ J} \end{aligned}$$

for Silicon $E_g = 1.1 \text{ eV}$

$$\lambda(\mu\text{m}) = \frac{1.24}{1.1} = 1.1 \mu\text{m} \quad \square$$

We need Near-Infrared Light, not even visible light!

= 1100 nm

To promote electrons from VB to CB Silicon , the wavelength of the photons must 1.1 μm (11 00 nm) or less

3-Thermal Energy :

$$\text{Thermal energy} = k \times T = 1.38 \times 10^{-23} \text{ J/K} \times 300 \text{ K} = 25 \text{ meV}$$

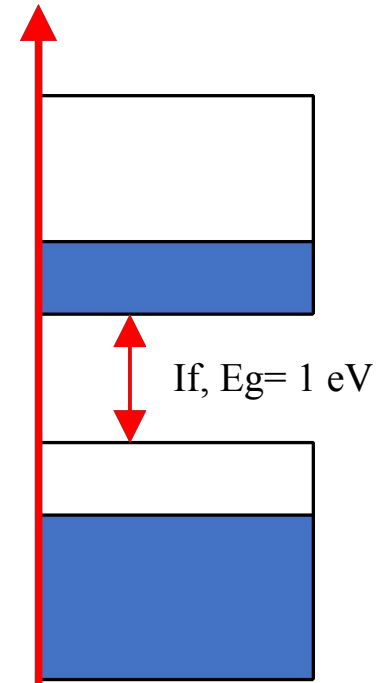
$$\text{Excitation rate} = \text{constant} \times \exp(-E_g / kT)$$

Although the thermal energy at room temperature, RT , is very small, *i.e.* **25 meV**, a few electrons can be promoted to the **CB**.

Electrons can be promoted to the CB by means of thermal energy.

This is due to the exponential increase of excitation rate with increasing temperature.

Excitation rate is a strong function of temperature.



Is it possible to excite the electron from valance band?

Fermi-Dirac distribution

The Fermi-Dirac distribution explains why semiconductor conductivity increases with temperature by showing how increased thermal energy broadens the distribution, allowing more electrons to occupy higher energy states in the conduction band, thereby creating more free charge carriers (electrons and holes). This increased carrier concentration leads to greater electrical conductivity, a phenomenon most evident in intrinsic (undoped) semiconductors.

1. Fermi-Dirac distribution function is:

$$f(E) = \frac{1}{1 + e^{(E-E_f)/(k_B T)}}$$

- E : energy of the state
- E_f : Fermi energy
- k_B : Boltzmann constant
- T : absolute temperature

2. At $T = 0$: step function.

3. At $T > 0$: smooth transition around E_f .

4. At $E = E_f$, always $f(E_f) = 0.5$.

What is Fermi-Dirac Distribution Function?

Now, if at any energy level we want to find the probability of finding an electron, we can use the Fermi Dirac distribution function. Let us talk in brief. Now, there are two types of Quantum Particles- Fermions and boson. Boson obeys Bose Einstein distribution function but here we need to talk about fermions in detail. Fermions obey Fermi Dirac distribution function. Fermi-Dirac are the two scientists who found this useful method. If the temperature increases from 0 K then energy level also changes and then electrons move to different energy level above the fermi energy. So there we cannot find the exact position of electrons, so we find the probability of finding electrons at a particular energy level E i.e P(E) through this fermi function

The probability that electrons will occupy a certain energy level at a certain temperature is given by the Fermi-Dirac distribution function:

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

Fermi-Dirac Statistics

$$F(E) = 1 / (1 + e^{(E - E_F)/kT})$$

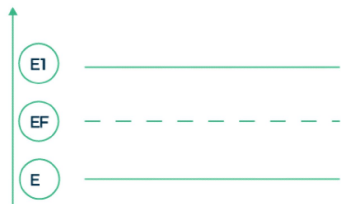
where F(E) = Probability in energy level, E,

E = energy level in which we are finding,

E_F = Fermi energy level,

K = Boltzmann's constant = 1.38 × 10⁻²³ J/k = 8.625 × 10⁻⁵ eV/k ,

T = absolute zero temperature.



•Fermi Level vs. Energy States:

- The Fermi level is a statistical concept, not a physical energy state itself, and it's defined as the energy level at which there is a 50% probability of finding an electron at a given temperature.

By applying Fermi-Dirac statistics,

E₁ > E_F , F(E₁) = 0 % , T = 0K (Here E₁ is an energy level above fermi energy level)

E₂ < E_F , F(E₂) = 100% , T = 0K (Here E₂ is the energy level below fermi energy level)

E = E_F , F(E_F) = 50% , T > 0K (at fermi energy)

Fermi-Dirac distribution function cannot provide the probability of fermi level at 0K. It is undefined.

Fermi Dirac Distribution Function, Fermi Level

The Probability that an electron occupies in an energy level E is

$$f(E) = \frac{1}{1 + e^{(E-E_F)/KT}}$$

$K \rightarrow$ Boltzmann's const, $T \rightarrow$ Temp in Kelvin
 $E_F \rightarrow$ Fermi level Energy

Fermi energy level $\rightarrow$ Highest filled energy level at absolute zero temp. (0K)

Fermi Level in conductors

Case 1 : at temp $T=0K$

(i) $E < E_F$ $E - E_F = -ve$

$$f(E) = \frac{1}{1 + e^{-ve/0}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + 0} = 1$$

$e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty}$

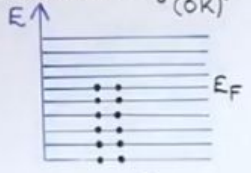
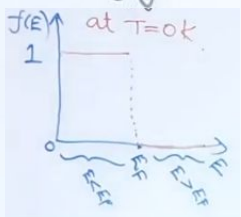
$f(E) = 1 \Rightarrow$ all energy levels below E_F are occupied

(ii) $E > E_F$ $E - E_F = +ve$

$$f(E) = \frac{1}{1 + e^{+ve/0}} = \frac{1}{1 + e^{\infty}} = \frac{1}{1 + \infty} = \frac{1}{\infty} = 0$$

$f(E) = 0 \Rightarrow$ All energy levels above E_F are empty.

(iii) $E = E_F$, $E - E_F = 0$

$$f(E) = \frac{1}{1 + e^0} = \text{Indeterminate}$$



Fermi Dirac Distribution Function, Fermi Level

The Probability that an electron occupies in an energy level E is

$$f(E) = \frac{1}{1 + e^{(E-E_F)/KT}}$$

$K \rightarrow$ Boltzmann's const, $T \rightarrow$ Temp in Kelvin
 $E_F \rightarrow$ Fermi level Energy

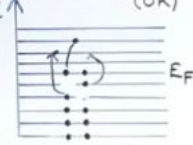
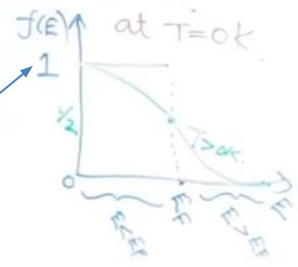
Fermi energy level $\rightarrow$ Highest filled energy level at absolute zero temp. (0K)

Fermi Level in conductors

Case 2 : Temp $T > 0K$

$E = E_F$ $E - E_F = 0$

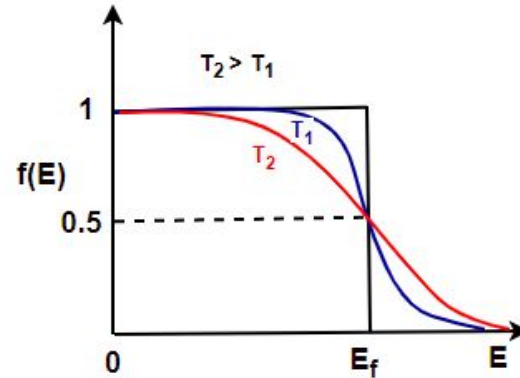
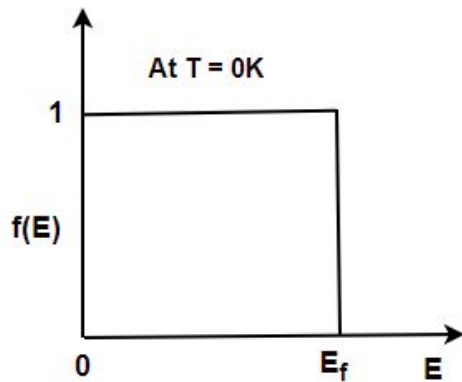
$$f(E) = \frac{1}{1 + e^{0/KT}} = \frac{1}{1 + e^0} = \frac{1}{1 + 1} = \frac{1}{2} = 0.5$$

$$f(E) = \frac{1}{2} = 50\%$$



Probability varies 1-0

<https://www.youtube.com/watch?v=Ww9wcs3yNWI>

<https://www.youtube.com/watch?v=WfNj79UWcJw>



•**Effect of temperature:** As the temperature increases, the distribution "broadens". This means: Some electrons gain enough thermal energy to jump from lower energy levels to higher energy levels. More energy states above the Fermi level become occupied by electrons. More energy states below the Fermi level become empty

This plot is showing the **Fermi–Dirac distribution function** $f(E)$, which gives the probability that an electronic state of energy E is occupied by an electron at a given temperature T .

Left Plot: At $T = 0\text{ K}$

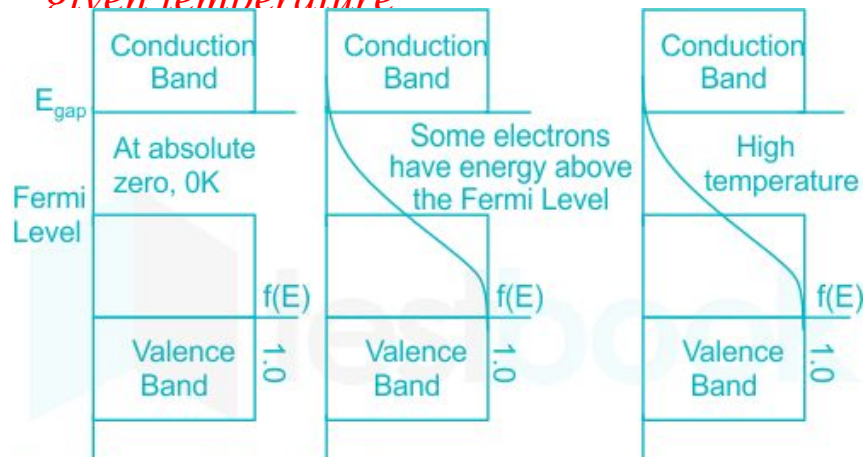
- At absolute zero, all electronic states with energy less than the Fermi energy (E_f) are completely filled ($f(E) = 1$).
- All states with $E > E_f$ are empty ($f(E) = 0$).
- So the distribution is a **step function**: sharp cutoff at $E = E_f$.

This means:

- Electrons fill the lowest energy states first.
- The highest filled state at $T = 0$ is called the **Fermi level** E_f .

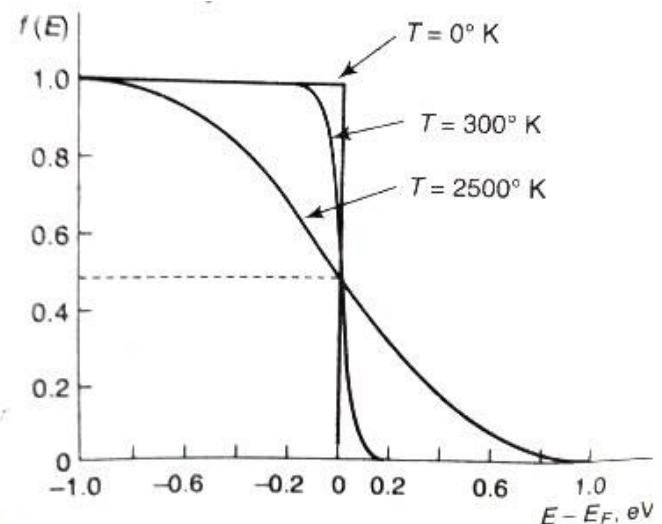
The [Fermi-Dirac](#) distribution doesn't state there's a probability of finding an electron *in* the forbidden gap; instead, it gives the **probability of finding an electron at a given energy state**, which is a crucial factor in determining the material's behavior. While there are no allowed electron states within the forbidden gap, the Fermi level's position—which is the energy level with a 50% probability of occupation—can cross this region, thus affecting the likelihood of electrons being present in the conduction and valence bands

•Probability Across the Band Gap: The Fermi-Dirac distribution function still describes how probability changes across the band gap, even though the probability of finding a state is zero within the forbidden region. It helps to determine how many electrons are available in the conduction band and how many holes are in the valence band at any given temperature



No electrons can be above the valence band at 0K, since none have energy above the Fermi level and there are no available energy states in the band gap.

At high temperatures, some electrons can reach the conduction band and contribute to electric current.



The Fermi-Dirac distribution explains conductivity by describing how electron energy states are occupied at different temperatures. At absolute zero (0 K), electrons fill all states below the [Fermi energy](#), leaving higher states empty. As temperature increases, the distribution broadens, giving electrons enough energy to occupy these higher states, particularly in the conduction band, which increases the availability of charge carriers and thus enhances conductivity.

The Role of the Fermi-Dirac Distribution

- **Electron Probability:** The Fermi-Dirac function is a statistical model that shows the probability of an electron occupying a specific energy state at a given temperature.
- **Fermi Energy (E_F):** This is the highest energy level that electrons can occupy at absolute zero.

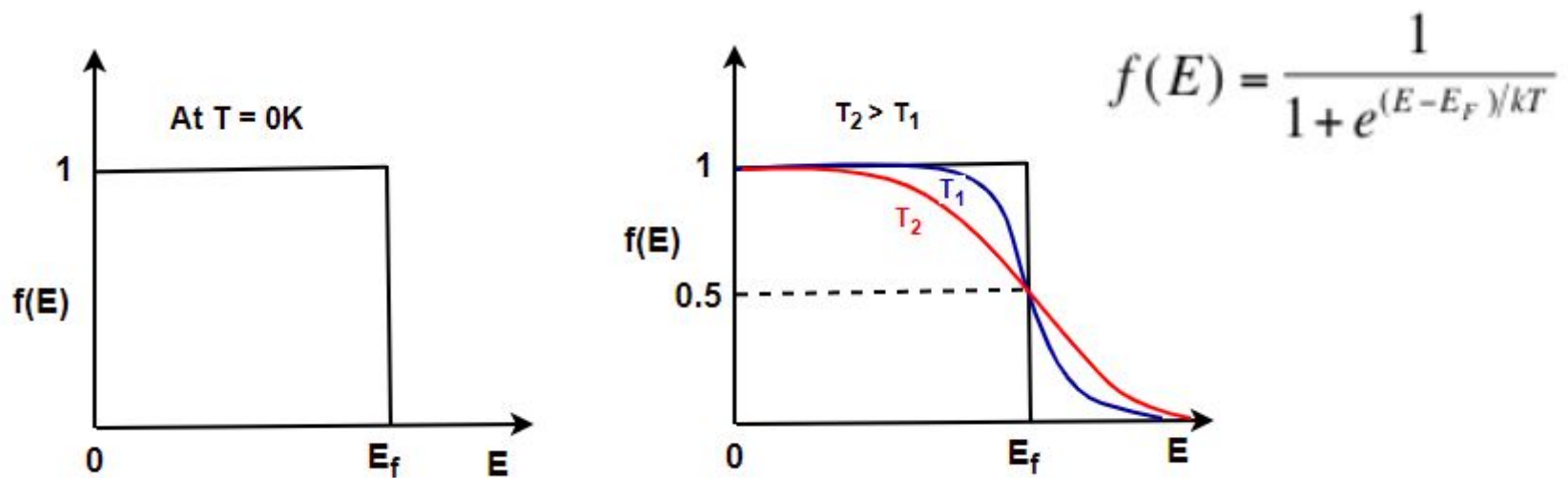
How Temperature Affects Electron Distribution

- **At 0 K:** The distribution acts like a step function; all energy states below E_F are filled with electrons, and all states above E_F are empty.
- **At Higher Temperatures:** With increasing temperature, electrons gain energy. The Fermi-Dirac distribution broadens, meaning:
 - More states *above* the Fermi level become occupied by electrons.
 - More states *below* the Fermi level become unoccupied.

Impact on Conductivity

- **More Free Electrons:** The increased occupation of higher energy levels, especially those in the [conduction band](#), results in a greater number of free electrons available to move and carry electrical current.
- **Increased Conductivity:** The higher density of these mobile electrons directly contributes to an increase in the material's electrical conductivity.

In summary, the Fermi-Dirac distribution provides a framework to understand how temperature influences electron behavior, directly correlating with the observed increase in conductivity as more electrons become available to conduct electricity.



Right Plot: At $T > 0\text{ K}$

- As temperature increases, the sharp step function **smears out**.
- At E_f , the probability becomes $f(E_f) = 0.5$, regardless of temperature.
- Some electrons gain enough thermal energy to move into states above E_f , leaving vacancies below E_f .

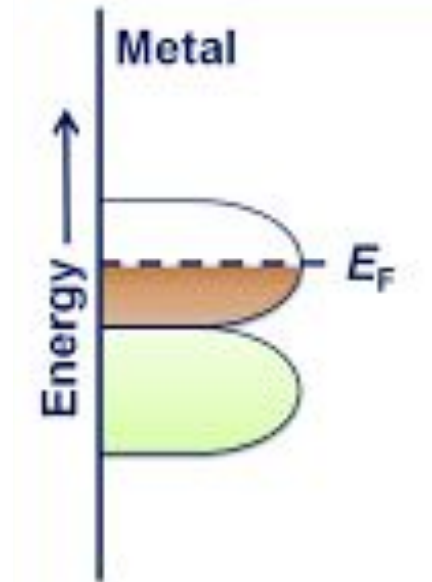
Here:

- Blue curve: Lower temperature (T_1) $\rightarrow$ the smearing is small, close to the step.
- Red curve: Higher temperature ($T_2 > T_1$) $\rightarrow$ more smearing, larger number of electrons excited to higher energy levels.

How to connect this Fermi–Dirac distribution to the physical behavior of metals and semiconductors.

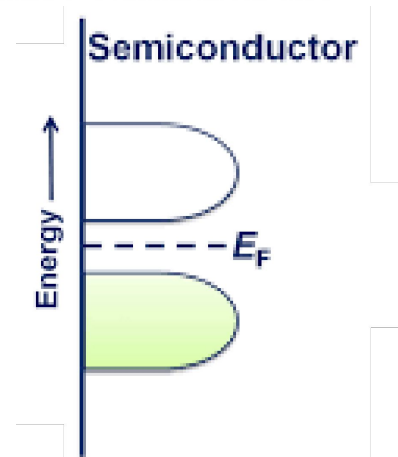
◆ Metals

- In metals, the Fermi energy E_f lies inside the conduction band.
- At $T = 0K$:
 - All states below E_f are filled.
 - States above E_f are empty.
- At $T > 0K$:
 - Only electrons **within a few $k_B T$ of E_f** are thermally excited.
 - These electrons can move into nearby empty states above E_f .
- Why important?
 - This is why only electrons near the Fermi level contribute to **electrical conductivity** and **heat capacity** in metals.
 - Electrons deep below E_f remain stuck (no nearby empty states).



◆ Semiconductors

- In semiconductors, the **Fermi level lies in the band gap**:
 - Near the middle of the gap for intrinsic (pure) semiconductors.
 - Closer to the conduction band (n-type) or valence band (p-type) in doped semiconductors.
- At $T = 0K$:
 - Conduction band empty, valence band full → **no conduction**.
- At $T > 0K$:
 - Some electrons gain enough energy to jump from the valence band across the gap to the conduction band.
 - Probability of occupation in the conduction band is given by $f(E)$.
 - The number of excited electrons (and holes) grows rapidly with temperature.
- **Why important?**
 - Explains why semiconductors are **insulators at low T** but **conductors at higher T**.



How come the fermi level in a semiconductor is in the forbidden gap but still has 1/2 probability of being occupied?

When you see that Probability of occupying is 1/2 but there is no electron or holes it seems there is some inconsistency but that's not the case.

Actually the no. of electron/holes present at any energy level is defined by the product of **probability of occupancy** and the **density of states**.

Now, **probability of occupancy** by electron is given by Fermi-Dirac distribution function, $f(E) = \frac{1}{1 + e^{\{(E-E_f)/kT\}}}$, so at $E = E_f$ energy we get $f(E) = \text{probability of occupancy of electron} = 1/2$

But the **density of states** (represented by N_c or N_v) is not available within forbidden energy gap, it is present only in valence band (for holes, N_v) and conduction band (for electrons, N_c).

So, No. of electrons in E_f present in forbidden energy gap = Probability of occupancy * Density of states = $(1/2) * 0 = 0$ hence you can't find any electrons at that energy.

1. What the Fermi–Dirac distribution really means

The function

$$f(E) = \frac{1}{1 + e^{(E-E_f)/(k_B T)}}$$

gives the probability that an available state of energy E is occupied.

● But notice:

If there is **no state at energy E** (density of states $D(E) = 0$), then the probability $f(E)$ doesn't matter, because you cannot occupy a state that does not exist.

👉 So in the **band gap** of a semiconductor, where $D(E) = 0$, the Fermi–Dirac distribution may still give some value (like 0.5 at $E = E_f$), but there are no states $\rightarrow$ no electrons can exist there.

◆ 2. How conduction starts just above 0 K

- At $T = 0K$:
 - Valence band full, conduction band empty → no conduction.
- At $T > 0K$ (just above 0 K):
 - A tiny fraction of electrons in the valence band can absorb thermal energy and jump across the band gap E_g into the conduction band.
 - These electrons do **not pass through the forbidden gap**—they make a quantum transition directly from a valence-band state to a conduction-band state.
- The number of such electrons is proportional to:

$$n \propto e^{-\frac{E_g}{2k_B T}}$$

which is extremely small at low T .

◆ 3. Why it looks confusing

- The Fermi–Dirac curve in your plot looks smooth and “nonzero” everywhere.
- But in reality, the **electron occupancy** is given by:

$$n(E) = f(E) \cdot D(E)$$

where $D(E)$ = density of states.

- Inside the band gap, $D(E) = 0$.
 - So even though $f(E)$ is between 0 and 1, $n(E) = 0$.
 - This resolves the apparent paradox.

Role of quantum mechanics

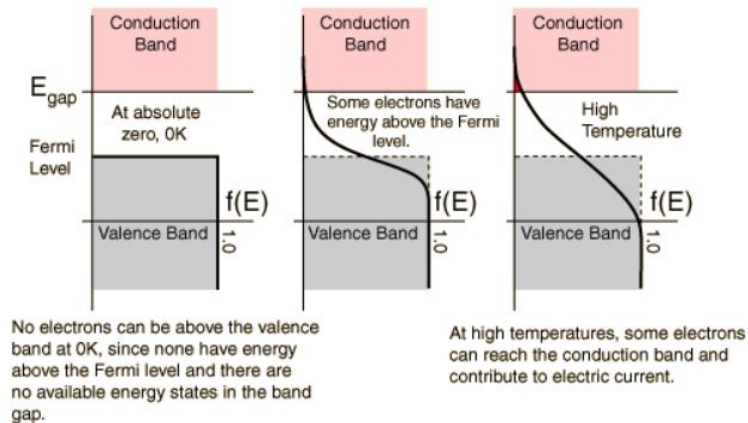
- In quantum mechanics, an electron is described by a wavefunction that belongs to an allowed energy band.
- When the electron absorbs energy E_g , the **entire wavefunction transforms** from a valence-band state into a conduction-band state.
- This is a **quantum transition**, not a gradual move.

The Fermi-Dirac distribution is a fundamental statistical concept in quantum mechanics that describes how a collection of identical particles, called fermions (like electrons), are distributed among available energy states

The probability that electrons will occupy a certain energy level at a certain temperature is given by the Fermi-Dirac distribution function:

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

where T is temperature, k is Boltzmann's constant (1.38×10^{-23} Joules per Kelvin), E_F is the Fermi energy level of a given material and E is the allowed energy state whose probability we are trying to find. This equation just means that as temperature increases, electrons are more likely to be found in the higher energy states. When the temperature is near absolute zero, we see that $f(E)$ becomes 1, giving that nearly all the electrons are below the Fermi level (see below). At higher temperatures, energy levels in the conduction band have a higher probability of occupation and levels in the valence band are more likely to be empty. There is also an equation which allows us to solve for the Fermi level of a material using our knowledge of the band gap of that material, however we will not discuss that equation in detail here.



Above we see that the distribution "smears" as the temperature rises. A huge difference between a conductor and semiconductor is that increasing temperature usually hinders current flow in conductors, while improving conductivity in semiconductors. This is due to the band gap and Fermi level of semiconductors.

Some points about Fermi levels:

- They allow us to make calculations as to the density of electrons and holes in a material, or the relative amounts of each, depending on the temperature. This is crucial to our understanding of current flow through semiconductors.
- In conductors (i.e. metals), electrons only partially fill the valence band and the valence and conduction bands are very close or overlap, thus electrons become conductive (free) very easily. On the other hand, insulators and semiconductors have Fermi levels lying in the forbidden band gap and have full valence bands, therefore insulators have electrons with nowhere to go (hence non-conductive) and semiconductors become conductive only at certain temperatures (when more electrons can be found in the higher energy states due to thermal excitation).
- The Fermi level and band gap in a solid largely determine its electrical properties. In a perfect semiconductor (in the absence of impurities/dopants), the Fermi level lies close to the middle of the band gap

Why the conductivity increases with temperature for semiconductor but decreases for metal?

Why is the Fermi-Level sometimes located in the band gap between conduction band and valence band although there are no energy levels allowed in between, but the probability of finding an electron at the fermi level is 0.5?

Even in the absence of energy states within the band gap, the Fermi level, a statistical measure, exists to indicate the probability of an electron occupying a given energy level, not the existence of an actual state

- Fermi Level vs. Energy States:**

- The Fermi level is a statistical concept, not a physical energy state itself, and it's defined as the energy level at which there is a 50% probability of finding an electron at a given temperature.*

- Fermi-Dirac Distribution:** The Fermi level is related to the Fermi-Dirac distribution function, which describes the probability of an energy state being occupied by an electron.

- Semiconductors and the Band Gap:** In semiconductors, the Fermi level typically lies within the band gap (the region between the valence band and the conduction band) because the probability of finding an electron in the band gap is not zero, even though there are no energy states available there.

- Thermodynamic Equilibrium:** The Fermi level helps describe the thermodynamic equilibrium of a material, where the probability of an electron occupying a state is determined by the material's temperature and the energy of the state.

- Doping and Fermi Level Shift:** Doping a semiconductor (adding impurities) can shift the Fermi level, either towards the conduction band (n-type doping) or towards the valence band (p-type doping).

- Importance of Fermi Level:** The Fermi level is important for understanding the electrical and thermal properties of solids, as it helps determine the conductivity and other transport phenomena.

What is Fermi energy (E_F) level?

The Fermi Level is the highest filled energy level in the energy band at zero kelvin. The Fermi Level is the energy level in the energy band diagram for which the probability of occupancy (in other words we can say the presence of majority charge carriers) becomes half. The Fermi Level and Fermi Energy will be changing depending on the type of material. i.e., different materials will have different Fermi Levels and corresponding Fermi Energy.

So according to the Fermi Energy definition, it is the highest occupied energy level of any material at absolute zero. To be more specific, all the electrons in the material will occupy the energy level at or below that material's Fermi Level at 0K. The concept of the Fermi Level comes from Fermi-Dirac statistics. Electrons are Fermions (the particles with odd half-integral spin) and by the Pauli exclusion principle cannot exist in identical energy states. The energy level corresponds to the Fermi Energy known as the Fermi Level. The Fermi Level can also be described as the maximum energy level of an electron at 0 Kelvin at which it can reach. Because all electrons are in the lowest energy state at absolute zero temperature, the Fermi Level falls between the valence and conduction bands. The Fermi Level can be thought of as a sea of Fermions (or electrons) above which no electrons exist due to a lack of energy at 0 Kelvin. As the solids are heated and electrons are added to or removed from them, the Fermi Level changes.

Now, the question arose: why do we need Fermi Level or Fermi Energy? Suppose we increase the temperature of the system by 0K to say 100K then, then some of the electrons lying below the Fermi Level may get excited to certain energy levels, and to determine the density of electrons at that temperature, we introduce the Fermi Dirac distribution. As the temperature of material increases above absolute zero, the probability of electrons existing in an energy level greater than the Fermi Energy increases and hence there will not be a constant highest occupied energy level.

Fermi Level in the Band Gap

- 1. Temperature Effects:** At absolute zero (0 K), the Fermi level lies within the valence band for insulators and semiconductors. However, as temperature increases, thermal excitation can allow some electrons to jump from the valence band into the conduction band.
- 2. Position of the Fermi Level:** In intrinsic semiconductors (pure materials without dopants), the Fermi level is typically located near the middle of the band gap at higher temperatures, reflecting an equal probability of occupancy of the valence and conduction bands. In doped semiconductors, the Fermi level shifts towards the conduction band in n-type materials (donor doping) and towards the valence band in p-type materials (acceptor doping).
- 3. Probability Interpretation:** The statement that the probability of finding an electron at the Fermi level is 0.5 refers to the Fermi-Dirac distribution function. At the Fermi level, this function indicates that there is a 50% chance that a state at that energy level is occupied by an electron.

- **Fermi-Dirac Distribution:** The probability $f(E)$ of finding an electron at energy E is given by:

$$f(E) = \frac{1}{1 + e^{(E - E_F)/(kT)}}$$

- At $E = E_F$, $f(E_F) = 0.5$.

- The Fermi level can be located in the band gap because it represents a statistical average of the energy levels in the material. Even though there are no allowed states in the gap, the Fermi level reflects the balance of electrons and holes in the system.
- The Fermi level's position is influenced by temperature and doping, affecting the distribution of electrons across the available energy states.

In conclusion, the Fermi level's position in the band gap is a reflection of the statistical nature of electron occupancy in a solid, rather than the presence of actual energy states in that region.

By definition, there are no allowed electronic states *within* the band gap. So, if the Fermi level falls inside the band gap, it does *not* mean there are electrons physically sitting at that energy level. The Fermi level (E_F) is **not** a physical energy level where electrons reside. Instead, it is a **theoretical construct** and a **crucial reference point** with profound importance in solid-state physics and semiconductor devices.

Why is the Fermi Level So Important? (Even in the Band Gap)

Even though there are no states *at* E_F in the gap, its position is critically important for five key reasons:

1. The "Filling Line" and Carrier Concentration:

Think of the Fermi level as a kind of "water level" or "filling line" for electrons. The Fermi-Dirac distribution $f(E)$ tells you how filled the states at any other energy are. For states far below E_F (e.g., in the valence band), $f(E) \approx 1$. They are almost completely full. For states far above E_F (e.g., in the conduction band), $f(E) \approx 0$. They are almost completely empty. The exact position of E_F within the band gap precisely determines the number of electrons that have enough thermal energy to jump from the valence band to the conduction band. If E_F is closer to the conduction band (CB), it means it's easier to excite electrons into the CB (because $f(E)$ for states near the CB is higher). This results in a higher electron concentration (n). If E_F is closer to the valence band (VB), it means it's easier for holes to be created in the VB (because $f(E)$ for states near the VB is lower). This results in a higher hole concentration (p). In a pure (intrinsic) semiconductor, E_F lies near the middle of the gap, making $n = p$.

2. The Driving Force for Current

The Fermi level is directly related to the electrochemical potential of the electrons. A difference in Fermi level between two points is what drives electron flow, just as a difference in water level (pressure) drives water flow. Electrons flow from a region of high E_F to a region of low E_F . Holes flow from a region of low E_F to a region of high E_F . This is the fundamental principle behind all semiconductor device operation (diodes, transistors, etc.)

3. Determining Contact Potentials and Band Bending

When two different materials (e.g., a metal and a semiconductor, or a p-type and n-type semiconductor) are brought into contact, electrons will flow until the Fermi levels align across the interface. This flow creates charged regions and bends the energy bands. This band bending creates the built-in electric fields that are essential for devices like: PN Junctions (the heart of diodes and transistors), Schottky Diodes and Solar Cells.

4. Classifying Materials

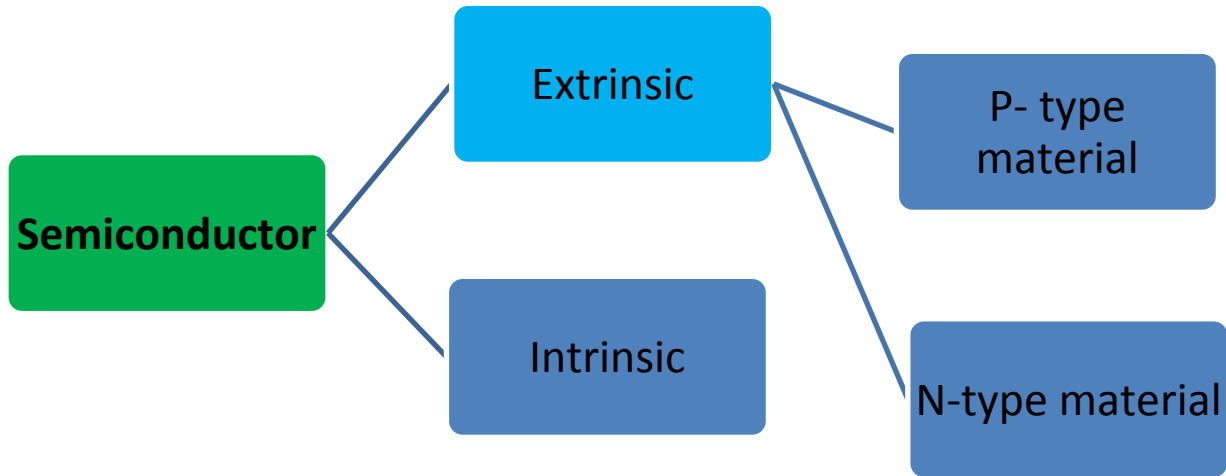
The position of E_F relative to the bands provides a fundamental classification: Conductor (Metal): E_F lies within a partially filled band. Lots of states are available immediately above and below E_F , allowing easy movement of electrons. Insulator: E_F is in a large band gap. The probability of an electron reaching the conduction band is vanishingly small. Semiconductor: E_F is in a small band gap. A measurable number of electrons can be thermally excited across the gap. Its position can be easily shifted by doping.

5. Doping

This is where the concept becomes most practical. We intentionally move the Fermi level by adding impurities (dopants).

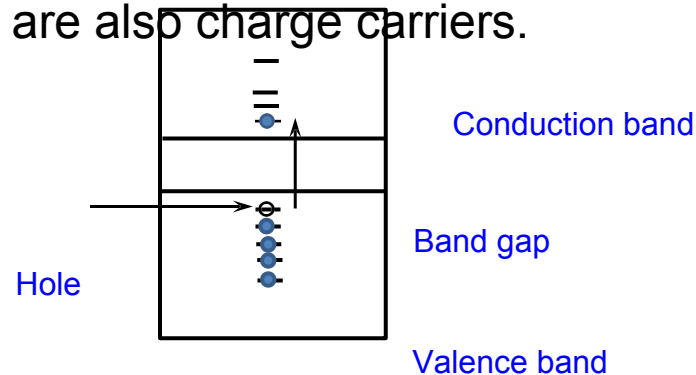
- n-type doping: Add atoms with extra electrons (e.g., Phosphorus in Silicon). These donors create energy states just below the conduction band. Electrons easily jump from these states to the CB, which raises the Fermi level closer to the conduction band.
- p-type doping: Add atoms with a missing electron (e.g., Boron in Silicon). These acceptors create states just above the valence band. Electrons from the VB easily jump to these states, creating holes, which lowers the Fermi level closer to the valence band.

TYPES OF SEMICONDUCTORS



Intrinsic Semiconductors

- ❑ Semiconductors like Si and Ge have relatively narrow band gap generally below 2 eV.
- ❑ Therefore, it is possible to excite electrons from the valence to the conduction band. This is called intrinsic semi conductivity.
- ❑ Every electron that is excited to the conduction band leaves behind a hole in the valence band.
- ❑ An electron can move in to a *hole* under an electrical potential and thus holes are also charge carriers.



Intrinsic semiconductor

- An intrinsic semiconductor is a pure semiconductor which does not have any doping agent (an impurity added to crystal lattice).
- In an intrinsic semiconductors, the number of charge carrier depends upon property of the material not on impurity present.
- Has little current conduction capability at room temperature.

Intrinsic semiconductors are components with four valence electrons per atom, i.e. elements that happen in “Group-IV” of the periodic table, such as germanium (Ge) and silicon (Si). Each atom has four valence electrons. Each of these valence electrons can create a covalent bond with one of the valence electrons in a neighboring atom. All of the valence electrons are included in a covalent bond. Rigorously speaking, this is not the case: depending on the temperature, several electrons can “break” their covalent bonds and take part in conduction. However, it is possible to significantly increase a semiconductor’s conducting ability by adding small quantities of an impurity to the semiconductor, in a manner designated doping. The impurity, which is added to the intrinsic semiconductor is named the dopant. A doped semiconductor is committed to as an extrinsic semiconductor.

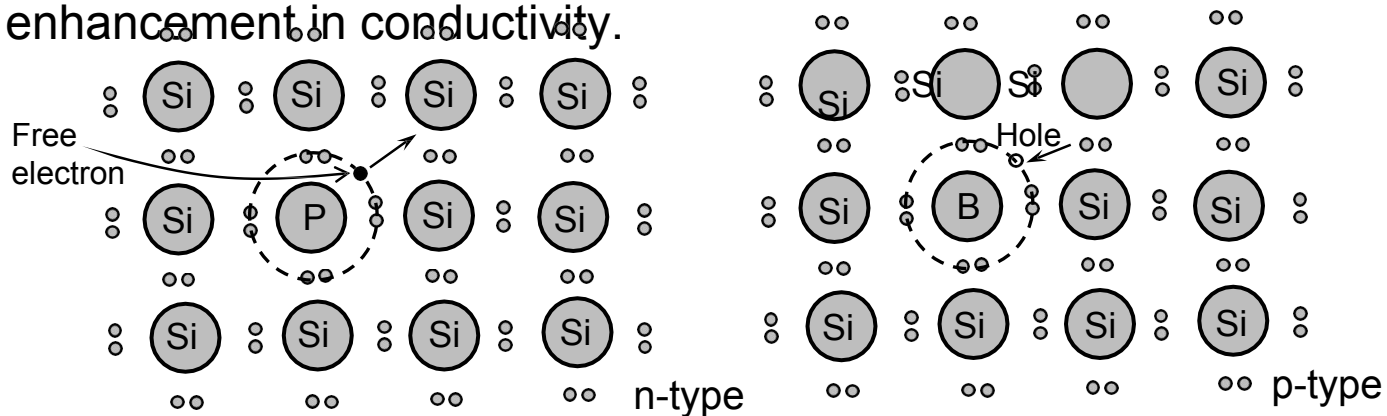
Semiconductors

- **Extrinsic Semiconductors**

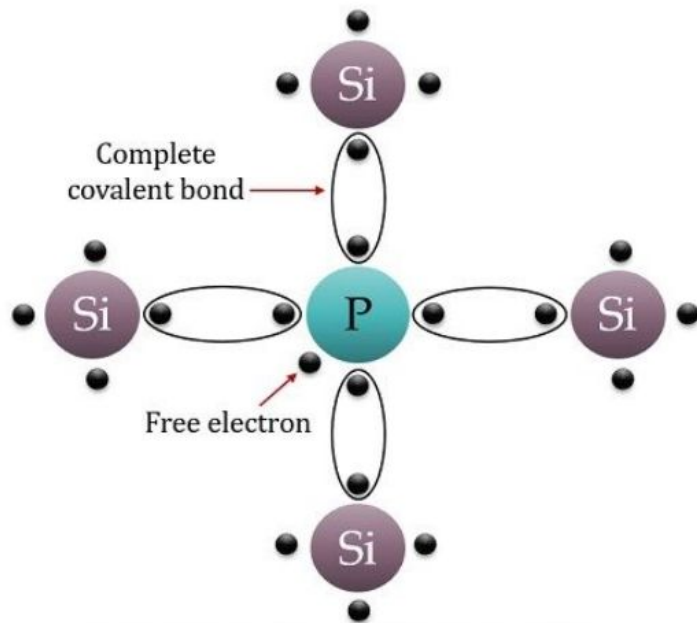
- An extrinsic semiconductor is a semiconductor whose electrical conductivity can be increased by adding the trace amounts of other elements such as impurities in the material.
- There are two types of impurities added to Ge and Si Crystal.
 - Pentavalent material - it is made up of atoms which have five valence electrons.
 - Trivalent material – it is materials has three valence electrons.

Extrinsic Semiconductors

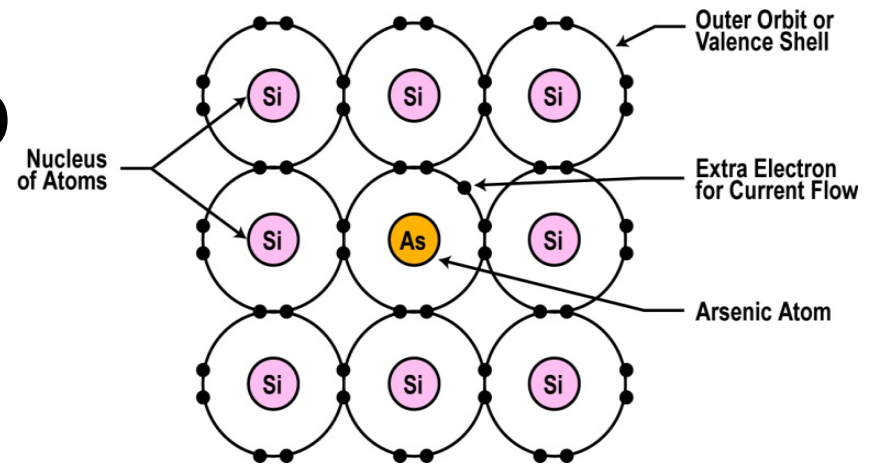
- ❑ The conductivity is enhanced by adding impurity atoms (dopant) in extrinsic semi conductors . All semi conductors for practical purposes are extrinsic.
- ❑ A higher valence dopant e.g. P (5+) in Si (4+) creates an extra electron (n-type) while a lower valence dopant like B (3+) creates a hole (p-type) as shown in the atomic bonding model below.
- ❑ This increases the charge carrier concentration and hence the enhancement in conductivity.



N-Type Semico



- Si = Intrinsic semiconductor atom
- P = Pentavalent impurity atom



N-Type Semiconductor

N-type extrinsic semiconductors are created when group V elements like phosphorus, antimony, bismuth, etc., are doped to a complete semiconductor crystal. These are named so because doping these elements will create an additional electron in the atom's valence shell.

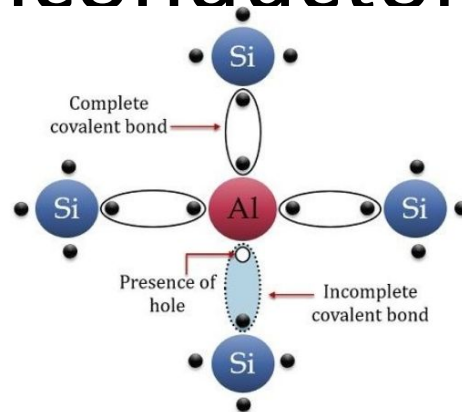
Formation of n-type semiconductor (Reference: electronicsdesk.com)

Semiconductors

- **Another Way to Dope**

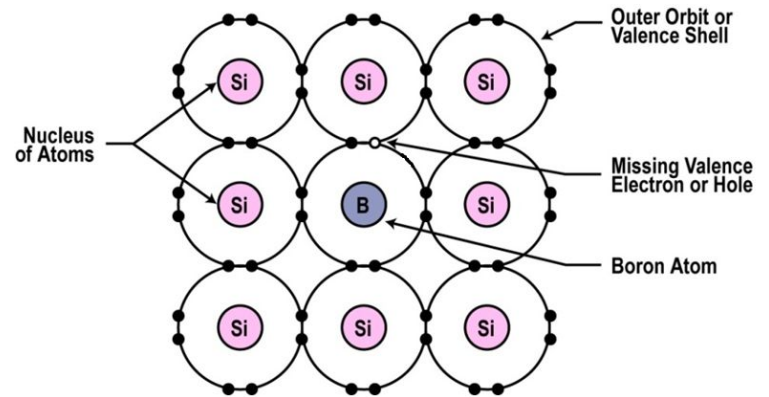
- You can also dope a semiconductor material with an atom such as boron that has only 3 valence electrons.
- The 3 electrons in the outer orbit do form covalent bonds with its neighboring semiconductor atoms as before. But one electron is missing from the bond.
- This place where a fourth electron should be is referred to as a hole.
- The hole assumes a positive charge so it can attract electrons from some other source.
- Holes become a type of current carrier like the electron to support current flow.

P-type Semiconductors



- Si = Intrinsic semiconductor atom
- Al = Trivalent impurity atom

Formation of p-type semiconductor (Reference: electronicsdesk.com)



P-Type Semiconductor

P-type semiconductors are produced by doping an intrinsic semiconductor with an electron acceptor element throughout manufacture. The term *p-type* regards the positive charge of a hole. As opposed to *n-type* semiconductors, *p-type* semiconductors have a bigger hole concentration than electron concentration.

In *p-type* semiconductors, cavities are the majority carriers and electrons are the minority carriers. A typical *p-type* dopant for silicon is aluminum, boron, or gallium. For *p-type* semiconductors, the Fermi level is underneath the intrinsic Fermi level and lies closer to the valence band than the conduction band.

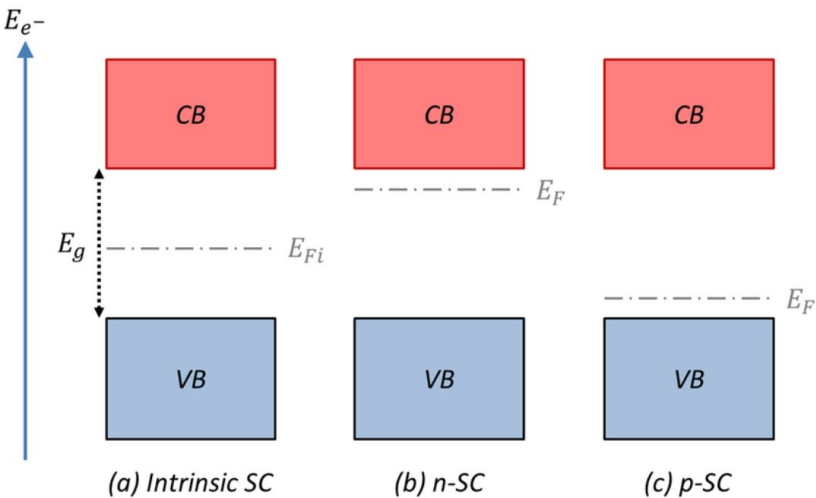
Types of Semiconductor Materials:

The silicon doped with extra electrons is called an “N type” semiconductor.
 “N” is for negative, which is the charge of an electron.

Silicon doped with material missing electrons that produce locations called holes is called “P type” semiconductor.
 “P” is for positive, which is the charge of a hole.

N type as well as P type semiconductor is electrically neutral.

Parameter	P-Type	N-Type
Impurity doped	Trivalent impurity	Pentavalent impurity
Also known as	Acceptor atom because of presence of additional hole.	Donor atom due to the existence of additional electron.
Doped group	Group III elements. For eg - boron, gallium, indium, aluminium etc.	Group V elements. for eg - arsenic, antimony, bismuth, phosphorus etc.
Majority carriers	Holes	Electrons
Minority carriers	Electrons	Holes
Conductivity	Due to presence of holes.	Due to presence of electrons.
Presence of fermi level	Fermi level appears closer to the valence than the conduction band.	Fermi level is present nearer to the conduction band than the valence ba
Concentration of electrons	Low	Very high as compared to p type semiconductor
Concentration of holes	High	Comparatively less than p type semiconductor.



A schematic diagram of Fermi level position within the band gap for different types of semiconductors
(Reference: [researchgate.net](https://www.researchgate.net))

What is Fermi energy (E_F) level?

The Fermi level represents the highest occupied energy level at absolute zero temperature (0 Kelvin), and the Fermi-Dirac distribution helps determine the probability of an energy level being occupied at a given temperature.

Key Differences between P-Type and N-Type Semiconductor:

In what follows, we itemized the key differences between p-type and n-type semiconductors:

1. A p-type semiconductor is created when group III elements are doped to a complete semiconductor material. As opposite, an n-type semiconductor is created when group V elements are doped to an intrinsic semiconductor.
2. As components like gallium, boron, indium, etc., are doped to form a p-type semiconductor; therefore, it creates an additional hole, thus also known as acceptor atom. On the contrary, components like bismuth, arsenic, antimony, etc., are doped to have an n-type semiconductor, creating an additional electron, thus also termed as donor atom.
3. Another factor that makes critical differences between p-type and n-type semiconductors is that in p-type semiconductors, holes are the majority carriers. While in the case of an n-type semiconductor, electrons behave as majority carriers.
4. The minority carriers in the case of the p-type semiconductor are electrons, whereas, in that of n-type semiconductors, these are holes.
5. The conductivity of the device depends mainly on the majority of carriers. Hence in the p-type semiconductor, holes are subject to current conduction. On the contrary, in the case of an n-type semiconductor, electrons are responsible for current conduction.
6. In the case of a p-type semiconductor, the fermi-level appears closer to the valence band than that of the conduction band. On the contrary, the Fermi level in the case of an n-type semiconductor exists near the conduction band.
7. The collection of holes is more than electrons in the case of p-type semiconductor material. This is while in n-type semiconductors, the concentration of electrons is more significant than gaps.

Band Theory of Solids
from quantum mechanical point of view

Are electrons in a specific quantum state in the valence or conduction band?

Within a band:

- Each electron occupies a specific allowed state labeled by k and n .
- Pauli exclusion principle still applies: only one electron (or two with opposite spins) per state.
- The band is like a continuum of closely spaced energy levels, and electrons fill up from the lowest energies.

Yes — each electron occupies a specific quantum state within the band. *But there's an important twist: These quantum states are delocalized over the entire crystal.*

In a solid:

- *Electrons are not bound to a single atom like they are in an isolated atom.*
- *Instead, they occupy **quantum states that extend over many atoms** — in fact, often the **entire crystal**.*
- These extended states are described by **Bloch wavefunctions** — they look like a **plane wave modulated by a periodic function** matching the crystal lattice.

Mathematically, the electron's wavefunction in a crystal is

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$$

where:

- $\mathbf{k}$ is the crystal momentum (a quantum number),
- n is the band index (which band — valence or conduction, etc.),
- $u_{n\mathbf{k}}(\mathbf{r})$ is a periodic function with the periodicity of the lattice.

Is the electron "spread out"?

Yes — the quantum state is **delocalized**, meaning:

- *The probability of finding the electron is **distributed over the entire crystal** (especially in a perfect crystal with no impurities (doping)).*
- *This delocalization is **what allows conduction**: electrons can respond to electric fields, move, and carry current.*

Development of Free Electron Theory

1. The classical free electron theory [Drude and Lorentz]

It is a macroscopic theory, in which the electrons move freely in lattice and it obeys the laws of classical mechanics. *Here the electrons are assumed to move in a constant potential.*

2. The quantum free electron theory [Sommerfeld Theory]

It is a microscopic theory, according to this theory the electrons in lattice moves in a constant potential and *it obeys law of quantum mechanics.*

3. Brillouin Zone Theory [Band Theory]

Bloch developed this theory in which the electrons move in a periodic potential provided by periodicity of crystal lattice. *It explains the mechanisms of conductivity on the basis of energy bands and hence band theory.* The electron theory explains the structure and properties of solids through their electronic structure. It explains the binding in solids, behavior of conductors and insulators, ferromagnetism, electrical and thermal conductivities of solids, elasticity, cohesive and repulsive forces in solids etc.

Classical free electron theory of metals

This theory was developed by Drude and Lorentz during 1900 and hence is also known as Drude-Lorentz theory. According to this theory, a metal consists of electrons which are free to move about in the crystal like molecules of a gas in a container. Mutual repulsion between electrons is ignored and hence potential energy is taken as zero. Therefore, the total energy of the electron is equal to its kinetic energy.

In 1900, P. Drude suggested that the metal crystals consist of positive ion cores with their valence electrons free to move among these positive metal ions. These valence electrons are also known as conduction electrons since they are responsible for the conduction of electricity in metals. The Coulomb's force of attraction between these positive ions and negatively charged electrons do not permit these free electrons to leave the metal surface. The potential field due to these ion's cores is supposed to be uniform and hence potential energy of electrons may be taken as constants (which is taken as zero for convenience). Hence we have to deal only with the kinetic energy of the electrons. The mutual repulsion among the electrons is neglected.

Postulates of Classical free electron theory:

1. **All the metals are composed of atoms.** Each atom have central nucleus around which there are revolving electrons.
2. **The electrons are free to move in all possible directions** about the whole volume of metals.
3. **The free electrons move in a constant potential field.** Hence the potential energy of the electrons is constant.
4. In the absence of an electric field the electrons move in random directions making collisions from time to time with positive ions which are fixed in the lattice or other free electrons. All the collisions are elastic i.e.; no loss of energy.
5. When an external field is applied the free electrons are slowly drifting towards the positive potential.
6. Since the electrons are assumed to be a perfect gas they obey classical kinetic theory of gasses.
7. Classical free electrons in the metal obey Maxwell-Boltzmann statistics.

Merits of Classical Free Electron Theory:

- 1) *It is used to verify Ohm's law.*
- 2) The electrical and thermal conductivities of metals can be explained.
- 3) It is used to derive Wiedemann- Franz law
- 4) It is used to explain the optical properties of metals

Drawbacks of Classical free electron theory:

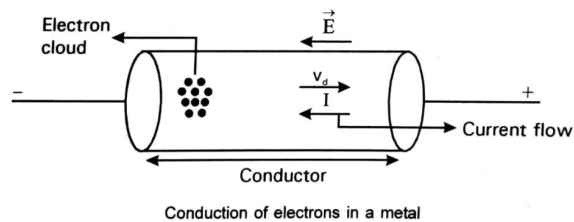
- 1) **According to this theory, $\kappa/\sigma = L \cdot T$, a constant (Wiedemann-Franz law) for all temperatures. But this is not true at low temperatures.**
- 2) The theoretically predicted value of specific heat of a metal does not agree with the experimentally obtained value.
- 3) This theory fails to explain ferromagnetism, superconductivity, photoelectric effect, Compton Effect and black body radiation.
- 4) It is a macroscopic theory.

The Wiedemann–Franz law, a fundamental principle in solid-state physics, states that the ratio of a metal's electronic thermal conductivity (κ , kappa) to its electrical conductivity (σ , sigma) is proportional to temperature (T), described by the equation $\kappa/\sigma = L \cdot T$, where L is the Lorenz number.

Quantum Free Electron Theory:

The quantum free electron theory, developed by **Sommerfeld**, explains the behavior of electrons in metals by applying quantum mechanics, including concepts like quantized energy levels and the Fermi-Dirac distribution to improve upon the classical free electron theory

In 1928, Sommerfeld suggested that electron should be treated quantum mechanically in the light of Pauli's exclusion principle which states that no two particles can have all the quantum numbers identical. *He used Fermi-Dirac statistics rather than Maxwell-Boltzmann statistics and succeeded in explaining many experimental facts. It overcomes many of the drawbacks of classical theory.*



- Main concept:** The theory builds upon the concept of free electrons within a metal, treating them as if they are moving freely within a potential well, similar to a gas.

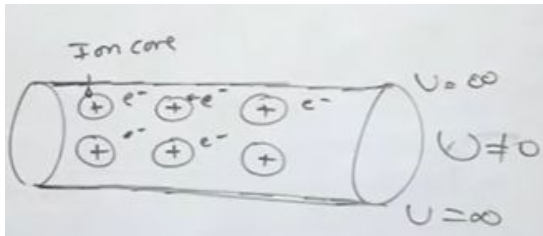
- Quantum Nature:** *Unlike the classical model, the quantum free electron theory acknowledges the quantized nature of electron energy levels,*

- Fermi-Dirac Statistics:** *The theory uses the Fermi-Dirac distribution to describe how electrons occupy these energy levels, taking into account their spin-half nature and the Pauli exclusion principle (no two electrons can occupy the same quantum state).*

- Fermi Level:** The Fermi level represents the highest occupied energy level at absolute zero temperature (0 Kelvin), and the Fermi-Dirac distribution helps determine the probability of an energy level being occupied at a given temperature.

- Electrical Conductivity:** The theory provides an expression for electrical conductivity based on quantum concepts, including the Fermi velocity and mean free path of electrons.

- Improvements over Classical Theory:** *The quantum free electron theory provides a more accurate explanation of various phenomena, including electrical conductivity, heat capacity, and thermal properties of metals, compared to the classical model.*

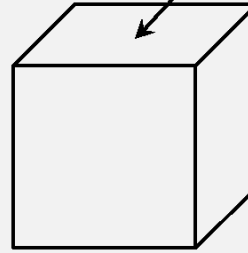


Assumptions of Quantum Free Electron Theory

- 1) Valence electrons move freely in a constant potential within the boundaries of metal and is prevented from escaping the metal at the boundaries (high potential). Hence the electron is trapped in a potential well.
- 2) The distribution of electrons in various allowed energy levels occurs as per Pauli Exclusion Principle.
- 3) The attraction between the free electrons and lattice ions and the repulsion between electrons themselves are ignored.
- 4) The distribution of energy among the free electrons is according to Fermi-Dirac statistics.
- 5) The energy values of free electrons are quantized.
- 6) To find the possible energy values of electron Schrodinger time independent wave equation is applied. The problem is similar to that of particle present in a potential box.

7) Energy of electron is $E_n = \frac{n^2 h^2}{8mL^2}$
 , Where $n=1, 2, 3, \dots$

The Free-Electron Model



}

$$\hat{H}\Psi = E\Psi$$

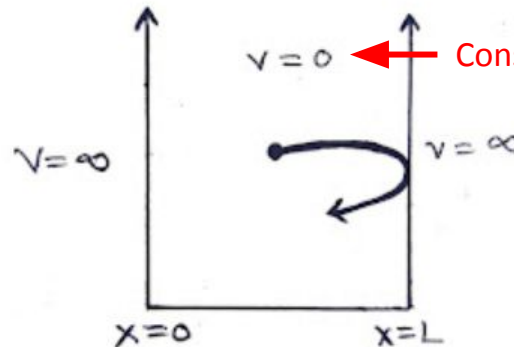
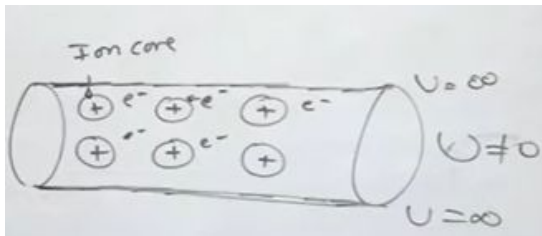
$$E_{n_x n_y n_z} = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2) \quad n_x = 1, 2, 3, \dots \quad n_y = 1, 2, 3, \dots \quad n_z = 1, 2, 3, \dots$$

Free Electron Gas in one-dimensional Box

Consider an electron having mass m confined to move in a one-dimensional rectangular box of length L i.e., The electron moves along a straight line say along the x -axis, let the wall of the box be rigid and non-penetrable. Let the electron can travel along the x -axis between $x=0$ and $x=L$.

The potential V is defined as

$$V = \begin{cases} 0 & \text{for } 0 < x < L \\ \infty & \text{for } x < 0 \text{ and } x > L \end{cases}$$



The Schrodinger's wave equation for the wave-function of the electron moving along the x -axis in a straight line may be written as

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\Psi = 0$$

For free electron inside the box $V=0$, so the above eq. becomes

$$\frac{d^2\Psi}{dx^2} + \frac{2mE}{\hbar^2}\Psi = 0$$

Where E is the Kinetic-energy of the electron

The Schrodinger's wave equation for the wave-function of the electron moving along the x-axis in a straight line may be written as

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For free electron inside the box $V=0$, so the above eq. becomes

$$\frac{d^2\Psi}{dx} + \frac{2mE}{\hbar^2}\Psi = 0$$

Where E is the Kinetic-energy of the electron

Or

$$\frac{d^2\Psi}{dx} + K^2\Psi = 0 \quad \dots(1)$$

Where

$$K^2 = \frac{2mE}{\hbar^2} \quad \dots(2)$$

The general solution of Eq. (1) may be written as

$$\Psi(x) = A \sin Kx + B \cos Kx \quad \dots(3)$$

Where A and B are the constants to be determined by the boundary conditions.

Boundary condition

The wave function $\Psi=0$ at $x=0$ and $x=L$

1. At $x=0$, $\Psi=0$, so eq. (3) becomes

$$0 = 0 + B$$

$$B = 0$$

So eq. (3) becomes

$$\Psi = A \sin Kx \quad \dots(4)$$

2. At $x=L$, $\Psi=0$, so eq. (4) becomes

$$0 = A \sin Kx \quad (A \text{ is constant so can't equal to } 0)$$

$$\sin KL = \sin n\pi$$

$$KL = n\pi$$

$$K = \frac{n\pi}{L}$$

$$K^2 = \frac{n^2 \pi^2}{L^2} \quad \dots(5)$$

For nth state eq. (4) becomes

$$\Psi = A \sin \frac{n\pi x}{L} \quad \dots(6)$$

Using eq. (2), eq. (5) can be written as

$$\frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{L^2}$$

Or

$$E = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

Since E depends on 'n', so let us denote the energy of the electron by E_n

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \dots(7)$$

The number n is called the quantum number. We find the lowest energy is obtained when $n=1$

From eq. (7), we have

$$E_1 = \frac{\hbar^2 \pi^2}{2mL^2}$$

i.e.,

$$E_n = n^2 E_1 \quad \dots(8)$$

Now, the spacing between the energy levels can be found by

$$\begin{aligned} E_{n+1} - E_n &= (n+1)^2 E_1 + n^2 (n+1)^2 E_1 \\ &= (2n+1) E_1 \end{aligned}$$

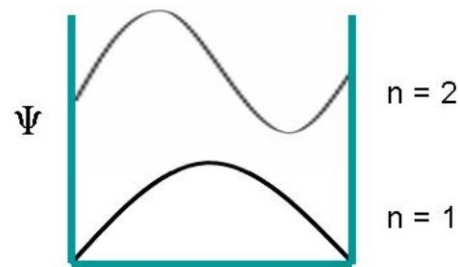
This is an important result that tells us:

1. The energy of a particle is quantized and
2. The lowest possible energy of a particle is **NOT** zero. This is called the **zero-point energy** and means the particle can **never be at rest** because it always has some kinetic energy.

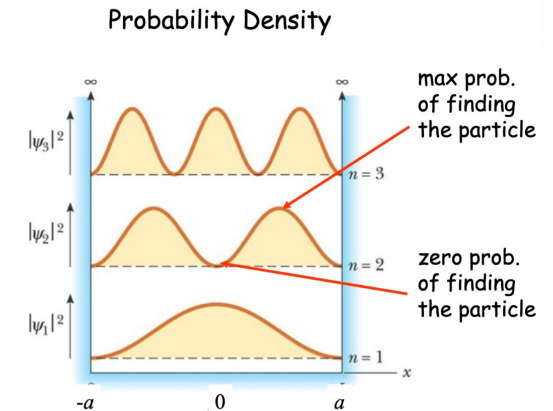
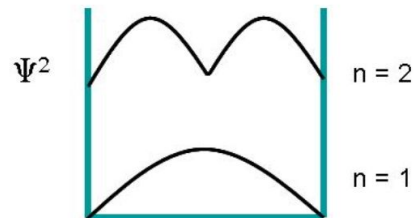
This is also consistent with the [Heisenberg Uncertainty Principle](#): if the particle had zero energy, we would know where it was in both space and time.

What does all this mean?

The wavefunction for a particle in a box at the $n = 1$ and $n = 2$ energy levels look like this:



The probability of finding a particle a certain spot in the box is determined by squaring ψ . The probability distribution for a particle in a box at the $n = 1$ and $n = 2$ energy levels looks like this:



Notice that the number of **nodes** (places where the particle has zero probability of being located) increases with increasing energy n . Also note that as the energy of the particle becomes greater, the quantum mechanical model breaks down as the energy levels get closer together and overlap, forming a continuum. This continuum means the particle is free and can have any energy value. At such high energies, the classical mechanical model is applied as the particle behaves more like a continuous wave. Therefore, the particle in a box problem is an example of [Wave-Particle Duality](#).

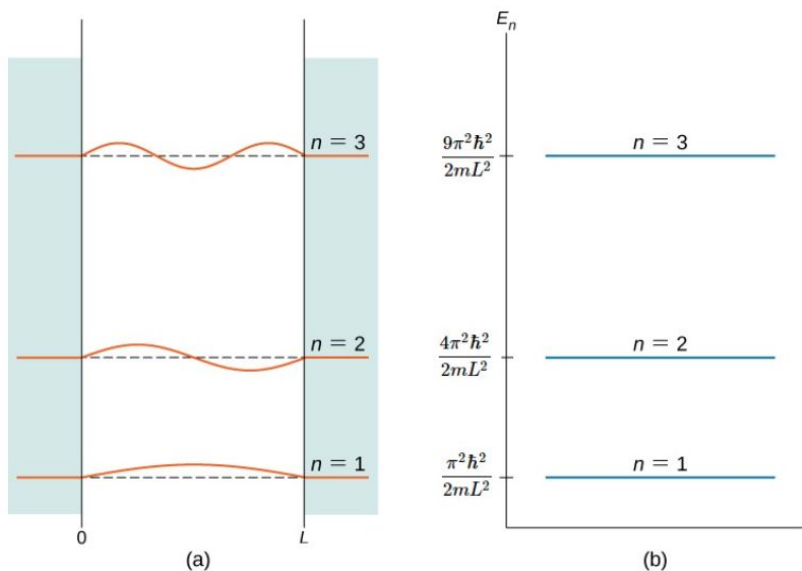
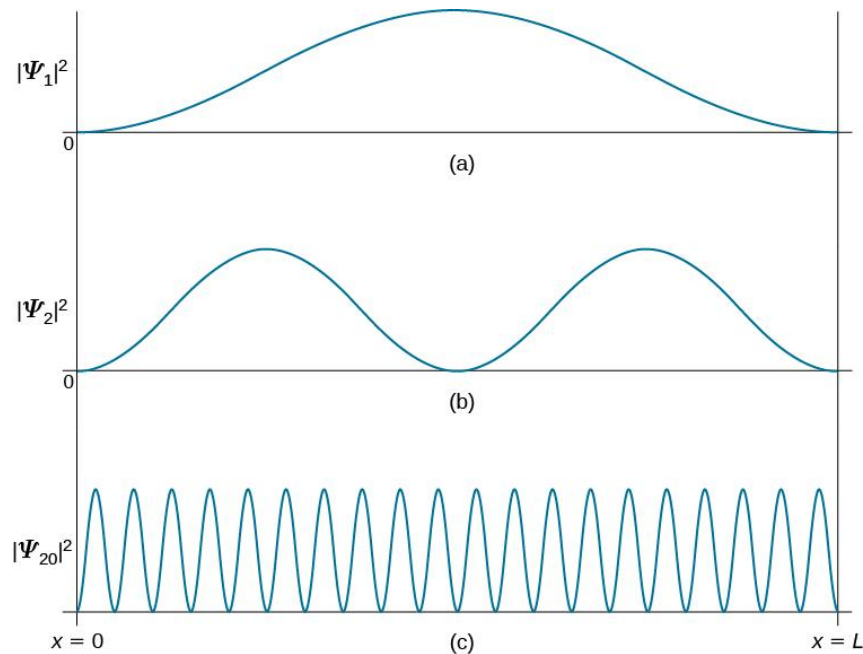


Figure 3.5.2: A plot of $\psi_n(x)$ for the first four wavefunctions. (CC-BY 4.0; OpenStax).



The probability density distribution $|\psi_n(x)|^2$ for a quantum particle in a box for: (a) the ground state, $n=1$; (b) the first excited state, $n=2$; and, (c) the nineteenth excited state, $n=20$.

https://chem.libretexts.org/Courses/Pacific_Union_College/Quantum_Chemistry/03%3A_The_Schrodinger_Equation_and_a_Particle_in_a_Box/3.05%3A_The_Energy_of_a_Particle_in_a_Box_is_Quantized#:~:text=Nodes%20and%20Curvature,in%20more%20complex%20quantum%20systems.

https://chem.libretexts.org/Courses/Pacific_Union_College/Quantum_Chemistry/03%3A_The_Schrodinger_Equation_and_a_Particle_in_a_Box/3.05%3A_The_Energy_of_a_Particle_in_a_Box_is_Quantized#:~:text=Nodes%20and%20Curvature,in%20more%20complex%20quantum%20systems.

Merits of quantum free electron theory

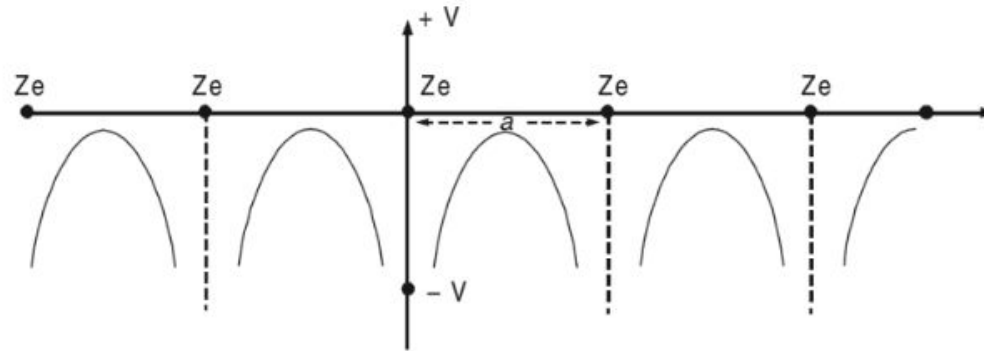
1. *It successfully explains the electrical and thermal conductivity of metals.*
2. It can explain the Thermionic phenomenon.
3. *It explains temperature dependence of conductivity of metals.*
4. It can explain the specific heat of metals.
5. It explains magnetic susceptibility of metals.
6. It can explain photo electric effect, Compton Effect and black body radiation etc.
7. It gives the correct mathematical expression for the thermal conductivity of metals.

Demerits of quantum free electron theory

1. *It is unable to explain the metallic properties exhibited by only certain crystals.*
2. It is unable to explain why the atomic arrays in metallic crystals should prefer certain structures only.
3. *This theory fails to distinguish between metal, semiconductor and Insulator.*
4. It also fails to explain the positive value of Hall Co-efficient.
5. *According to this theory, only two electrons are present in the Fermi level and they are responsible for conduction which is not true.*

Bloch's theorem

A crystalline solid consists of a lattice which is composed of a large number of positive ion cores at regular intervals and the conduction electrons move freely throughout the lattice. The variation of potential inside the metallic crystal with the periodicity of the lattice is explained by Bloch theorem. The periodic potential $V(x)$ may be defined by means of the lattice constant 'a' as $V(x) = V(x + a)$



Conclusion: From the above result it is clear that the energy spectrum of an electron in a periodic potential consists of allowed and forbidden energy bands. The regions corresponding to complex values of λ represent the allowed energy bands. The regions corresponding to real values of λ represent their forbidden energy bands.

In three dimensions the Bloch Theorem is expressed as

$$\psi_k(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} u_k(\vec{r})$$

Periodic potential in a crystal. The potential energy of an electron at a distance r from an atomic nucleus of charge $+Ze$ is given by

$$V = - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

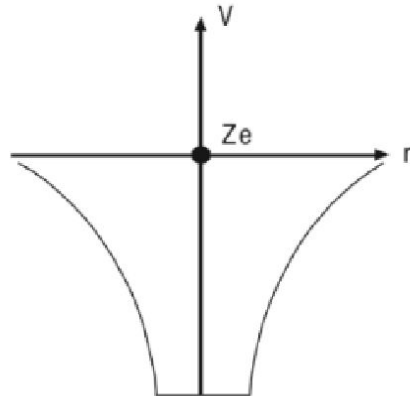


Fig. 6.5

The variation of potential energy V with distance r for an isolated system is shown in Fig. 6.5.

When a crystal is formed a number of such atomic nuclei are brought close to each other. The potential of an electron in such a case is the sum of the potential energies due to individual nuclei. The variation of potential energy with distance in a *one dimensional* case is shown in Fig. 6.6.

The atomic nuclei have a positive charge and the electrostatic potential energy of an electron in the field of positive charge is *attractive, i.e., negative*. This is why the potential energy is shown along the $-V$ axis. To a reasonable approximation the nuclei (or ion cores) have been considered at rest and the potential

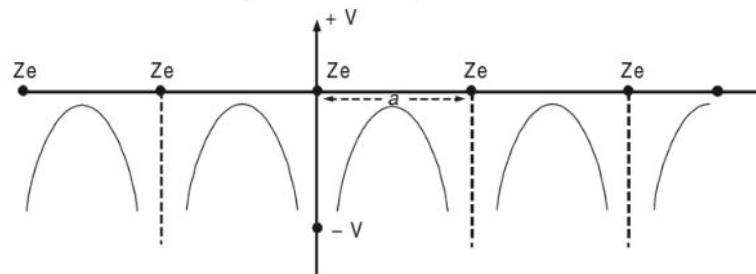
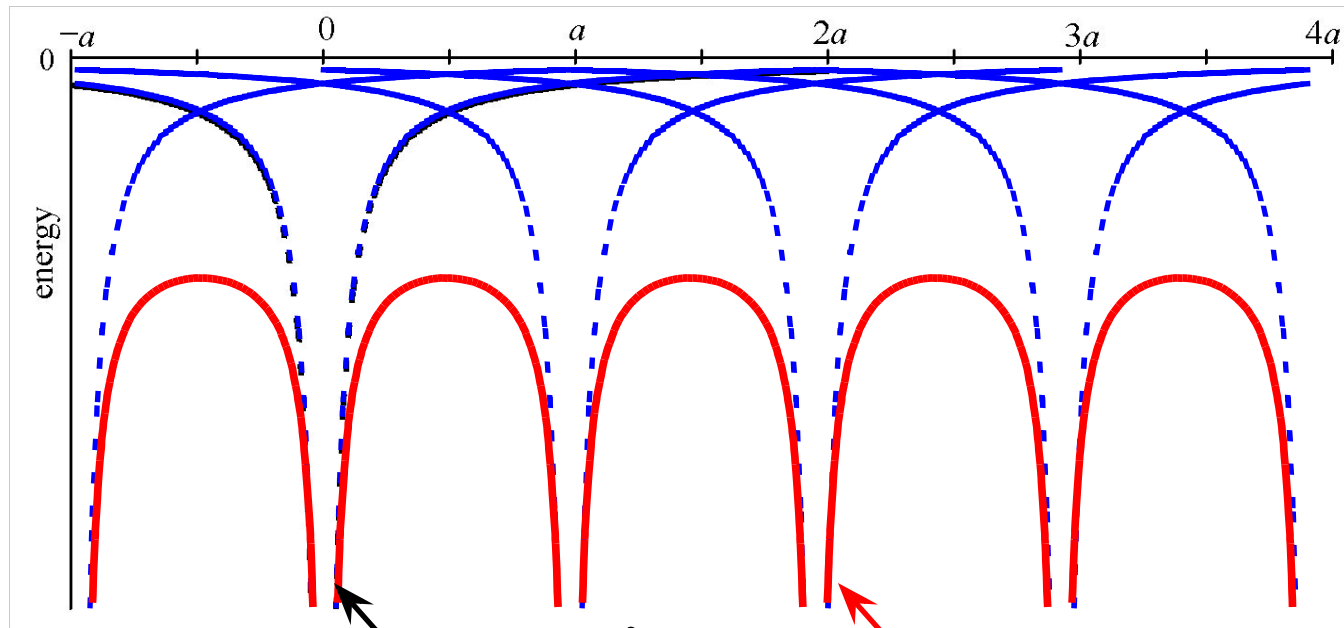


Fig. 6.6

experienced by an electron in the crystal is assumed to be periodic with the period equal to the lattice constant a .

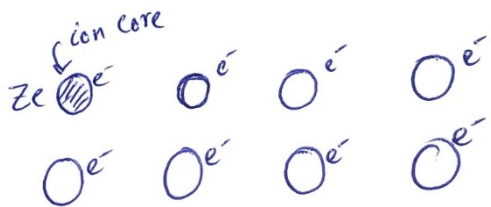
The Band Theory of Solids

Add an electric potential to represent the atomic ions.



$$V(x) = \frac{-Ze^2}{4\pi\epsilon_0 x}$$

sum



The potential is minimum at the positive ion sites and maximum between the two ions. The one-dimensional Schrödinger equation corresponding to this can be written as

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\Psi=0 \quad \text{where } \hbar = h / 2 \pi$$

$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - V(x))\Psi=0 \text{ -----(1)}$$

The periodic potential $V(x)$ may be defined by means of the lattice constant 'a' as

$$V(x) = V(x + a) \text{ ----- (2)}$$

The one-dimensional solution of the above Schrödinger equation is

$$\Psi(x) = e^{ikx} \cdot U_k(x) \text{ -----(3)}$$

In the above equation $U_k(x)$ is called “modulating function”. Because free electron wave is modulated by $U_k(x)$ is periodic with the periodicity at the crystal lattice. Let us now consider a linear chain of atoms of length L in one-dimensional case with „N“ number of atoms in the chain. Then

$$U_k(x) = U_k(x + Na) \text{ ----- (4)}$$

Where 'a' is lattice distance.

From equation (3) and (4), we have $\psi(x+Na) = e^{ik(x+Na)} \cdot U_k(x+Na)$

$$= e^{ikNa} \cdot U_k(x) e^{ikx}$$

$$\psi(x+Na) = \psi(x) \cdot e^{ikNa} \text{ ----- (5)}$$

Applying the condition of

$$\psi(x) = e^{ikx} \cdot U_k(x) \text{ ----- (3)}$$

$$U_k(x) = U_k(x+Na) \text{ ----- (4)}$$

This is referred to as Bloch condition. Similarly, the complex conjugate of eq.(5) can be written as

$$\psi^*(x+Na) = \psi^*(x) \cdot e^{-ikNa} \text{ ----- (6)}$$

From eq(5) and eq(6)

$$\psi(x+Na) \cdot \psi^*(x+Na) = \psi(x) \cdot \psi^*(x)$$

$$|\psi(x+Na)|^2 = |\psi(x)|^2 \text{ ----- (7)}$$

It means that the

$$P(x) = P(x+a)$$

□ Each electron in a crystalline solid "belongs" to each and every atom forming the solid

This means that the electron is not localized around one particular atom and the probability of finding the electron is same throughout the crystal.

Hence in eq(5),

$$e^{ikNa} = 1 \quad \leftarrow \text{(We can write it from equation 5)}$$

$$kNa = 2\pi n, \text{ where } n \text{ is an integer}$$

$$k = (2\pi n) / Na$$

$$k = (2\pi n) / L$$

Where L is the length of the chain of atoms and $n = \pm 1, \pm 2, \pm 3, \pm 4, \dots$

When $n = N/2$, we have $k = \pi/a$. This is the edge of first Brillouin zone.

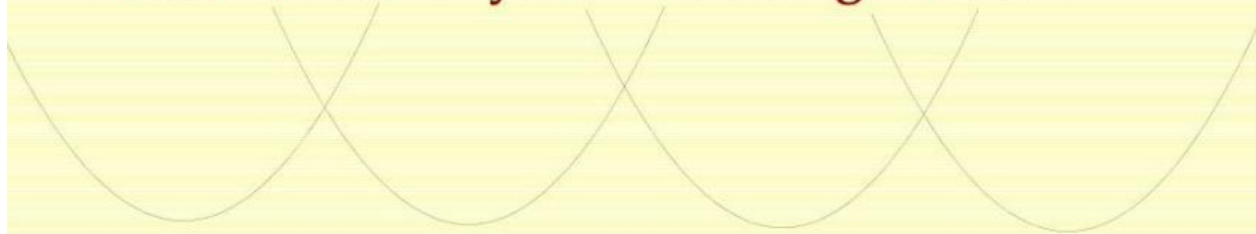
*Remember that in Bloch's theorem, the quantum number k (the wave vector) is not the real momentum of the electron. It's called the **crystal momentum** ($p = \hbar k$), and it's a quantum number that labels the state.*

Consequence of Bloch's Theorem

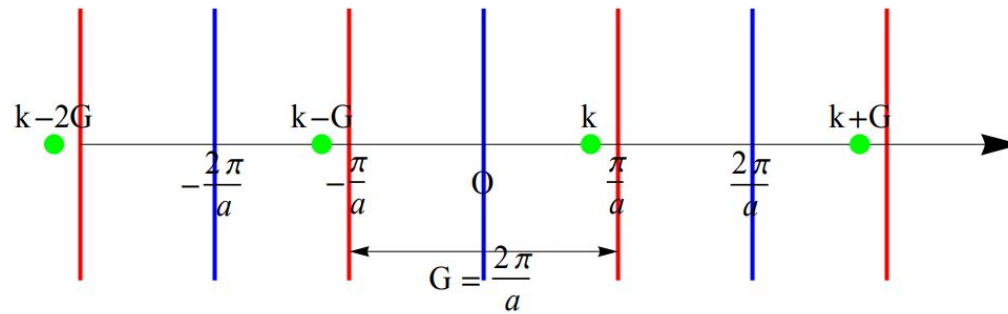
Probability of finding the electron

$$P(x) = P(x + a)$$

- Each electron in a crystalline solid “belongs” to each and every atom forming the solid



- Very accurate for metals where electrons are free to move around the crystal!



The first Brillouin zone is defined as $|k| \leq \frac{\pi}{a}$. There are N states in the first Brillouin zone.

When the spin of electron is taken into account, there are $2N$ states in the first Brillouin zone. Suppose that the number of electrons per unit cell is $n_c (= 1, 2, 3, \dots)$. Then the number of the total electrons is $n_c N$.

- (a) $n_c = 1$. So there are N electrons. $N/2N = 1/2$ (band-1: half-filled)
- (b) $n_c = 2$. $2N/2N = 1$ (band-1: filled).
- (c) $n_c = 3$. $3N/2N = 1.5$ (band-1: filled, band-2: half-filled).
- (d) $n_c = 4$. $4N/2N = 2$ (band-1: filled, band-2: filled).

When there are even electrons per unit cell, bands are filled. Then the system is an insulator. When there are odd electrons per unit cell, bands are not filled. Then the system is a conductor.

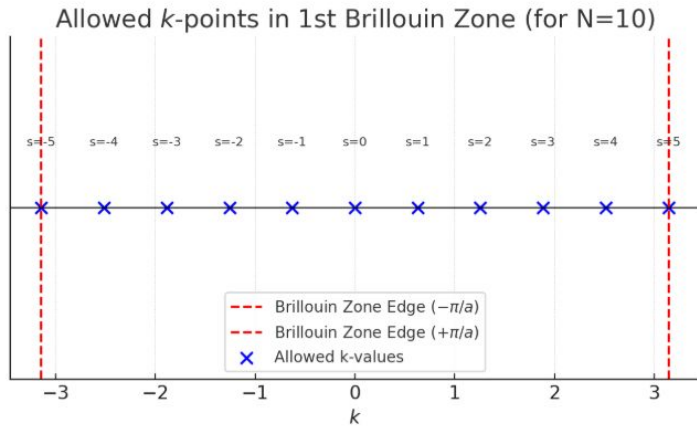
Physical and Mathematical Consequences

- 1.Redundancy: The wavevectors $\mathbf{k}$ and $\mathbf{k}+\mathbf{G}$ do **not** represent two different quantum states. They are two different *labels* for the **exact same physical state** of the electron.
- 2.Band Index: This is why energy bands are periodic in reciprocal space: $E_n(\mathbf{k}) = E_n(\mathbf{k} + \mathbf{G})$. The energy is the same for both labels.
- 3.Restricting $\mathbf{k}$ -space: *Because of this redundancy, we don't need to consider all possible $\mathbf{k}$ -vectors from negative to positive infinity. We can restrict ourselves to a finite region where every possible $\mathbf{k}$ is uniquely defined. This region is called the **First Brillouin Zone**.*
1. Any wavevector $\mathbf{k}'$ outside the first Brillouin Zone can be written as $\mathbf{k}' = \mathbf{k} + \mathbf{G}$, where $\mathbf{k}$ is the corresponding wavevector *inside* the first Brillouin Zone.
- 4.The Reduced Zone Scheme: This is the practice of "folding" all energy bands from higher Brillouin zones back into the first Brillouin Zone. In this scheme, the different bands (n=1, 2, 3...) arise precisely because for a single $\mathbf{k}$ *inside* the first zone, there are multiple *equivalent* $\mathbf{k}'$ points *outside* it ($\mathbf{k}' = \mathbf{k} + \mathbf{G}_1$, $\mathbf{k}' = \mathbf{k} + \mathbf{G}_2$, etc.), each with their own energy. When we fold them back, we get multiple energy values $E_1(\mathbf{k})$, $E_2(\mathbf{k})$, $E_3(\mathbf{k})$... for the same $\mathbf{k}$.

Analogy (*read it first! it helps you to understand it better*)

Imagine labeling rooms in an infinite hotel where every room is identical.

- You could give each room a unique number: ...-2, -1, 0, 1, 2, 3... (**This is like using all $\mathbf{k}$ -space**).
- But you realize the hotel is periodic. Room 5 is identical to Room 0, Room 6 is identical to Room 1, etc.
- So, a much smarter way is to only label one set of unique rooms, say Rooms 0, 1, 2, 3, 4, and then note that any other room (e.g., Room 13) is equivalent to one in this set (Room 13 $\equiv$ Room 3, because $13 = 3 + (2*5)$). **This set of 5 rooms is the First Brillouin Zone.**
- $\mathbf{k}$ is the unique label (0,1,2,3,4).
- $\mathbf{G}$ is what allows you to "shift" between equivalent labels (e.g., +10, -5, +15).
- $\mathbf{k}+\mathbf{G}$ (e.g., $3+10=13$) is just another label for the same physical room as $\mathbf{k}=3$.



- The **red dashed lines** mark the edges of the **1st Brillouin zone** ($-\pi/a$ to $+\pi/a$).
- The **blue crosses** are the **allowed k -points** (quantized values).
- Each point is labeled by its integer index s .

So s simply enumerates these allowed wavevectors.
For $N = 10$, you see 11 allowed values (from $s = -5$ to $s = +5$).

Note that the restriction for the value of s arises from the fact that $\psi_{k+G}(x) = \psi_k(x)$

$$k = \frac{2\pi s}{L} = \frac{2\pi s}{Na} = \frac{\pi}{a} \left(\frac{2s}{N} \right),$$

where

$$-\frac{N}{2} \leq s \leq \frac{N}{2}.$$

$$k = \frac{2\pi s}{L} = \frac{2\pi s}{Na} = \frac{\pi}{a} \left(\frac{2s}{N} \right),$$

where:

- k = allowed wavevector,
- $L = Na$ = total length of the crystal (with N unit cells, each of lattice constant a),
- s = an integer (or sometimes half-integer, depending on boundary conditions).

Meaning of s in this context:

- s is an **integer index** that labels the discrete allowed values of the wavevector k .
- It comes from imposing **periodic boundary conditions** (Born-von Karman conditions):

$$\psi(x + L) = \psi(x),$$

which quantizes k in steps of $2\pi/L$.

Thus:

$$k_s = \frac{2\pi}{L} s \quad \text{with } s = 0, \pm 1, \pm 2, \dots, \pm \frac{N}{2}.$$

So s labels the quantum number of the allowed k -states inside the Brillouin zone.

In total, there are N distinct values of k , corresponding to the N atoms (or unit cells).

Proof of Bloch theorem

For an electron moving in a one dimensional periodic potential the potential energy is given by the relation

$$V(x) = V(x + a)$$

where a is the period equal to lattice constant. In this case Schrödinger's equation becomes

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] \psi = 0 \quad \dots (ii)$$

With reference to the solution of this equation there is an important theorem known as *Bloch theorem* or *Floquet's theorem* which states that there exist solutions of the form

$$\psi(x) = e^{\pm ikx} \mu_k(x) \quad \dots (iii)$$

where

$$\mu_k(x) = \mu_k(x + a) \quad \dots (iv)$$

Thus the solutions are plane waves of the type $e^{\pm ikx}$ modulated by the function $\mu_k(x)$ which has the same periodicity as the lattice constant.

The wave function of the type $\psi(x) = e^{\pm ikx} \mu_k(x)$ is called *Bloch function*. The wave vector $\vec{k}$ gives the direction of Bloch wave.

Proof. Suppose $f(x)$ and $g(x)$ are two real and independent solutions of equation (ii). Then the general solution is

$$\psi(x) = A f(x) + B g(x) \quad \dots (v)$$

where A and B are arbitrary constants. As the potential is periodic, i.e., $V(x) = V(x + a)$, therefore, the functions $f(x + a)$ and $g(x + a)$ must also be the solutions of equation (ii). Since a differential equation of the second order can have only two independent solutions, the functions $f(x + a)$ and $g(x + a)$ must be able to be expressed in terms of functions $f(x)$ and $g(x)$, i.e.,

$$f(x + a) = \alpha_1 f(x) + \alpha_2 g(x) \quad \dots (vi)$$

$$\text{and} \quad g(x + a) = \beta_1 f(x) + \beta_2 g(x) \quad \dots (vii)$$

where $\alpha_1, \alpha_2, \beta_1$ and β_2 are real functions of energy E .

According to Eq. (v), the general solution of Schrodinger's equation is

$$\psi(x) = A f(x) + B g(x)$$

$$\text{and} \quad \psi(x + a) = A f(x + a) + B g(x + a)$$

According to Eq. (vi) and (vii), we have

$$\begin{aligned} \psi(x + a) &= A\alpha_1 f(x) + A\alpha_2 g(x) + B\beta_1 f(x) + B\beta_2 g(x) \\ &= (A\alpha_1 + B\beta_1)f(x) + (A\alpha_2 + B\beta_2)g(x) \end{aligned} \quad \dots (viii)$$

Now select A and B such that

$$A\alpha_1 + B\beta_1 = \lambda A \quad \dots (ix)$$

$$\text{and} \quad A\alpha_2 + B\beta_2 = \lambda B \quad \dots (x)$$

where λ is a constant. Substituting in Eq. (viii), we get

$$\begin{aligned} \psi(x + a) &= \lambda A f(x) + \lambda B g(x) \\ &= \lambda (A f(x) + B g(x)) = \lambda \psi(x) \end{aligned} \quad \dots (xi)$$

Equations (ix) and (x) will give non-zero values of A and B if the determinant of their coefficient vanishes, *i.e.*,

$$\begin{vmatrix} \alpha_1 - \lambda & \beta_1 \\ \alpha_2 & \beta_2 - \lambda \end{vmatrix} = 0$$

or $\lambda^2 - (\alpha_1 + \beta_2) \lambda + \alpha_1 \beta_2 - \alpha_2 \beta_1 = 0$... (xii)

We shall first prove that $\alpha_1 \beta_2 - \alpha_2 \beta_1 = 1$ as given below.

As $f(x)$ and $g(x)$ are two real and independent solutions of Eq. (ii), we have

$$\frac{d^2 f(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] f(x) = 0$$
 ... (xiii)

and $\frac{d^2 g(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] g(x) = 0$... (xiv)

Multiplying Eq. (xiii) by $g(x)$ and Eq. (xiv) by $f(x)$ and subtracting former from the latter, we get

$$f(x) \frac{d^2 g(x)}{dx^2} - g(x) \frac{d^2 f(x)}{dx^2} = 0$$

or $f(x) \frac{dg(x)}{dx} - g(x) \frac{df(x)}{dx} = \text{a constant}$ If the double derivative of x is zero, then the first derivative is always a constant

The left hand side is called Wronskian, $W(x)$, of the solution and is constant in this case.

$$\therefore W(x) = f(x) \frac{dg(x)}{dx} - g(x) \frac{df(x)}{dx}$$

Now from Eq. (vi) and (vii), we get

$$\begin{cases} f(x+a) = \alpha_1 f(x) + \alpha_2 g(x) & \dots (vi) \\ g(x+a) = \beta_1 f(x) + \beta_2 g(x) & \dots (vii) \end{cases}$$

$$\begin{aligned} W(x+a) &= f(x+a) \frac{dg(x+a)}{dx} - g(x+a) \frac{df(x+a)}{dx} \\ &= f(x+a) \left[\beta_1 \frac{df(x)}{dx} + \beta_2 \frac{dg(x)}{dx} \right] - g(x+a) \left[\alpha_1 \frac{df(x)}{dx} + \alpha_2 \frac{dg(x)}{dx} \right] \\ &= [\alpha_1 f(x) + \alpha_2 g(x)] \left[\beta_1 \frac{df(x)}{dx} + \beta_2 \frac{dg(x)}{dx} \right] \\ &\quad - [\beta_1 f(x) + \beta_2 g(x)] \left[\alpha_1 \frac{df(x)}{dx} + \alpha_2 \frac{dg(x)}{dx} \right] \\ &= (\alpha_1 \beta_2 - \alpha_2 \beta_1) \left[f(x) \frac{dg(x)}{dx} - g(x) \frac{df(x)}{dx} \right] \\ &= (\alpha_1 \beta_2 - \alpha_2 \beta_1) W(x) \quad \text{(see the last equation in the previous page)} \quad \dots (xv) \end{aligned}$$

Thus,

$$W(x+a) = (\alpha_1 \beta_2 - \alpha_2 \beta_1) W(x)$$

But $W(x+a)$

$$= W(x) = \text{a constant} \quad \text{(see the last equation in the previous page)}$$

$\therefore$

$$\alpha_1 \beta_2 - \alpha_2 \beta_1 = 1$$

Hence, Eq. (xii) becomes

$$\lambda^2 - (\alpha_1 + \beta_2) \lambda + 1 = 0 \quad \dots (xvi)$$

$$\begin{aligned} \lambda_1 &= \frac{(\alpha_1 + \beta_2) + \sqrt{(\alpha_1 + \beta_2)^2 - 4}}{2} \\ \lambda_2 &= \frac{(\alpha_1 + \beta_2) - \sqrt{(\alpha_1 + \beta_2)^2 - 4}}{2} \end{aligned}$$

The quantity $(\alpha_1 + \beta_2)$ is a real function of energy E . There are two roots λ_1 and λ_2 of this quadratic equation, so there will be two functions $\psi_1(x)$ and $\psi_2(x)$ which have the property $\psi_1(x+a) = \lambda \psi(x)$. The product of the two roots $\lambda_1 \lambda_2 = 1$.

We consider the following cases:

(i) For energy ranges such that $(\alpha_1 + \beta_2)^2 < 4$. In such a case Eq. (xvi), will have complex roots. Since $\lambda_1 \lambda_2 = 1$ these roots will be complex conjugate of each other. Therefore, we write

$$\lambda_1 = e^{ika} \quad \text{and} \quad \lambda_2 = e^{-ika}$$

where k is real.

The corresponding functions $\psi_1(x)$ and $\psi_2(x)$ will then have the property

$$\psi_1(x+a) = e^{ika} \psi_1(x) \quad \dots (xvii)$$

and

$$\psi_2(x+a) = e^{-ika} \psi_2(x) \quad \dots (xviii)$$

Thus in general

$$\psi(x+a) = e^{\pm ika} \psi(x) \quad \dots (xix)$$

It can be seen that a function having the property given by Eq. (xix) is the Bloch function of the type given in Eq. (iii)

$$\text{i.e.,} \quad \psi(x) = e^{\pm ikx} \mu_k(x)$$

by replacing x by $(x+a)$, then we get

$$\begin{aligned} \psi(x+a) &= e^{\pm ik(x+a)} \mu_k(x+a) \\ &= e^{\pm ika} e^{\pm ikx} \mu_k(x) \end{aligned}$$

because

$$\mu_k(x+a) = \mu_k(x) \quad \text{Also } e^{\pm ikx} \mu_k(x) = \psi(x)$$

$$\therefore \psi(x+a) = e^{\pm ika} \psi(x) = \lambda \psi(x) \quad \dots (xx)$$

where

$$\lambda = e^{\pm ika}$$

We, therefore, find that Eq. (xix) and Eq. (xx) are the same. This proves Bloch theorem.

Hence Bloch Theorem is proved.

$$\begin{aligned} \lambda_1 &= \frac{(\alpha_1 + \beta_2) + \sqrt{(\alpha_1 + \beta_2)^2 - 4}}{2} \\ \lambda_2 &= \frac{(\alpha_1 + \beta_2) - \sqrt{(\alpha_1 + \beta_2)^2 - 4}}{2} \end{aligned}$$

Note! See slide 66

$$\psi(x) = e^{\pm ikx} \mu_k(x) \quad \dots (iii)$$

(ii) For the energy range such that $(\alpha_1 + \beta_2)^2 > 4$. In such a case Eq. (xvi) will have *real* roots λ_1 and λ_2 , which may be taken as

$$\lambda_1 = e^{\mu a} \quad \text{and} \quad \lambda_2 = e^{-\mu a}$$

where μ is real. The corresponding solutions to the Schrödinger's equation are

$$\psi_1(x) = e^{\mu x} \mu(x) \quad \text{and} \quad \psi_2(x) = e^{-\mu x} \mu(x) \quad \dots (xxi)$$

Though mathematically valid, these wave functions are not allowed because these functions become infinite at $+\infty$ or $-\infty$, *i.e.*, they are not bounded functions.

The roots $\lambda_1 = e^{ika}$ and $\lambda_2 = e^{-ika}$ (*allowed roots*) as well as the roots $\lambda_1 = e^{\mu a}$ and $\lambda_2 = e^{-\mu a}$ (forbidden roots) are functions of $(\alpha_1 + \beta_2)$, hence of energy E . The allowed roots correspond to the *allowed energy regions* and the forbidden (or disallowed) roots are associated with *forbidden energy regions*. This means that the energy spectrum of an electron moving in a periodic potential consists of allowed and forbidden energy regions or bands.

Repetitive slide for proof of Bloch theorem

Statement of Bloch theorem: Bloch theorem states that, the solutions of Schrödinger wave equation for an electron moving in a periodic potential are the plane waves modulated by a function having the same periodicity as that of the lattice.

Explanation: For an electron moving in a one-dimensional periodic potential, $V(x)$, the Schrödinger equation is written as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] = 0 \text{----- (1)}$$

Where

$$V(x) = V(x + a) \text{----- (2)}$$

and a is lattice constant.

According to the Bloch theorem, the solution of the above equation must be of the form

$$\psi(x) = e^{\pm iKx} u_K(x) \text{----- (3)}$$

Where $u_K(x) = u_K(x + a)$ ----- (4)

K =wave vector. u_K is a function having complete periodicity of the lattice.

The functions of the type of equation (3) are known as Bloch function

For three dimensional case it is given as $\psi(\vec{r}) = e^{\pm i\vec{K}\cdot\vec{r}} u_K(\vec{r})$

Proof: Let $g(x)$ and $f(x)$ be two real and independent solutions of the second order differential equation (1).

So the general solution can be written as

$$\psi(x) = A f(x) + B g(x) \text{----- (5)}$$

Since $V(x) = V(x + a)$, the functions $f(x + a)$ and $g(x + a)$ must also be the solutions of equation (1). But only two independent solutions are possible for second order linear differential equation. So $f(x + a)$ and $g(x + a)$ must be expressed as a linear combination of $f(x)$ and $g(x)$ as

$$\left. \begin{aligned} f(x + a) &= \alpha_1 f(x) + \alpha_2 g(x) \text{----- (a)} \\ g(x + a) &= \beta_1 f(x) + \beta_2 g(x) \text{----- (b)} \end{aligned} \right\} \text{--- (6)}$$

Where $\alpha_1, \alpha_2, \beta_1$ and β_2 are the real functions of energy E .

Now we may write another solution from equation (5) using equation (6) as

$$\begin{aligned} \psi(x + a) &= A f(x + a) + B g(x + a) \\ &= (A\alpha_1 + B\beta_1)f(x) + (A\alpha_2 + B\beta_2)g(x) \text{--- (7)} \end{aligned}$$

the property given in equation (6) we should choose A and B such that

$$\left. \begin{aligned} A\alpha_1 + B\beta_1 &= \lambda A \text{----- (a)} \\ A\alpha_2 + B\beta_2 &= \lambda B \text{----- (b)} \end{aligned} \right\} \text{----- (8)} \quad \text{Where } \lambda \text{ is a constant.}$$

So equation (7) may be written as

$$\psi(x+a) = \lambda[Af(x) + Bg(x)] = \lambda\psi(x) \text{ ----- (9)}$$

The equations (8) has non-zero values of A and B only if the determinant of their co-efficient vanish.

$$\text{So } \begin{vmatrix} \alpha_1 - \lambda & \beta_1 \\ \alpha_2 & \beta_2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (\alpha_1 - \lambda)(\beta_2 - \lambda) - \alpha_2\beta_1 = 0$$

$$\Rightarrow \alpha_1\beta_2 - (\alpha_1 + \beta_2)\lambda + \lambda^2 - \alpha_2\beta_1 = 0 \text{ ---- (10)}$$

Taking first order derivative of equations (6) we get

$$f'(x+a) = \alpha_1 f'(x) + \alpha_2 g'(x)$$

$$g'(x+a) = \beta_1 f'(x) + \beta_2 g'(x)$$

$$\text{Now, } f(x+a) g'(x+a) - g(x+a) f'(x+a)$$

$$= [\alpha_1 f(x) + \alpha_2 g(x)][\beta_1 f'(x) + \beta_2 g'(x)] - [\beta_1 f(x) + \beta_2 g(x)][\alpha_1 f'(x) + \alpha_2 g'(x)]$$

$$= \alpha_1\beta_1 f(x)f'(x) + \alpha_1\beta_2 f(x)g'(x) + \alpha_2\beta_1 g(x)f'(x) + \alpha_2\beta_2 g(x)g'(x)$$

$$- \alpha_1\beta_1 f(x)f'(x) - \alpha_2\beta_1 f(x)g'(x) - \alpha_1\beta_2 g(x)f'(x) - \alpha_2\beta_2 g(x)g'(x)$$

$$= [f(x)g'(x) - g(x)f'(x)](\alpha_1\beta_2 - \alpha_2\beta_1) \text{ ----- (11)}$$

Due to periodic property we may write

$$f(x+a) g'(x+a) - g(x+a) f'(x+a) = f(x) g'(x) - g(x) f'(x)$$

Putting this value in equation (11) we get

$$f(x) g'(x) - g(x) f'(x) = [f(x)g'(x) - g(x)f'(x)](\alpha_1\beta_2 - \alpha_2\beta_1)$$

$$\Rightarrow (\alpha_2\beta_1 - \alpha_1\beta_2) = 1$$

Putting this value in equation (10) we get

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + 1 = 0$$

$$\Rightarrow \lambda = \frac{(\alpha_1 + \beta_2) \pm \sqrt{(\alpha_1 + \beta_2)^2 - 4}}{2}$$

It gives two possible values λ_1 and λ_2 of λ such that $\lambda_1\lambda_2 = 1$. Thus there are two functions $\psi_1(x)$ and $\psi_2(x)$ which satisfy equation (9).

Now we consider the following two cases

<https://www.studocu.com/in/document/kolhan-university/physics/bloch-the-orem-lecture-notes-2/7409004>

Case 1: For energy range in which $(\alpha_1 + \beta_2)^2 > 4$

In this case the roots of λ are real and reciprocal of each other. Their solutions do not represent a wave function and thus there are no electronic state in the energy region corresponding to λ_1 and λ_2 .

Case 2: For energy range in which $(\alpha_1 + \beta_2)^2 < 4$

In this case the roots of λ are complex. $\lambda_1 \lambda_2 = 1$ show that the roots will be complex to each other.. so we write $\lambda_1 = e^{iKa}$ and $\lambda_2 = e^{-iKa}$

Where K is real.

The corresponding wave functions $\psi_1(x)$ and $\psi_2(x)$ then have the property

$$\psi_1(x+a) = e^{iKa} \psi_1(x)$$

$$\psi_2(x+a) = e^{-iKa} \psi_2(x)$$

In general we write

$$\psi(x+a) = e^{\pm iKa} \psi(x) \text{ ----- (12)}$$

Now we replace (x) by $(x+a)$ in equation (3) and get

$$\psi(x+a) = e^{\pm iK(x+a)} u_K(x+a)$$

Using equation (4) it becomes $\psi(x+a) = e^{\pm iKa} e^{\pm iKx} u_K(x)$

$$\Rightarrow \psi(x+a) = e^{\pm iKa} \psi(x) \text{ ----- (13)}$$

Equations (12) and (13) are the same.

Hence Bloch Theorem is proved.

Conclusion: From the above result it is clear that the energy spectrum of an electron in a periodic potential consists of allowed and forbidden energy bands. The regions corresponding to complex values of λ represent the allowed energy bands. The regions corresponding to real values of λ represent their forbidden energy bands.

In three dimensions the Bloch Theorem is expressed as

$$\boxed{\psi_k(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} u_k(\vec{r})}$$

Significance wave vector $\vec{k}$: the wave vector $\vec{k}$ has several properties

1. Under a crystal lattice translation which carries $\vec{r}$ to $\vec{r} + \vec{T}$ we have

$$\psi_k(\vec{r} + \vec{T}) = e^{i\vec{k} \cdot \vec{T}} e^{i\vec{k} \cdot \vec{r}} u_k(\vec{r} + \vec{T}) = e^{i\vec{k} \cdot \vec{T}} \psi_k(\vec{r})$$

Because $u_k(\vec{r} + \vec{T}) = u_k(\vec{r})$. Thus $e^{i\vec{k} \cdot \vec{T}}$ is the phase factor by which a block function is multiplied when a crystal translation $\vec{T}$ is made.

2. The quantity $\vec{k}$ enters in the conservation laws that govern collision processes in crystal.

**How Bloch theorem creates the band gap,
explain me mathematically**

How Bloch theorem creates the band gap, Explain me mathematically.

The band gap is not directly calculated from the Bloch theorem itself, but rather is a concept that arises from applying the Bloch theorem to the Schrödinger equation for electrons in a periodic crystal potential. The Bloch theorem, which describes electron wavefunctions in a periodic lattice, leads to the formation of energy bands (allowed energy ranges) separated by band gaps (forbidden energy ranges). To calculate the actual energy values of these bands and gaps, you need to solve the Schrödinger equation using the [Bloch functions](#) as solutions and incorporate the specific periodic potential of the crystal.

1. Schrödinger equation in a periodic potential

We start with the time-independent Schrödinger equation:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r})$$

with a **periodic potential**:

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$$

2. Bloch's theorem form of solution

Bloch's Theorem: It states that the solutions to the [Schrödinger equation](#) for an electron in a periodic potential (like a crystal lattice) are **Bloch functions**. These functions have the form

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_k(\mathbf{r})$$

where $u_k(\mathbf{r})$ has the lattice periodicity:

$$u_k(\mathbf{r} + \mathbf{R}) = u_k(\mathbf{r})$$

3. Expand periodic functions in Fourier series

Since both $V(\mathbf{r})$ and $u_k(\mathbf{r})$ are periodic, they can be written as sums of **reciprocal lattice vectors**:

- For potential:

$$V(\mathbf{r}) = \sum_G V_G e^{iG \cdot \mathbf{r}}$$

V_G = Fourier component of the crystal potential, determines **coupling strength**.

- For periodic part of wavefunction:

$$u_k(\mathbf{r}) = \sum_G C_{k+G} e^{iG \cdot \mathbf{r}}$$

C_{k+G} = coefficient of the plane wave with momentum $k + G$ in the expansion of the Bloch state, describes the **weight (mixing)** of that component

So the total wavefunction is:

$$\psi_k(\mathbf{r}) = \sum_G C_{k+G} e^{i(k+G) \cdot \mathbf{r}}$$

This shows: **Bloch wave = superposition of plane waves with momenta differing by reciprocal vectors.**

What is C_{k+G} ?

what is V_G ?

- When you expand the **periodic potential** $V(x)$ of the crystal in a Fourier series, you get

$$V(x) = \sum_G V_G e^{iGx}$$

where the sum runs over all reciprocal lattice vectors $G = 2\pi n/a$ in 1D.

- The Fourier coefficient is
$$V_G = \frac{1}{a} \int_0^a V(x) e^{-iGx} dx.$$
- Physically: V_G tells you **how strongly the periodic lattice couples electron states whose wavevectors differ by G .**
 - If $V_G = 0$, there's no mixing $\rightarrow$ no band gap.
 - The band gap size is proportional to $2|V_G|$.

- Bloch's theorem says the electron wavefunction can be written as a sum of plane waves shifted by reciprocal lattice vectors:

$$\psi_k(x) = \sum_G C_{k+G} e^{i(k+G)x}.$$

- Here:
 - C_{k+G} is the **expansion coefficient (amplitude)** of the plane-wave component with momentum $k + G$.
 - For a free electron (no lattice potential), only one term is nonzero (say $C_k = 1$), but in a real crystal, many coefficients C_{k+G} are nonzero because the periodic potential mixes them.

4. Plugging into Schrödinger equation

Substituting into Schrödinger equation gives a set of coupled equations for the coefficients:

$$\left(\frac{\hbar^2 |k + G|^2}{2m} - E \right) C_{k+G} + \sum_{G'} V_{G-G'} C_{k+G'} = 0$$

This is the **central equation of the nearly free electron model**.

5. Degeneracy at Brillouin zone edge

Now, focus on what happens at the **Brillouin zone boundary**.

- Take 1D for simplicity. Reciprocal lattice vector:

$$G = \frac{2\pi}{a}$$

- At $k = G/2 = \pi/a$:
 - Two plane waves $\exp(ikx)$ and $\exp(i(k - G)x)$ have **same energy** (degenerate):

$$E_0 = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 (k-G)^2}{2m}$$

- But the periodic potential couples these two waves. So we must solve a **2×2 matrix problem**:

$$\begin{bmatrix} E_0 - E & V_G \\ V_G^* & E_0 - E \end{bmatrix} \begin{bmatrix} C_k \\ C_{k-G} \end{bmatrix} = 0$$

6. Energy splitting (band gap)

The eigenvalues of this matrix are:

$$E = E_0 \pm |V_G|$$

So instead of one degenerate energy, we get **two energies separated by**:

$$\Delta E = 2|V_G|$$

This is the **band gap** at the Brillouin zone boundary.

- **Origin of the band gap:**

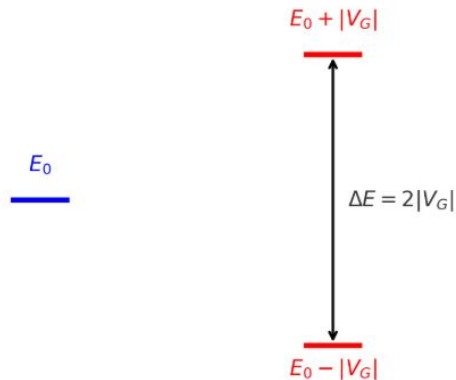
Degenerate free-electron states at the Brillouin zone boundary (k and $k - G$) are mixed by the periodic potential.

- The mixing lifts the degeneracy, giving two energies separated by

$$\Delta E = 2|V_G|$$

- Thus, a **forbidden energy gap** opens up at the zone edge.

Origin of Band Gap at Brillouin Zone Edge



- On the left (blue): a **single degenerate free-electron energy level** at the Brillouin zone edge.
- On the right (red): when the periodic potential couples the two degenerate states, the degeneracy is **lifted**, splitting into two levels:
 $E_0 + |V_G|$ and $E_0 - |V_G|$
- The separation is the **band gap**:
 $\Delta E = 2|V_G|$

This is the direct mathematical outcome of solving the 2×2 coupled equations at the zone boundary.

ΔE is the energy gap at the zone boundary from Bloch theorem

Yes, within the [band theory of solids](#), ΔE represents the energy gap at the [zone boundary](#), specifically the [Brillouin zone boundary](#), where the electron's allowed energy levels (bands) are separated by forbidden energy regions. Bloch's theorem explains the formation of these [energy bands and band gaps](#) in a periodic potential, which arises from the constructive and destructive interference of electron waves reflected by the crystal lattice at these boundaries.

- **Zone boundary:** The Brillouin zone boundary corresponds to specific wave vectors (k) where the electron waves are strongly reflected by the periodic lattice.
- **Energy gap (ΔE):** At the zone boundary, Bragg-like reflections cause energy discontinuities, resulting in an energy gap (ΔE) where no electron states can exist.

Physical origin of gaps (why they appear)

- Bloch states are of the form $\psi_k(x) = e^{ikx}u_k(x)$ with u_k periodic.
- The periodic potential couples plane waves differing by reciprocal-lattice vectors $G = 2\pi n/a$ (so k couples to $k + G$).
- At Brillouin-zone boundaries (e.g. $k = \pm\pi/a$) free-electron states that would be degenerate are mixed by the potential and split in energy — the splitting is a band gap.
- In the nearly-free-electron (two-state) picture the gap at a zone boundary is about $\Delta \approx 2|V_G|$, where V_G is the Fourier coefficient of the periodic potential at that G .

Kronig-Penney Model

Kronig-Penney Model

- An effective way to understand the energy gap in semiconductors is to model the interaction between the electrons and the structure of the crystal where atoms or groups of atoms are arranged on nodes of a 3D periodic lattice and vibrate around their equilibrium positions
- **R. de L. Kronig** and **W. G. Penney** developed a useful one-dimensional model of the electron lattice interaction in 1931.
- The model assumes that when an electron is near the positive ion site, potential energy is taken as zero. Outside the well, that is in between two positive ions potential energy is assumed to be V_0 . The model is one of the simplest periodic potentials to exhibit a band gap

Approach 1

Kronig and Penney treated a simplest example for one dimensional periodic potential. In this model it is assumed that the potential energy of an electron has the form of a periodic array of square wells as shown in Fig

BEHAVIOUR OF AN ELECTRON IN A PERIODIC POTENTIAL -THE KRONIG PENNEY MODEL (QUALITATIVE TREATMENT)

Kronig and Penney treated a simplest example for one dimensional periodic potential. In this model it is assumed that the potential energy of an electron has the form of a periodic array of square wells as shown in Fig. 1.17.

Here we have two regions. viz.

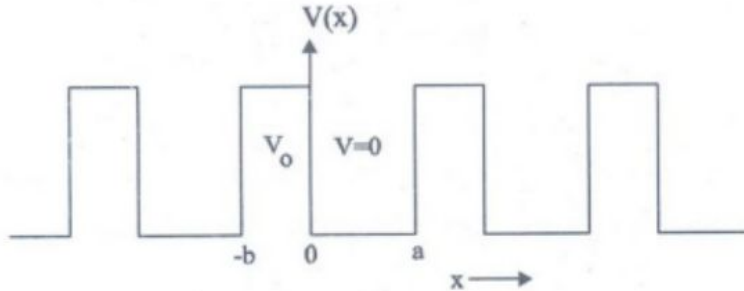
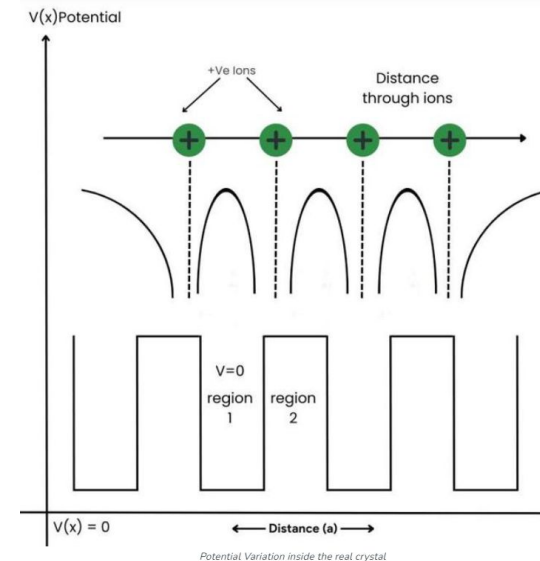
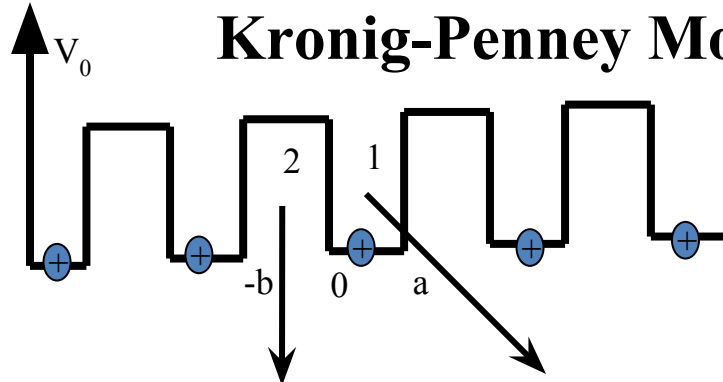


Fig. 1.17



The **total periodicity** of the potential is $a + b$, corresponds to the **lattice constant** a_{lattice}

Kronig-Penney Model



for region (2)

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}[E - V_0]\psi = 0$$

$$v(x) = v_0, -b < x < 0$$

$V(x) = V_0$, if the electron is outside the atom

for region (1)

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}[E]\psi = 0$$

$$v(x) = 0, 0 < x < a$$

$V(x) = 0$, if the electron is inside the atom or very close to the atom

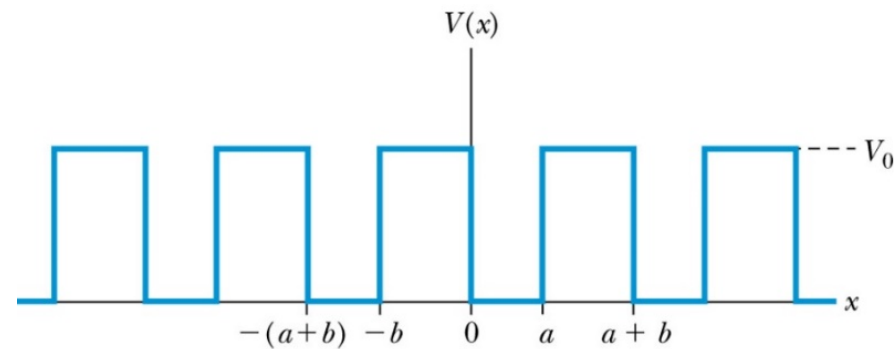
$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \dots (1)$$

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \dots (2)$$

where

$$\alpha^2 = \frac{8\pi^2mE}{h^2}$$

$$\beta^2 = \frac{8\pi^2m}{h^2}[V_0 - E]$$



• **Potential Barriers:** High potential (width b).

• **Potential Wells:** Zero potential (width a).

The pattern repeats infinitely: ...well-barrier-well-barrier

The values of a and b are chosen to reflect the material's atomic structure, and $a+b$ is the total periodicity of the potential, which corresponds to the lattice constant in a crystal.

*The **total periodicity** of the potential is $a + b$, corresponds to the **lattice constant** a_{lattice}*

The values of a and b are **arbitrary** in the basic Kronig-Penney model.

Because the Bloch theorem defines the lattice constant a as this periodicity, we directly equate:

In Kronig-Penney model, Lattice Constant $\equiv a_{\text{lattice}} = a_{\text{well}} + b_{\text{barrier}}$

What should be the value of a and b ?

The values of a and b are **arbitrary** in the basic Kronig-Penney model, but they should be chosen to reflect a reasonable physical model for a crystalline solid.

- a : This is the width of each individual potential well. It can be related to the spacing between atoms or atomic groups in a crystal.
- b : This is the width of the potential barrier between the wells. It reflects the distance between atoms or atomic groups where the potential is high.

If you're modeling a simple material with a regular atomic arrangement (e.g., a simple cubic or face-centered cubic structure), a and b can be chosen based on the width of the potential wells (where the atoms or ions are located) and the width of the potential barriers (regions where the potential is high, and no atoms are present).

The **total periodicity** of the potential, which is $a + b$, corresponds to the **lattice constant** a_{lattice} in the context of solid-state physics. This is the distance between two neighboring atoms or repeating units in the crystal lattice.

Thus, the **lattice constant** in the Kronig-Penney model is given by:

$$a_{\text{lattice}} = a + b$$

2. How are a and b determined?

In the real world, the values of a and b are determined by the physical characteristics of the material being modeled. These values are typically obtained from experimental measurements of the crystal lattice.

Here's how they are generally determined:

a. From Experimental Data (X-ray Diffraction, etc.):

- The **lattice constant** a_{lattice} is typically determined using techniques like **X-ray diffraction** or **neutron diffraction**. These methods can measure the spacing between the repeating units of the crystal lattice.
- Once the lattice constant a_{lattice} is known, you can choose values for a and b such that:

$$a_{\text{lattice}} = a + b$$

Understanding $V(x) = V_0$ and $V(x) = 0$

- **When $V(x) = V_0$:**

This corresponds to the **barrier regions** in the Kronig-Penney model where the potential energy is higher. These barriers represent regions **outside the atoms** or more precisely, regions where the electron experiences a repulsive potential due to the positively charged atomic cores and the lack of favorable bonding or electron cloud overlap.

So, $V(x) = V_0$ is usually interpreted as the electron being in the **interstitial region between atoms**, where the potential energy is relatively high.

- **When $V(x) = 0$:**

This corresponds to the **well regions**, where the potential energy is lower. These wells are typically interpreted as regions **inside or near the atoms**, where the electron feels the attractive potential of the atomic nucleus and the electron cloud.

Hence, $V(x) = 0$ can be thought of as the electron being **inside or near the atom**, in a potential well where the electron is more likely to be found due to lower potential energy.

Summary:

	Physical Interpretation
	<i>Electron in the barrier region (outside the atom) with higher potential energy</i>
	<i>Electron in the well region (inside or near the atom) with lower potential energy</i>

Important note:

- This is a **simplified, idealized model**. In reality, electrons are not strictly "inside" or "outside" atoms in such a binary way, but the model captures the essential physics of periodic potential variations and how they influence electron behavior in solids.
- The Kronig-Penney model abstracts the atomic potentials into simple rectangular wells and barriers to make the problem analytically solvable and to understand band structure formation.

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \dots (1)$$

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \dots (2)$$

Now you have to solve these two equations using Bloch's periodic potential

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

For both the region, the appropriate solution suggested by Bloch is of the form,

$$\Psi(x) = e^{ikx} u_x(x) \dots (3)$$

Differentiating equation (3) and substituting it in equation (1) and (2) and then further solving it under the boundary conditions, we get

$$\frac{P \sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos(ka) \dots (4)$$

where $P = \frac{mV_0 b a}{\hbar^2}$ is called as Scattering power of the potential barrier, which is the measure of the strength with which the electrons are attracted by the positive ions.

In equation (4), there are only two variables (i.e) α and k . We know $\cos ka$ can take values only from -1 to 1, Therefore the left hand side of equation (4) must also fall in this range. A plot is made between the LHS of equation (4) and αa for a value of $P = 3\pi/2$ (arbitrary) as shown in Fig. 1.18.

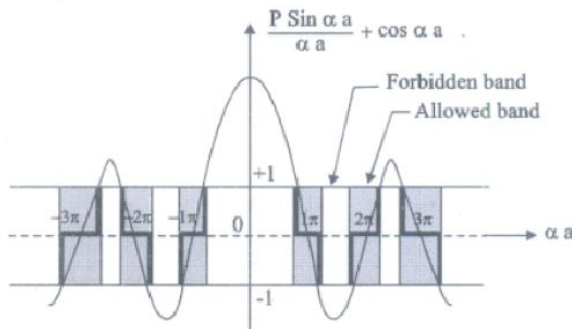


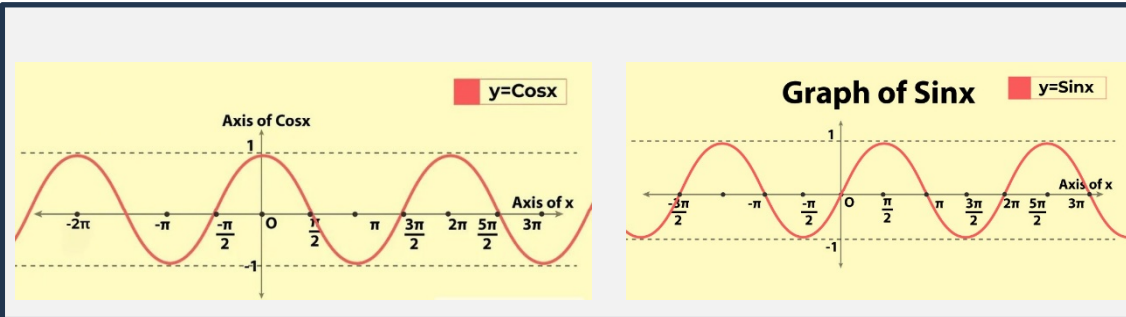
Fig. 1.18

Here,

- a = lattice periodicity (distance between barriers/wells).
- k = Bloch wavevector.
- $\alpha = \sqrt{2mE/\hbar^2}$ is the free-particle wavevector inside the well region.

P is a dimensionless measure of the barrier strength (related to the barrier height V_0 and width b). $P = V_0 b 2ma/\hbar^2$

Note!



On substituting the value of α and β

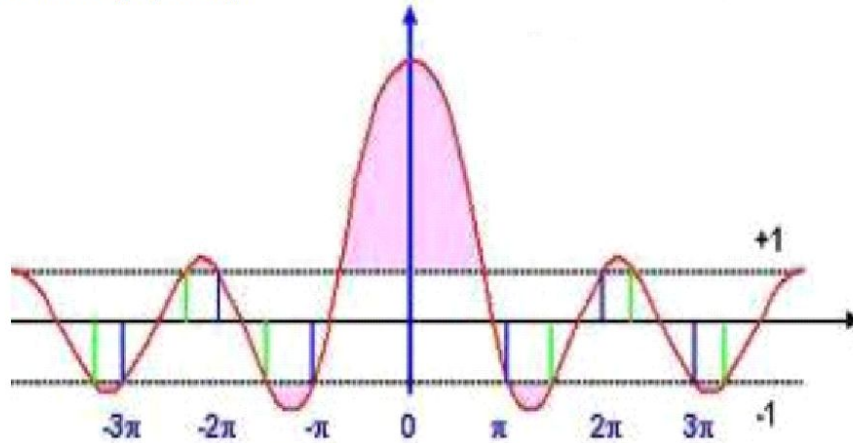
We get

$$p \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

Here $p = \frac{m v_0 b a}{\hbar^2}$ is scattering power

And ' $v_0 b$ ' is known as barrier strength.

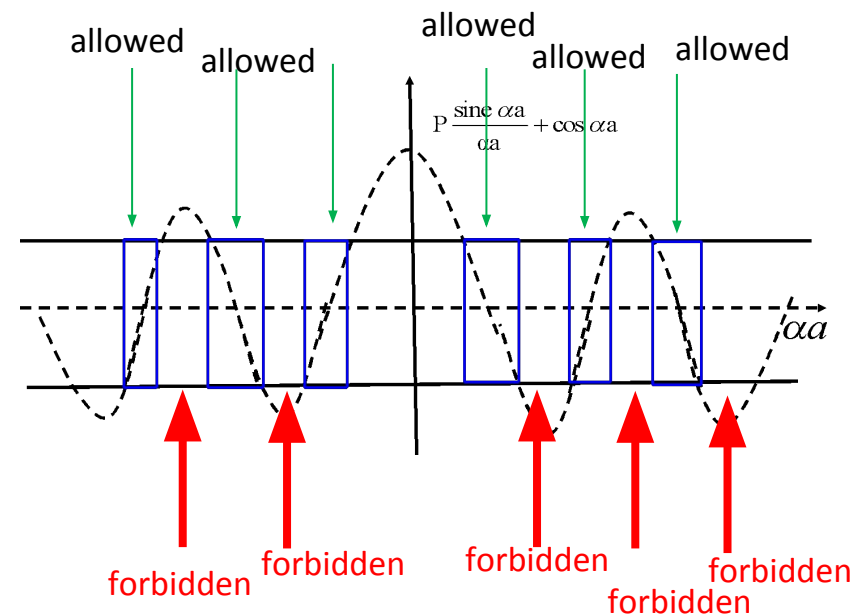
It can be solved graphically.



We notice that the only allowed energies are those for which

$$-1 \leq f(\alpha a) \leq 1$$

Whenever $f(\alpha a)$ is outside the domain $[-1, 1]$, there are no solutions.



*Energies where $f(\alpha a)$ lies outside this window are **forbidden states** → **band gaps***

what you saw in that plot is the mathematical origin of the multiple band / band-gap structure of real solids.

How This Equation Creates Bands and Gaps

The left-hand side, $\cos(ka)$, is always between -1 and 1 for a *real* wavevector k . For the equation to have a solution, the right-hand side must also be between -1 and 1.

$$-1 \leq \cos(\alpha a) + \frac{\lambda}{\alpha a} \sin(\alpha a) \leq 1$$

1. **Allowed Energies (Bands):** The ranges of energy E (which is proportional to α^2) for which the right-hand side of the equation lies between -1 and +1 correspond to **allowed energy bands**. For these energies, a real wavevector k exists, and the electron can propagate through the crystal.
2. **Forbidden Energies (Gaps):** The ranges of energy E for which the magnitude of the right-hand side is *greater than 1* correspond to **forbidden energy bands** or **band gaps**. For these energies, no propagating wave solution exists. The wavevector k becomes complex, leading to exponentially decaying waves.

Multiple allowed and forbidden regions

When you plot $f(\alpha a)$ vs. αa , you see oscillations that repeatedly cross in and out of the $[-1, 1]$ band:

- Where the curve lies **inside** $[-1, 1]$ $\rightarrow$ these correspond to **allowed energy bands**.
- Where the curve lies **outside** $[-1, 1]$ $\rightarrow$ these correspond to **forbidden bands (band gaps)**.

So the solid naturally develops multiple bands, separated by forbidden regions.

Physical meaning

- The **first allowed region** = **valence band** (lowest energy electrons).
- The **first gap** = **band gap**.
- The **second allowed region** = **conduction band**.
- Higher regions = higher conduction bands.

This is why the Kronig–Penney model is considered the simplest demonstration of **band formation in solids**.

$$(P/\alpha a)\sin \alpha a + \cos \alpha a = \cos ka$$

Where, $P = mV_0ab/\hbar$ is called scattering power

V_0b = barrier strength

$$\alpha^2 = 2mE/\hbar^2 \text{ and}$$

$k = 2\pi/\lambda$ is the wave vector

(known relationship in quantum mechanics)

$$\frac{P}{\alpha a} \sin \alpha a + \cos \alpha a$$

Remember!

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

Detail explanation for $k = 2\pi/\lambda$

The relationship $\lambda = \frac{2\pi}{k}$ connects the wavelength λ of a wave to its angular wave number k . Here's a breakdown of why this relationship holds:

1. Definitions:

- **Wavelength (λ):** This is the distance between successive peaks (or troughs) of a wave. It is a measure of how long one complete wave cycle is.

- **Angular wave number (k):** This is defined as the number of radians per unit distance. It is given by the formula:

$$k = \frac{2\pi}{\lambda}$$

This means k tells us how many radians correspond to one unit of distance.

2. Understanding k :

- In one complete wave cycle, the wave goes through 2π radians (from peak to peak or trough to trough). Hence, if you have a wavelength λ , the wave number k indicates how many radians fit into that distance.

3. Deriving the Relationship:

- Rearranging the wave number formula $k = \frac{2\pi}{\lambda}$ gives:

$$\lambda = \frac{2\pi}{k}$$

This shows that the wavelength is inversely proportional to the wave number.

4. Physical Interpretation:

- A larger wave number k means that there are more cycles in a given distance, which results in a shorter wavelength λ . Conversely, a smaller k means fewer cycles, leading to a longer wavelength.

In summary, $\lambda = \frac{2\pi}{k}$ illustrates the inverse relationship between wavelength and angular wave number, rooted in the definition of how waves propagate through space.

In the equation, $\cos(ka) = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$, a is the lattice constant or is the length of the potential well? In the beginning, it was mentioned that $a_{\text{lattice}} = a_{\text{well}} + b_{\text{barrier}}$. Please clarify it?

In the equation:
$$\cos(ka) = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$$

The constant a is indeed the lattice constant, which in the full Kronig-Penney model is defined as $a_{\text{well}} + b_{\text{barrier}}$.

Let's trace the logic one more time to make it crystal clear:

1. **General Kronig-Penney Model:** We have a repeating unit of a well (width a_{well}) and a barrier (width b_{barrier}). The periodicity is therefore Lattice Constant (a) = $a_{\text{well}} + b_{\text{barrier}}$.
2. **Delta-Function Simplification:** To derive the simpler equation above, we take the limit where the barrier becomes a delta function. This means:
 - The barrier width b_{barrier} shrinks to zero: $b_{\text{barrier}} \rightarrow 0$
 - The barrier strength is contained in the parameter P , which is proportional to the area under the barrier ($V_0 * b_{\text{barrier}}$).

3. **The Result:** In this limit, the lattice constant a remains the distance between adjacent delta-function barriers. Since $b_{\text{barrier}} \rightarrow 0$, the well width a_{well} effectively becomes equal to the lattice constant a .
- **Physically:** The lattice points (the locations of the delta-function barriers) are separated by a distance a .
 - **Mathematically:** The general dispersion relation simplifies to the one above, where a is the fundamental period of the potential, $V(x) = V(x + a)$.

So, your statement is 100% accurate: In that equation, a is the lattice constant, which is conceptually $a_{\text{well}} + b_{\text{barrier}}$ from the original physical picture.

The delta-function model is just a convenient mathematical limit where $b_{\text{barrier}} = 0$, making $a_{\text{well}} = a$.

Now, analyze the equation again,

After solving the Schrodinger time-independent wave equation for regions 1 and 2 :

$$d^2\Psi/dx^2 + 8\pi^2m/h^2[E - V(x)]\Psi = 0 \text{ ---(1)}$$

Region 1: ($V(x) = 0$) ($0 < x < a$)

Region 2: ($V(x) = V_0$) ($-b < x < 0$)

and applying Bloch theorem and boundary conditions: $\Psi = e^{ikx}u_k(x)$ --- (2)

We get [$P \sin \alpha a / \alpha a + \cos \alpha a = \cos ka$] --- (Final equation)

where P represents the strength with which ions attract the electrons toward each other.

Case 1: When $P \rightarrow 0$

If the barrier strength is zero, the potential is just flat (no periodic barriers at all).

We get $\cos \alpha a = \cos ka$

$$\alpha a = ka$$

$$\alpha = k \text{ (note it!)}$$

$$\alpha^2 = k^2$$

$$8\pi^2mE/h^2 = (2\pi/\lambda)^2$$

$$E = p^2/2m \text{ ---(3)}$$

From equation (3) we can say that the total energy of an electron is purely kinetic energy, and the electron is free to move inside the solid crystal. Hence we can conclude that it is the case for **Conductors**.

$$(\hbar = \frac{h}{2\pi})$$

$$k = 2\pi/\lambda$$

(known relationships in quantum mechanics)

Detail calculation when $p \rightarrow 0$

If $P \rightarrow 0$

$$\cos \alpha a = \cos ka$$

$$\alpha a = ka$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{2m}{\hbar^2} E = \left(\frac{2\pi}{\lambda}\right)^2 = \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{\hbar^2}{2m} \times \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{\hbar^2}{4\pi^2 \times 2m} \times \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{1}{2m} \left(\frac{h}{\lambda}\right)^2$$

$$E = \frac{p^2}{2m}$$

$\left. \begin{array}{l} P = \hbar k \\ P = \frac{h}{\lambda} \end{array} \right\}$
Crystal Momentum

Physical meaning

- With $P = 0$, there are **no barriers and no periodic potential**, so there are **no band gaps**.
- The electron is completely free, and the energy spectrum is continuous and parabolic.

- With $P = 0$, there are **no barriers and no periodic potential**, so there are **no band gaps**.
- The electron is completely free, and the energy spectrum is continuous and parabolic.

$$\longrightarrow \alpha = k$$

Remember!

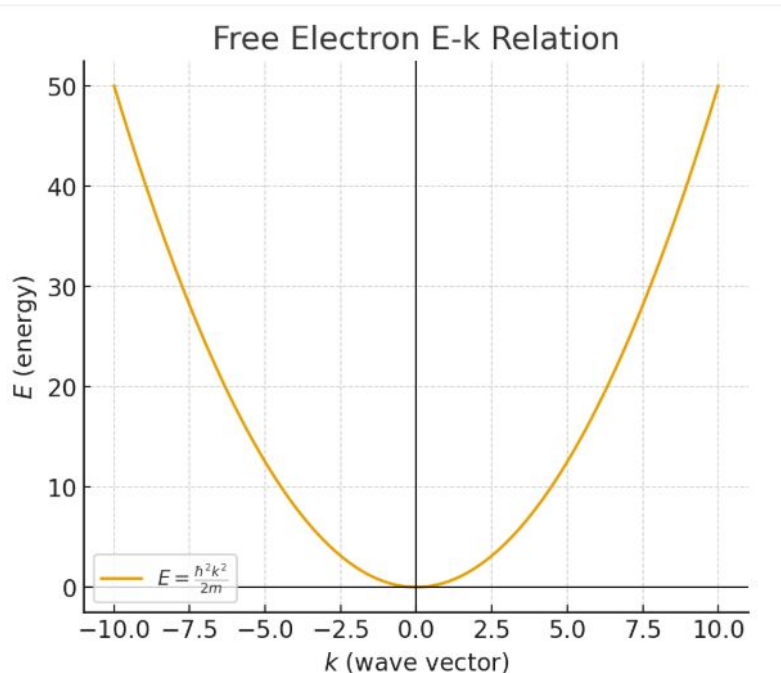
$$(\hbar = \frac{h}{2\pi})$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

Here's the sketch of the **free electron $E-k$ relation**:

It shows the parabolic dependence of energy on wave vector:



$$E = \frac{\hbar^2 k^2}{2m}$$

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

Nature of k (Wavevector):

- In an **infinite crystal** (or in free space with no boundary conditions), the wavevector k is **continuous**.
- **Important:** At $P=0$, there are no band gaps. These energies are just points on a continuous parabola.

From equation (3) we can say that the total energy of an electron is purely kinetic energy, and the electron is free to move inside the solid crystal. Hence we can conclude that it is the case for Conductors.

Remember the equation!

Case 2: $P \rightarrow \infty$ (very strong barrier)

- The second term

$$\frac{mP}{\hbar^2 \alpha} \sin(\alpha a) \text{ dominates.}$$

- For the equation to remain physically meaningful, the **right-hand side** must still lie between -1 and $+1$.

That means the only way to satisfy the condition when $P \rightarrow \infty$ is if

$$\sin(\alpha a) = 0 \quad \Rightarrow \quad \alpha a = n\pi, \quad n = 1, 2, 3, \dots \quad \sin(\alpha a) \approx 0.$$

But recall $\alpha = \sqrt{2mE/\hbar^2}$.

So the allowed energies are:

$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2.$$

$$E_n = \frac{n^2 h^2}{8ma^2}$$

as $P \rightarrow \infty$, the Kronig–Penney model reduces to isolated infinite wells of width a , and only discrete bound energy levels are allowed (no bands)

From equation (4) we can say that the energy values are discrete and the electron is bound in the potential well and hence it does not move. Hence we can conclude that it is the case for Insulators.

The concept of insulator, semiconductor and metal

So:

- $P = 0$: free electron $\rightarrow$ no band gaps. This corresponds to a metal with overlapping bands, because electrons can move without restriction
- $0 < P < \infty$: weak to strong periodic potential $\rightarrow$ bands with finite width and gaps. Weak to strong periodic potential $\rightarrow$ bands with finite width and gaps. Energy levels split into allowed bands separated by forbidden gaps. If it lies in a small gap (~ 1 eV) $\rightarrow$ semiconductor (electrons need some energy to jump the gap). If the gap is large ($\sim > 3$ eV) $\rightarrow$ insulator (electrons cannot jump easily). *Stronger $P \rightarrow$ gaps become wider.*

$P \rightarrow \infty$: impenetrable barriers $\rightarrow$ discrete energy levels like infinite wells. Electrons are fully localized like in isolated atoms or a strongly bound system. This is more of a model limit rather than a real material. Real solids rarely reach this limit. *Electron is confined $\rightarrow$ discrete energy levels like particle in a box.*

$0 < P < \infty$: weak to strong periodic potential

Again, the maximum and minimum values for Cosine function is ± 1

Now, $E = \frac{\alpha^2 \hbar^2}{2m}$

Apply, $\alpha = k$

Theoretically, you cannot substitute $\alpha=k$, when $0 < P < \infty$.
but, that equivalence holds true in the limiting case $P \rightarrow 0$

So,

Brillouin zones:

$$\cos ka = \pm 1$$

$$ka = n\pi$$

$$k = \pm \frac{n\pi}{a}$$

$$k = \pm \frac{\pi}{a} \rightarrow \text{First B.Z.}$$

$$k = \pm \frac{2\pi}{a} \rightarrow \text{Second B.Z.}$$

Remember! Slide 103

$$k = \pm \frac{n\pi}{a}$$

Remember! Slide 97 $(\hbar = \frac{h}{2\pi})$

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

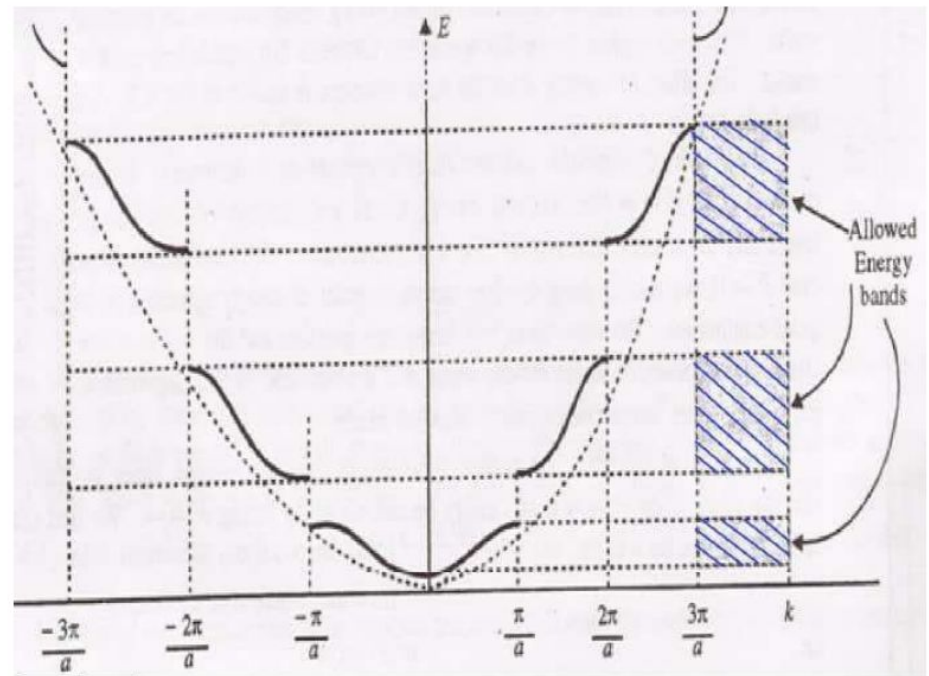
$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

We get, **Brillouin Zones (E-K curve):** a parabolic curve

Brillouin Zones

The Brillouin zone is a representation of permissive values of k of the electrons in one, two or three dimensions.

Thus, the energy spectrum of an electron moving in the presence of a periodic potential fields is divided into allowed zones and forbidden zones.



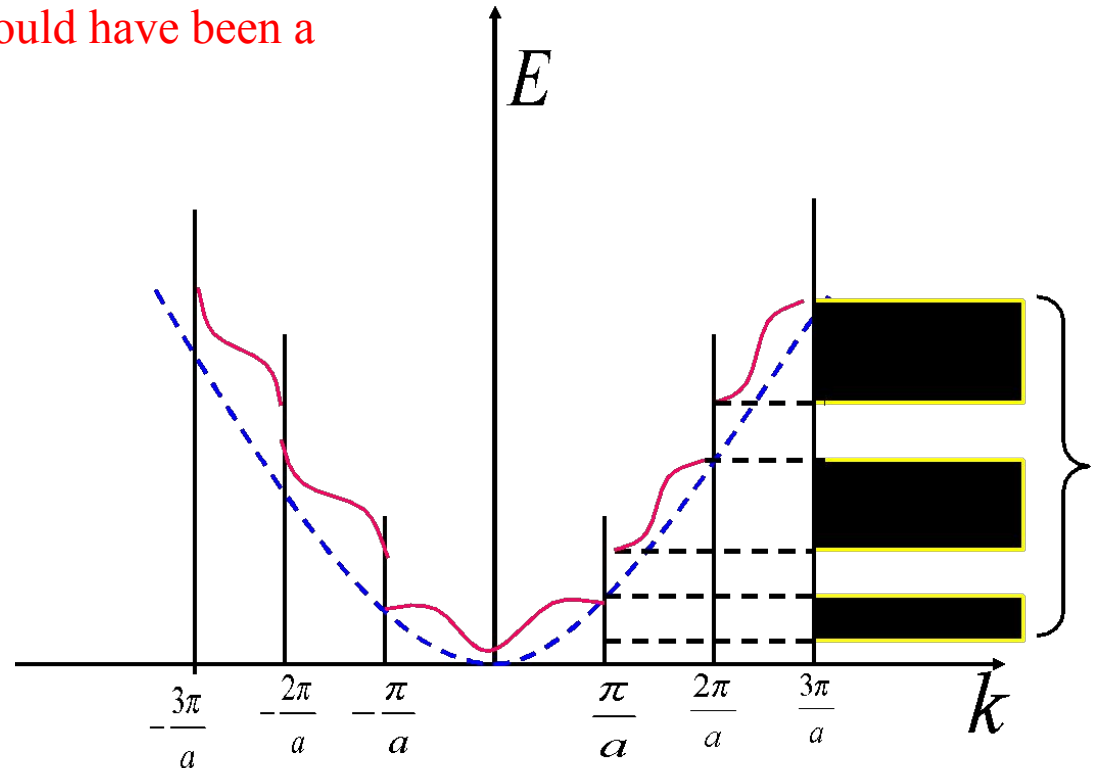
k should have a discrete values

Key points:

- Band gaps appear at **Brillouin zone boundaries**.
- Width of the gap $\propto$ **strength of periodic potential P**.

Otherwise, for $P=0 \rightarrow$ the curve would have been a smooth parabola (free electron)

E-k Diagram



The presence of the periodic potential filters out certain k-values $\rightarrow$ only those corresponding to allowed solutions survive.

- This is how continuous free-electron parabola splits into discrete allowed bands separated by forbidden gaps.

Physical picture of “discrete” k

- k itself is **not quantized into isolated values** (like in a finite box), rather:

Within an **allowed band**, k is continuous.

At the **Brillouin zone boundaries** ($k=\pm\pi/a, \pm2\pi/a, \pm3\pi/a, \dots$), **band gaps open**.

This periodic “cutting out” of k-regions makes the **spectrum discrete in energy**, not in k.

For $n = 1, 2, 3$:

- $E_1^\pm \approx \frac{\hbar^2}{2m} \left(\frac{\pi}{a}\right)^2 \pm \frac{P\hbar^2}{ma^2\pi}$
- $E_2^\pm \approx \frac{\hbar^2}{2m} \left(\frac{2\pi}{a}\right)^2 \pm \frac{P\hbar^2}{2ma^2\pi}$
- $E_3^\pm \approx \frac{\hbar^2}{2m} \left(\frac{3\pi}{a}\right)^2 \pm \frac{P\hbar^2}{3ma^2\pi}$

Conclusions

From the Fig. 1.18 the following conclusions can be made.

- (i) The energy spectrum has a number of allowed energy bands denoted by solid horizontal line separated by forbidden band gaps denoted by dotted lines.
- (ii) The width of allowed energy band (shaded portion) increases with the increase in αa .
- (iii) When P is increased, the binding energy of the electrons with the lattice points is also increased. Therefore the electron will not be able to move freely and hence the width of the allowed energy band is decreased. Especially for $P \rightarrow \infty$, then allowed energy band becomes infinitely narrow and the energy spectrum becomes a line spectrum as shown in Fig. 1.19.

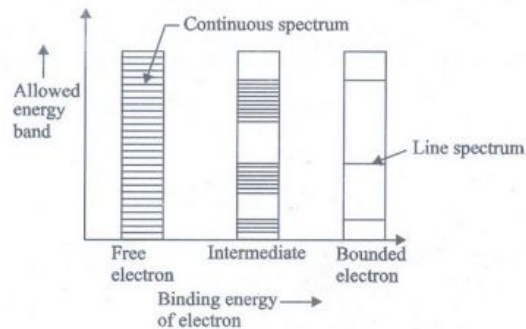


Fig. 1.19

- (iv) When P is decreased, the binding energy of the electron decreases and thus it moves freely over the lattice points and hence we get a wide range of allowed energy levels as shown in Fig. 1.19.
- (v) Thus by varying P from zero to infinity we get the energy spectra of all ranges.

Repeat slide

Conclusion from Kronig –Penny Model:

- 1). The Energy spectrum of E consists of an infinite number of allowed energy bands separated by intervals in which, there are no allowed energy levels. These are known as forbidden regions.
- 2). When α increases, the first term of e on LHS decreases, so that the width of the allowed energy bands is increased and forbidden energy regions become narrow.
- 3). The width of the allowed band decreases with the increase of p value. When $p \rightarrow \alpha$, the allowed energy regions become infinitely narrow and the energy spectrum becomes a line spectrum.

write the energy expression, when $k=n*\pi/a$ for a series of n values like $n=1,2,3$, in the context of Kronig penny model

The fundamental equation is:

$$\cos(ka) = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$$

where $\alpha = \frac{\sqrt{2mE}}{\hbar}$, and P is the dimensionless barrier strength.

Step 1: Substitute $k = \frac{n\pi}{a}$

If $k = \frac{n\pi}{a}$, then $ka = n\pi$. Therefore:

$$\cos(ka) = \cos(n\pi) = (-1)^n$$

Substituting this into the dispersion relation gives:

$$(-1)^n = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$$

This is the key equation that determines the energy E (through α) at the zone boundaries.

Step 2: Consider the Case of Vanishing Barriers ($P \rightarrow 0$)

This is the free-electron case, which gives us a reference point. If $P = 0$, the equation simplifies to:

$$(-1)^n = \cos(\alpha a)$$

The solutions to this are:

$$\alpha a = n\pi$$

(Since $\cos(n\pi) = (-1)^n$).

Substituting the definition of α :

$$\frac{\sqrt{2mE}}{\hbar} a = n\pi$$

Solving for E :

$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2$$

This is the familiar **free-electron parabolic energy** at the zone boundary. For $n = 1, 2, 3$:

- $E_1 = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2$
- $E_2 = \frac{\hbar^2}{2m} \left(\frac{2\pi}{a} \right)^2$
- $E_3 = \frac{\hbar^2}{2m} \left(\frac{3\pi}{a} \right)^2$

Important: At $P = 0$, there are no band gaps. These energies are just points on a continuous parabola.

Step 3: The General Case with Finite Barriers ($P > 0$)

For $P > 0$, the equation $(-1)^n = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$ cannot be solved algebraically for E . It must be solved numerically or graphically. However, we can understand what happens conceptually and find the energies at the **edges of the band gaps**.

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Graphical Interpretation

The right-hand side of the equation, $f(\alpha a) = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$, is a function that oscillates and whose amplitude decreases with αa .

1. **Band Edges Occur when $f(\alpha a) = \pm 1$:** The Brillouin zone boundary ($k = \frac{n\pi}{a}$) is where the band gaps open. The top of one band and the bottom of the next band *both* satisfy the condition that $\cos(ka) = \pm 1$.
2. **Two Solutions per n :** For a given integer n , there will be **two solutions** for αa (and hence two energies) that satisfy $f(\alpha a) = (-1)^n$. These are the **lower and upper band edges** for the gap that opens near the free-electron energy E_n .
 - The lower solution, $(\alpha a)_n^-$, gives the energy at the **top of the lower band** (valence band edge for gap # n).

- The higher solution, $(\alpha a)_n^+$, gives the energy at the **bottom of the upper band** (conduction band edge for gap #n).

Approximate Solutions for Small P

For a small potential barrier ($P \ll 1$), we can use perturbation theory. The band gap E_g that opens at $k = \frac{n\pi}{a}$ is approximately:

$$E_g^{(n)} \approx \frac{2P|V_n|}{n\pi}$$

where V_n is a Fourier component of the potential. For the delta-function model, this simplifies. The gap center is approximately at the free-electron energy E_n , and the gap width is:

$$E_g^{(n)} \approx \frac{2P}{a^2} \frac{\hbar^2}{2m} \frac{2}{n\pi} = \frac{2P\hbar^2}{ma^2n\pi}$$

Therefore, the two band edge energies for small P are:

$$E_n^\pm \approx \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2 \pm \frac{1}{2} E_g^{(n)} \approx \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2 \pm \frac{P\hbar^2}{ma^2n\pi}$$

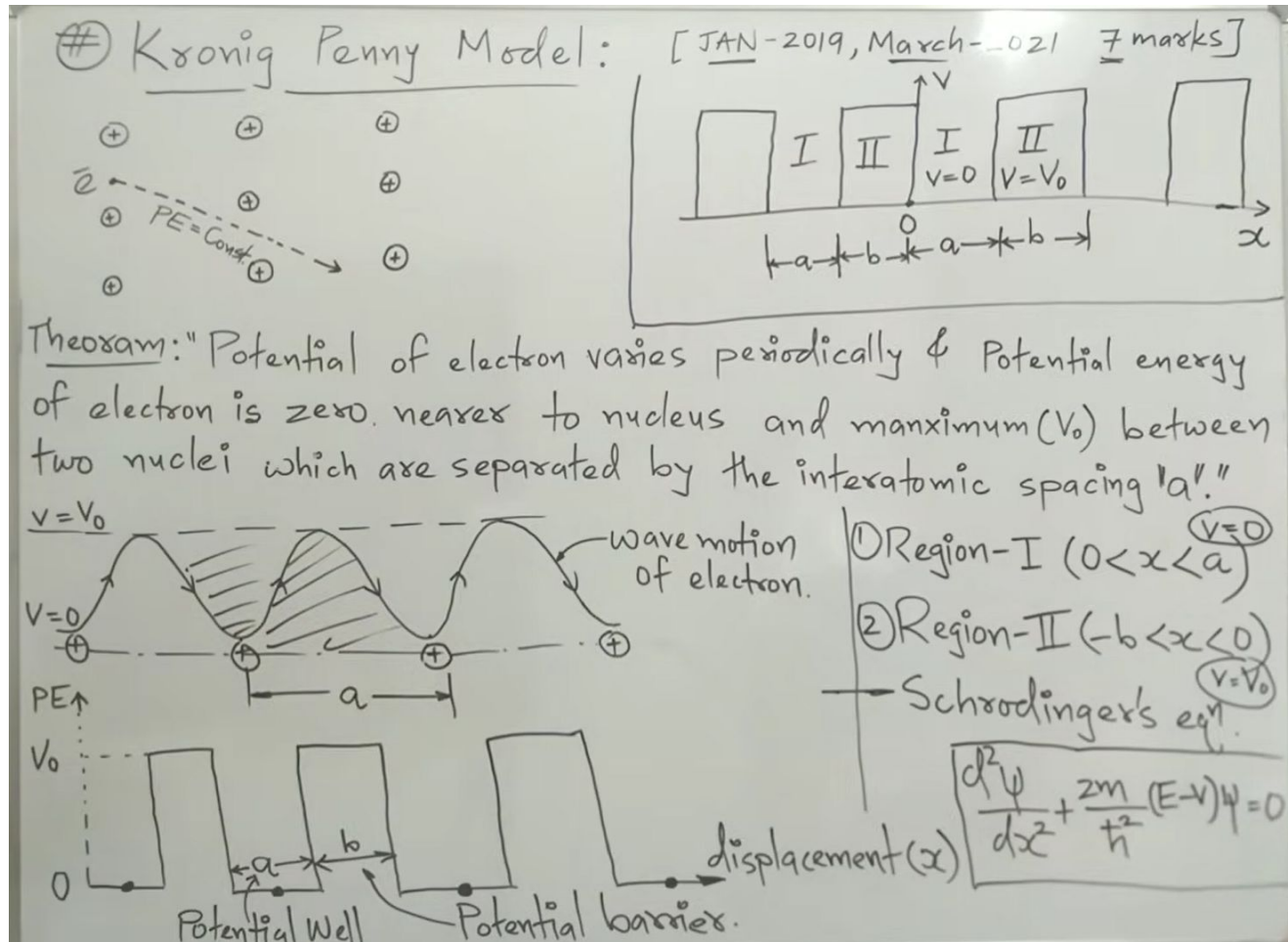
For $n = 1, 2, 3$:

- $E_1^\pm \approx \frac{\hbar^2}{2m} \left(\frac{\pi}{a}\right)^2 \pm \frac{P\hbar^2}{ma^2\pi}$
- $E_2^\pm \approx \frac{\hbar^2}{2m} \left(\frac{2\pi}{a}\right)^2 \pm \frac{P\hbar^2}{2ma^2\pi}$
- $E_3^\pm \approx \frac{\hbar^2}{2m} \left(\frac{3\pi}{a}\right)^2 \pm \frac{P\hbar^2}{3ma^2\pi}$

n	k-value	Free-Electron Energy (P=0)	Energy with Finite P (Band Edges)
1	$k = \frac{\pi}{a}$	$E_1 = \frac{\hbar^2}{2m} \left(\frac{\pi}{a}\right)^2$	E_1^- (Top of 1st Band) and E_1^+ (Bottom of 2nd Band)
2	$k = \frac{2\pi}{a}$	$E_2 = \frac{\hbar^2}{2m} \left(\frac{2\pi}{a}\right)^2$	E_2^- (Top of 2nd Band) and E_2^+ (Bottom of 3rd Band)
3	$k = \frac{3\pi}{a}$	$E_3 = \frac{\hbar^2}{2m} \left(\frac{3\pi}{a}\right)^2$	E_3^- (Top of 3rd Band) and E_3^+ (Bottom of 4th Band)

In conclusion: At $k = n\pi/a$, the energy is not a single value but is split by the band gap. There are two possible energies, defining the edges of the forbidden gap. The exact values must be found by solving the transcendental equation $(-1)^n = \cos(\alpha a) + P \frac{\sin(\alpha a)}{\alpha a}$ numerically for a given value of P .

Approach 2



https://www.youtube.com/watch?v=ZoFOpJT_cN0

Region-I ($0 < x < a$) $V=0$ [Potential Well]

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0 \quad \text{--- (1)}$$

Region-II ($-b < x < 0$) $V=V_0$ [Potential Barrier]

$$\frac{d^2\psi}{dx^2} + \frac{2m(E-V_0)}{\hbar^2} \psi = 0 \quad \text{--- (2)}$$

Assuming $\frac{2mE}{\hbar^2} = \alpha^2$
and $\frac{2m(V_0-E)}{\hbar^2} = \beta^2$

So, eqⁿ (1) & (2) becomes

$$\frac{d^2\psi}{dx^2} + \alpha^2 \psi = 0 \quad \text{--- (3)}$$

and $\frac{d^2\psi}{dx^2} - \beta^2 \psi = 0 \quad \text{--- (4)}$

general solⁿ of eqⁿ (3) & (4) is

$$\psi = e^{-ikx} u_k(x) \quad \text{(5) & (6)}$$

$$\frac{d\psi}{dx} \Rightarrow \frac{d^2\psi}{dx^2}$$

$$\left[\frac{d^2\psi}{dx^2} + \frac{2m(E-V)}{\hbar^2} \right] \psi = 0$$

By solving (5) & (6) with the help of matrix method

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos k a$$

Above eqⁿ is called as Kronig Penny eqⁿ for 1D motion of an electron.

Above eqⁿ we will apply for two cases and will find the energy of electron.

Case-I $P \rightarrow 0$ (Potential Well)

Case-II $P \rightarrow \infty$ (Potential barrier)

After first and 2nd derivative of ψ , and putting the values in equation 3 & 4, we get two different equation 5 & 6. Equation 5 & 6 are not written here.

https://www.youtube.com/watch?v=ZoFOpJT_cN0

case-I $P \rightarrow 0$

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$\cos \alpha a = \cos ka$$

$$\alpha a = ka$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{2mE}{\hbar^2} = \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{2\pi^2}{\lambda^2} \times \frac{\hbar^2}{m}$$

$$E = \frac{2\pi^2}{\lambda^2} \times \frac{h^2}{4\pi^2 \times m}$$

$$E = \frac{1}{\lambda^2} \times \frac{h^2}{2m}$$

$$E = \frac{m^2 v^2}{2m}$$

$$E = \frac{p^2}{2m}$$

Free electron energy when $P \rightarrow 0$

$$k = \frac{2\pi}{\lambda}$$

$$k^2 = \frac{4\pi^2}{\lambda^2}$$

$$\hbar = \frac{h}{2\pi}$$

$$\hbar^2 = \frac{h^2}{4\pi^2}$$

$$\lambda = \frac{h}{mv}$$

$$p = mv$$

case-II $P \rightarrow \infty$

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$\sin \alpha a = \frac{(\cos ka - \cos \alpha a) \alpha a}{P} \quad \checkmark \quad \alpha^2 = \frac{2mE}{\hbar^2}$$

$$\sin \alpha a = 0$$

$$\alpha a = \sin^{-1}(0)$$

$$\alpha a = 0, \pi, 2\pi, 3\pi, \dots$$

$$\alpha a = n\pi \quad (n=0, 1, 2, 3, \dots)$$

$$\alpha = \frac{n\pi}{a}$$

$$\alpha^2 = \frac{n^2 \pi^2}{a^2}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{a^2}$$

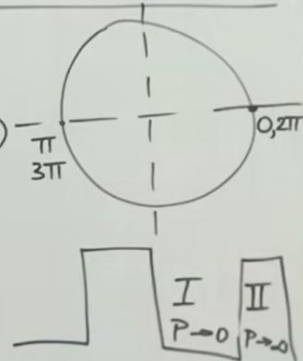
$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

$$E = \frac{n^2 \pi^2}{2m} \times \frac{h^2}{4\pi^2} \times \frac{1}{a^2}$$

$$E = \frac{n^2 h^2}{8ma^2}$$

$$\frac{d^2 \psi}{dx^2} + \frac{2m(E-V)}{\hbar^2} \psi = 0$$

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$



$$\left(\hbar = \frac{h}{2\pi} \right)$$

https://www.youtube.com/watch?v=ZoFOpJT_cN0

Approach 2: Result & Discussions continue.....

1. The energy spectrum of the electrons consists of energy bands allowed and forbidden.
2. The width of the allowed energy region or band increases with increasing values of P or energy.
3. In case $P \rightarrow \infty$, the allowed region becomes infinitely narrow.

Since as $P \rightarrow \infty$

$$\sin \alpha a = 0$$

i.e if $\alpha a = \pm n\pi$ with $n = 0,1,2,3, \dots$

and from Eq. (3), we obtain

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2 \quad \text{for } P \rightarrow \infty$$

$$E = \frac{\alpha^2 \hbar^2}{2m}$$

Remember!

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

$$(\hbar = \frac{h}{2\pi})$$

This is the result we obtain for a particle in a box of atomic dimensions with a constant potential i.e. electron tightly bound and tunneling through the barriers becomes improbable. This shows the case of an insulator.

In case when $P \rightarrow 0$, we obtain

$$\cos \alpha a = \cos Ka$$

$$\alpha = K$$

Hence
$$E = \frac{\hbar^2 K^2}{2m}$$

Remember!

where

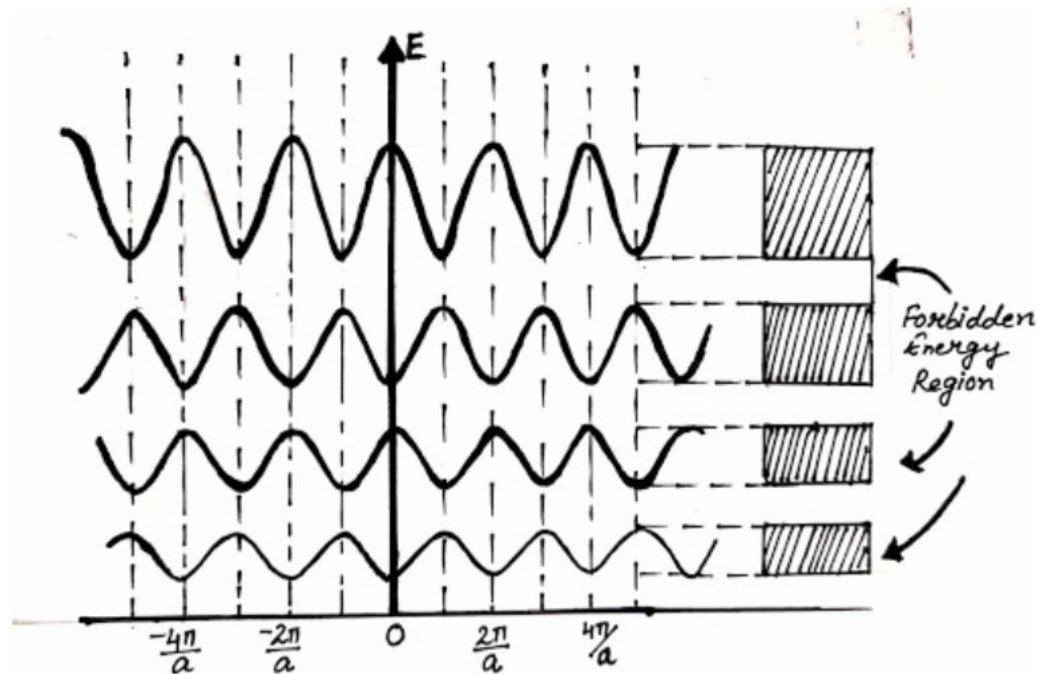
$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

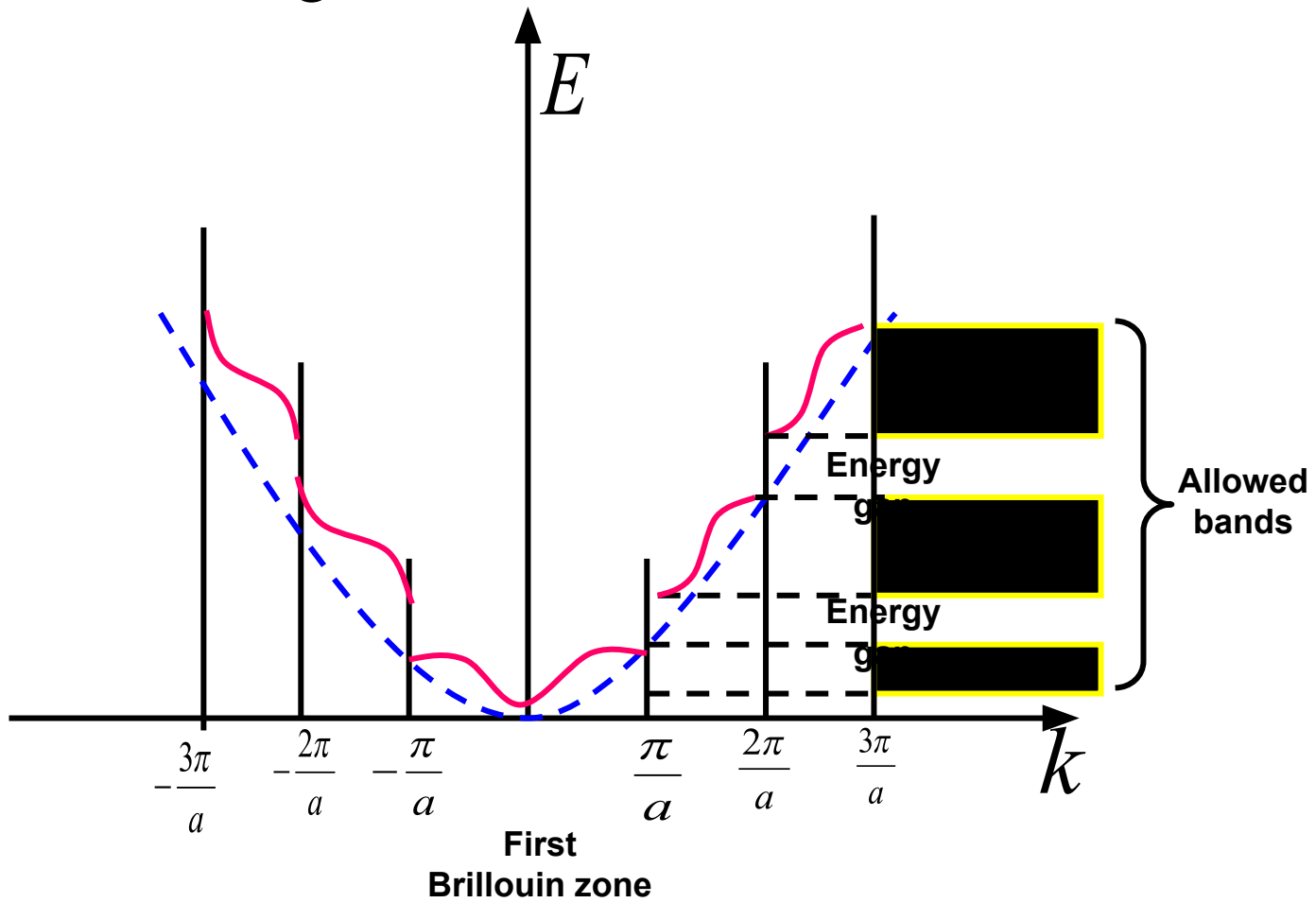
This represents the energy of a completely free electron for which any energy value is possible. This shows the case of the conductor.

4. E-K diagram

Right-hand side of $\cos Ka$ of eq. (15) is an even periodic function and its value does not change whether Ka is positive or negative. As a result, the total energy E of the electron is an even periodic function of K with a period $2\pi/a$. This periodic repetition of energy is shown in fig. A. This may be considered as obtained by the repetition of the region $-\frac{\pi}{a} < K < \frac{\pi}{a}$, this is the first Brillouin zone. This representation is known as the periodic zone scheme.



E-k Diagram



Conclusions

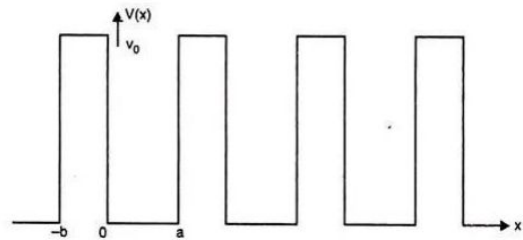
1. The motion of electrons in a periodic lattice is characterized by the bands of allowed energy separated by forbidden regions.
2. As the value of αa increases, the width of allowed energy bands also increases and the width of the forbidden bands decreases. i.e., the first term of equation decreases on the average with increasing αa .
3. Let us now consider the effect of varying barrier strength P . if $V_0 b$ is large, i.e. if p is large, the function described by the left hand side of the equation crosses $+1$ and -1 region as shown in figure. thus the allowed bands are narrower and the forbidden bands are wider.

Extra Slide

Approach 3

The Kronig-Penney model demonstrates that a simple one-dimensional periodic potential yields energy bands as well as energy band gaps. While it is an oversimplification of the three-dimensional potential and band structure in an actual crystal, it is an instructive tool to demonstrate how the band structure can be calculated for a periodic potential, and how allowed and forbidden energies are obtained when solving the Schrodinger equation.

The Kronig-Penney model is represented by the one-dimensional periodic potential as :



The potential is defined as $V(x) = V_0 \text{ } -b < x < 0$
 $= 0 \text{ } 0 < x < a$

The potential has a period of $c = a + b$.

The Schrodinger equation for this model is

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0 \quad 0 < x < a$$

$$\frac{d^2\psi}{dx^2} + \frac{2m(E - V_0)}{\hbar^2}\psi = 0 \quad -b < x < 0$$

Let

$$\alpha = \frac{(2mE)^{1/2}}{\hbar} \quad \beta = \frac{[2m(V_0 - E)]^{1/2}}{\hbar}$$

∴ The solutions of schrodinger Equations are

$$\psi_1 = Ae^{i\alpha x} + Be^{-i\alpha x} \quad 0 < x < a$$

and

$$\psi_2 = Ce^{\beta x} + De^{-\beta x} \quad -b < x < 0$$

Since the wave function has periodicity c

$$\therefore \psi(x + c) = e^{ikc}\psi(x)$$

Let $\psi_1(x) = \psi(x)$ for $-b < x < 0$
 and $\psi_2(x) = \psi(x)$ for $0 < x < a$,
 then following two boundary conditions are used

$$1.. \psi_1(0) = \psi_2(0) \text{ and } \frac{d\psi_1(0)}{dx} = \frac{d\psi_2(0)}{dx} \dots\dots\dots 2$$

Again , from Bloch’s theorem, $u(x) = u(x + a)$ therefore $u(x)$ and its first derivative must also be periodic.

Therefore, we have the two additional conditions

$$3.....u(-b) = u(a) \quad \text{and} \quad \frac{du(-b)}{dx} = \frac{du(a)}{dx} \dots\dots\dots 4$$

By using these four basic boundary condition

$$\text{We get} \quad A + B = C + D \quad \text{-----}(1)$$

$$\beta C - \beta D = i\alpha A - i\alpha B \text{-----}(2)$$

$$Ae^{i(\alpha-k)a} + Be^{-i(\alpha+k)a} = Ce^{-(\beta-ik)b} + De^{(\beta+ik)b} \text{-----}(3)$$

$$i(\alpha - k)Ae^{i(\alpha-k)a} - i(\alpha + k)Be^{-i(\alpha+k)a} = (\beta - ik)Ce^{-(\beta-ik)b} - (\beta + ik)De^{(\beta+ik)b} \text{-----}(4)$$

Four boundary conditions give four equations

For nontrivial solution of the above four equations

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ \beta & -\beta & i\alpha & -i\alpha \\ e^{(ik-\beta)b} & e^{(ik+\beta)b} & e^{i\alpha(a-k)} & e^{i\alpha(a+k)} \\ (\beta - ik)e^{(ik-\beta)b} & -(\beta + ik)e^{(ik+\beta)b} & i(\alpha - k)e^{i\alpha(a-k)} & -i(\alpha + k)e^{i\alpha(a+k)} \end{vmatrix} = 0$$

On solving the determinant, we get

$$\frac{\beta^2 + \alpha^2}{2\alpha\beta} \sin \beta b \sin \alpha a + \cosh \beta b \cos \alpha a = \cos K(a+b)$$

On substituting the value of α and β

We get

$$\frac{\beta^2 + \alpha^2}{2\alpha\beta} \sinh \beta b \sin \alpha a + \cosh \beta b \cos \alpha a = \cos K(a+b) \quad (17)$$

$$V_0 \rightarrow \infty, b \rightarrow 0$$

$$\text{If } b \rightarrow 0 \quad \sinh \beta b \rightarrow \beta b$$

$$\cosh \beta b \rightarrow 1$$

$$\frac{\beta^2 + \alpha^2}{2\alpha\beta} \times \beta b \sin \alpha a + \cos \alpha a = \cos \alpha a$$

$$\left(\frac{\beta^2 + \alpha^2}{2\alpha} \right) b \sin \alpha a + \cos \alpha a = \cos \alpha a \quad (18)$$

$$\text{Now, } \alpha^2 = \frac{2mE}{\hbar^2}$$

$$\beta^2 = \frac{2m}{\hbar^2} (V_0 - E)$$

$$\alpha^2 + \beta^2 = \frac{2mE}{\hbar^2} + \frac{2m}{\hbar^2} V_0 - \frac{2mE}{\hbar^2}$$

$$\alpha^2 + \beta^2 = \frac{2mV_0}{\hbar^2}$$

$$\frac{\alpha^2 + \beta^2}{2\alpha} = \frac{2mV_0}{2\alpha\hbar^2} = \frac{mV_0}{\alpha\hbar^2}$$

$$\frac{mV_0 b}{\alpha\hbar^2} \sin \alpha a + \cos \alpha a = \cos \alpha a$$

$$\frac{mV_0 ab}{\alpha\hbar^2} \sin \alpha a + \cos \alpha a = \cos \alpha a$$

$$\left(\frac{mV_0 ba}{\hbar^2} \right) \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos \alpha a$$

$$p \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos \alpha a$$

$$\text{Here } p = \frac{mV_0 ba}{\hbar^2} \text{ is scattering power}$$

And ' $V_0 b$ ' is known as barrier strength.

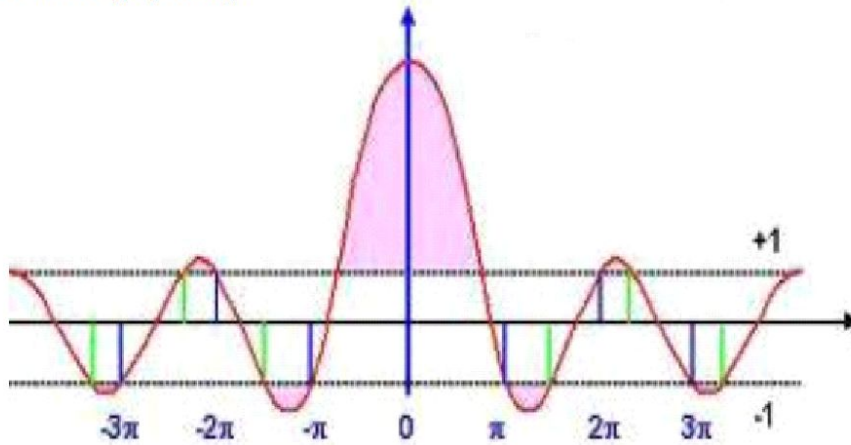
On substituting the value of α and β
We get

$$p \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

Here $p = \frac{mv_0 b a}{h^2}$ is scattering power

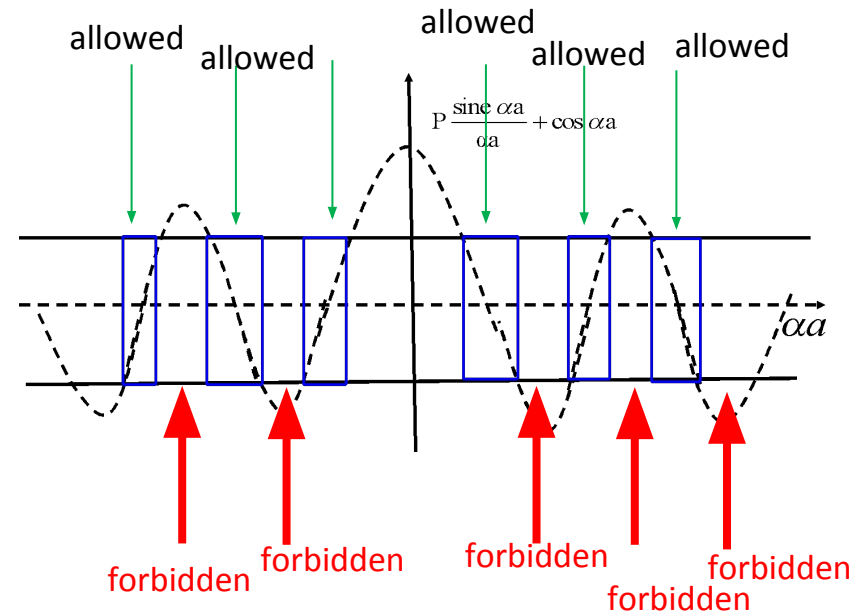
And ' $v_0 b$ ' is known as barrier strength.

It can be solved graphically.

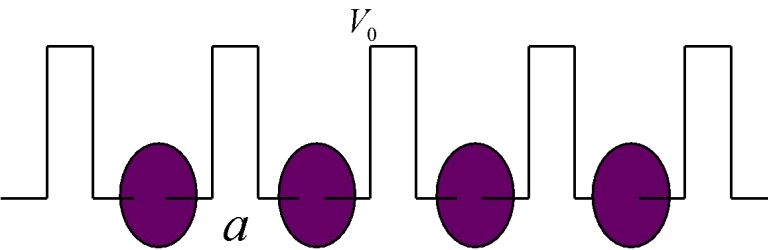
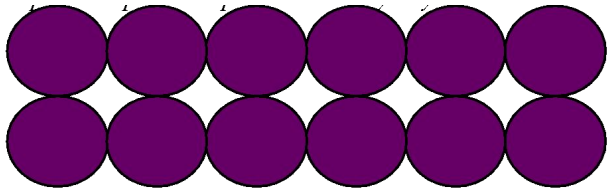


We notice that the only allowed energies are those for which
 $-1 \leq f(\alpha a) \leq 1$

Whenever $f(\alpha a)$ is outside the domain $[-1, 1]$, there are no solutions.

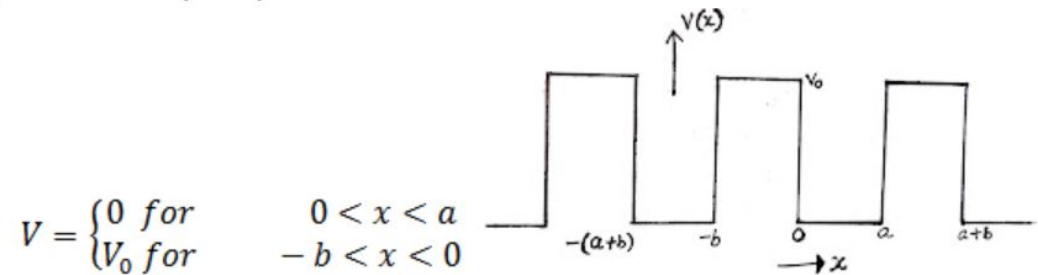


Approach 4



The Kronig-Penney Model

Kronig and Penney examined the behavior of electrons in a periodic potential by considering a relatively simple and one-dimensional model. It is assumed that the potential energy of an electron has the shape of a square well as shown in fig. The period of potential is $(a+b)$



This type of periodic potential is an approximation of the actual picture.

Now Schrodinger wave eq. for the two regions can be written as,

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}E\Psi = 0 \quad \text{for } 0 < x < a \quad \dots(1)$$

And

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\Psi = 0 \quad \text{for } -b < x < 0 \quad \dots(2)$$

It is assumed that E the energy of an electron is smaller than V_0

$$\text{Let } \frac{2mE}{\hbar^2} = \alpha^2 \quad \text{And } \frac{2m}{\hbar^2}(V_0 - E) = \beta^2 \quad \dots(3)$$

Where α and β are two real quantities.

So eq. (1) and (2) become

$$\frac{d^2\Psi}{dx^2} + \alpha^2\Psi = 0 \quad \text{for } 0 < x < a \quad \dots(4)$$

$$\frac{d^2\Psi}{dx^2} + \beta^2\Psi = 0 \quad \text{for } -b < x < 0 \quad \dots(5)$$

Since the potential is periodic, so the solution of eq. (4) and (5) must be of the form of Bloch functions i.e.

$$\Psi(x) = \exp(iKx) u_K(x) \quad \dots(6)$$

Where $u_K(x)$ is the periodic function x with period $(a+b)$

$$\therefore \frac{d\Psi}{dx} = iK \exp(iKx) u_K(x) + \exp(iKx) \frac{du_K}{dx}$$

And

$$\begin{aligned} \frac{d^2\Psi}{dx^2} = & -K^2 \exp(iKx) u_K(x) + iK \exp(iKx) \frac{du_K}{dx} + iK \exp(iKx) \frac{du_K}{dx} \\ & + \exp(iKx) \frac{d^2u_K}{dx^2} \end{aligned}$$

Or

$$\frac{d^2\Psi}{dx^2} = -K^2 \exp(iKx) u_K(x) + 2iK \exp(iKx) \frac{du_K}{dx} + \exp(iKx) \frac{d^2u_K}{dx^2} \dots(7)$$

Putting eq. (6) and (7) in eq. (4) and (5), we get

$$\frac{d^2u_1}{dx^2} + 2iK \frac{du_1}{dx} - (\alpha^2 - K^2)u_1 = 0 \text{ for } 0 < x < a \dots(8)$$

$$\frac{d^2u_2}{dx^2} + 2iK \frac{du_2}{dx} - (\beta^2 + K^2)u_2 = 0 \text{ for } -b < x < 0 \dots(9)$$

Where u_1 represents the value of $u_K(x)$ in the interval $0 < x < a$ and u_2 represents the value of in the interval $-b < x < 0$

The solutions of these equations are

$$u_1 = Ae^{i(\alpha-K)x} + Be^{-i(\alpha+K)x} \text{ for } 0 < x < a \quad \dots(10)$$

$$u_2 = Ce^{(\beta-iK)x} + De^{-(\beta+iK)x} \text{ for } -b < x < 0 \quad \dots(11)$$

Where A, B, C and D are arbitrary constants which can be determined by following boundary conditions.

$$[u_1]_{x=0} = [u_2]_{x=0} \quad \dots(12a)$$

$$\left[\frac{du_1}{dx}\right]_{x=0} = \left[\frac{du_2}{dx}\right]_{x=0} \quad \dots(12b)$$

$$[u_1]_{x=a} = [u_2]_{x=-b} \quad \dots(12c)$$

$$\left[\frac{du_1}{dx}\right]_{x=a} = \left[\frac{du_2}{dx}\right]_{x=-b} \quad \dots(12d)$$

Applying these boundary conditions to eqns. (10) and (11), we get

$$A + B = C + D \quad \dots(13a)$$

$$Ai(\alpha - K) - Bi(\alpha + K) = C(\beta - iK) - D(\beta + iK) \quad \dots(13b)$$

$$Ae^{i(\alpha-K)a} + Be^{-i(\alpha+K)a} = Ce^{(\beta-iK)b} + De^{-(\beta+iK)b} \quad \dots(13c)$$

$$\begin{aligned} Ai(\alpha - K)e^{i(\alpha-K)a} - Bi(\alpha + K)e^{-i(\alpha+K)a} \\ = C(\beta - iK)e^{(\beta-iK)b} - D(\beta + iK)e^{-(\beta+iK)b} \end{aligned} \quad \dots$$

(13d)

These eqns can be solved for non-zero value of A, B, C and D only if the determinant of the coefficient of A, B, C and D becomes zero.

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ i(\alpha - K) & -i(\alpha + K) & \beta - iK & -(\beta + iK) \\ e^{i(\alpha-K)a} & e^{-i(\alpha+K)a} & e^{(\beta-iK)b} & e^{-(\beta+iK)b} \\ i(\alpha - K)e^{i(\alpha-K)a} & -i(\alpha + K)e^{-i(\alpha+K)a} & (\beta - iK)e^{(\beta-iK)b} & -(\beta + iK)e^{-(\beta+iK)b} \end{vmatrix} = 0$$

On solving, we get

$$\frac{\beta^2 + \alpha^2}{2\alpha\beta} \sin \beta b \sin \alpha a + \cosh \beta b \cos \alpha a = \cos K(a + b) \quad \dots(14)$$

In order to make the situation more simple, Kronig and Penney considered the height of the potential barrier is very large. i.e

$V \rightarrow \infty$ and simultaneously the width of the barrier $b \rightarrow 0$ in such a way that the product $V_0 b$ remain finite.

$$\lim_{\beta b \rightarrow 0} \frac{\sin \beta b}{\beta b} = 1$$

Under these circumstances, eq. (14) becomes

$$\frac{mV_0 b}{\hbar^2 \alpha} \sin \alpha a + \cos \alpha a = \cos Ka$$

This eq. can be written more simply as

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos Ka \quad \dots(15)$$

Where $P = \frac{mV_0 ba}{\hbar^2} \quad \dots(16)$

Which is a measure of the area $V_0 b$ of the potential barrier. Large value of P means that given electron is more strongly bound to a particular potential well.

Now R.H.S of eq. (1.72) can assume values between +1 and -1 and hence only those value of αa are allowed which make L.H.S of this eq. lie between +1.

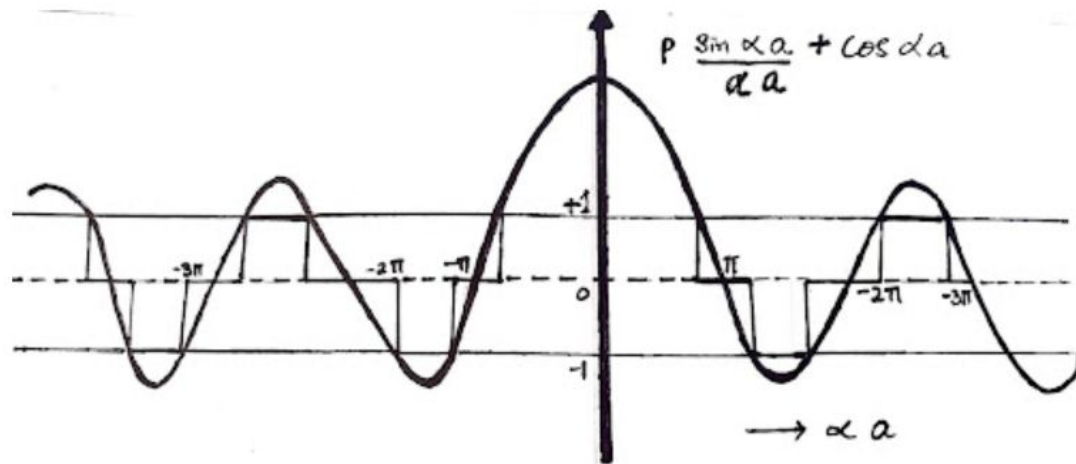


Fig shows left hand side of this equation as a function of αa for values $P = 3\pi/2$. Since α^2 is proportional to energy E , hence abscissa gives a measure of energy.

1. The energy spectrum of the electrons consists of energy bands allowed and forbidden.
2. The width of the allowed energy region or band increases with increasing values of P or energy.
3. In case $P \rightarrow \infty$, the allowed region becomes infinitely narrow.

Since as $P \rightarrow \infty$

$$\sin \alpha a = 0$$

i.e if $\alpha a = \pm n\pi$ with $n = 0, 1, 2, 3, \dots$

and from Eq. (3), we obtain

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2 \quad \text{for } P \rightarrow \infty$$

$$E = \frac{\alpha^2 \hbar^2}{2m}$$

Remember!

where

$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

$$(\hbar = \frac{h}{2\pi})$$

This is the result we obtain for a particle in a box of atomic dimensions with a constant potential i.e. electron tightly bound and tunneling through the barriers becomes improbable. This shows the case of an insulator.

In case when $P \rightarrow 0$, we obtain

$$\cos \alpha a = \cos Ka$$

$$\alpha = K$$

Hence
$$E = \frac{\hbar^2 K^2}{2m}$$

Remember!

where

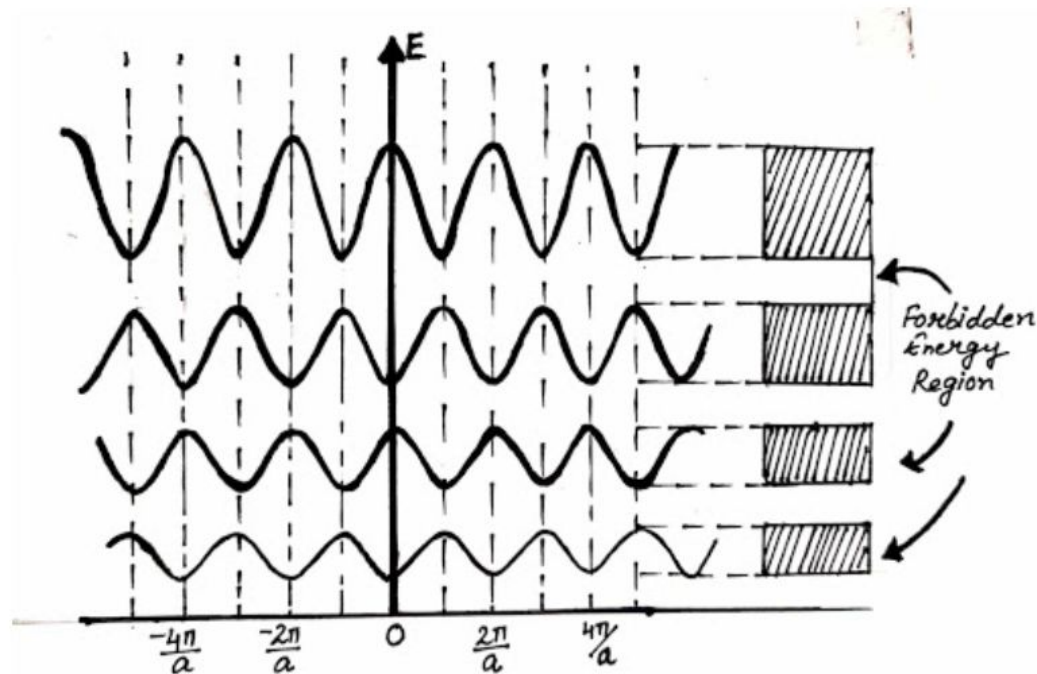
$$\alpha^2 = \frac{8\pi^2 m E}{h^2}$$

$$\beta^2 = \frac{8\pi^2 m}{h^2} [V_0 - E]$$

This represents the energy of a completely free electron for which any energy value is possible. This shows the case of the conductor.

4. E-K diagram

Right-hand side of $\cos Ka$ of eq. (15) is an even periodic function and its value does not change whether Ka is positive or negative. As a result, the total energy E of the electron is an even periodic function of K with a period $2\pi/a$. This periodic repetition of energy is shown in fig. A. This may be considered as obtained by the repetition of the region $-\frac{\pi}{a} < K < \frac{\pi}{a}$, this is the first Brillouin zone. This representation is known as the periodic zone scheme.



Conclusions

1. The motion of electrons in a periodic lattice is characterized by the bands of allowed energy separated by forbidden regions.
2. As the value of αa increases, the width of allowed energy bands also increases and the width of the forbidden bands decreases. i.e., the first term of equation decreases on the average with increasing αa .
3. Let us now consider the effect of varying barrier strength P . if $V_0 b$ is large, i.e. if p is large, the function described by the left hand side of the equation crosses $+1$ and -1 region as shown in figure. thus the allowed bands are narrower and the forbidden bands are wider.

Brillouin Zones

- Electrons in solids are permitted to be in allowed energy bands separated by forbidden energy gaps.
- The allowed energy band width increases with αa .
- The electron has allowed energy values in the region or zone extending from $k = (-\pi/a)$ to (π/a) , called the first Brillouin zone.
- Similarly, second Brillouin zone extends from

$$k = (-\pi/a) \text{ to } (-2\pi/a) \text{ and}$$

$$k = (\pi/a) \text{ to } (2\pi/a).$$